# INTEGRATED PHOTONIC GAS AND LIQUID SENSORS

# INTEGRATED PHOTONIC GAS AND LIQUID SENSORS

BY

RAGHI EL SHAMY, M.Sc.

A THESIS SUBMITTED TO THE DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

AND THE SCHOOL OF GRADUATE STUDIES

OF MCMASTER UNIVERSITY

IN PARTIAL FULFILMENT OF THE REQUIREMENTS

FOR THE DEGREE OF DOCTOR OF PHILOSOPHY

© Copyright by Raghi El Shamy, December 2024

All Rights Reserved

Doctor of Philosophy (2024) McMaster University

(Electrical and Computer Engineering) Hamilton, Ontario, Canada

TITLE: Integrated photonic gas and liquid sensors

AUTHOR: Raghi El Shamy

M.Sc., (Electrical Engineering)

Ain Shams University, Cairo, Egypt

SUPERVISOR: Xun Li

NUMBER OF PAGES: xxiv, 210

# Abstract

Chemical and biological detection is critical in various fields, yet conventional methods often suffer from limitations such as low sensitivity, restricted dynamic range, complex preparation, and bulky equipment. These shortcomings necessitate the development of an affordable, compact, simple, real-time, and accurate detection approach. Refractive index (RI) sensors are particularly well-suited for integrated on-chip optical sensing due to their high sensitivity to small-volume samples. Even minor changes in the refractive index over short distances can induce significant phase shifts in the propagating wave, enabling exceptionally high sensitivities. By leveraging CMOS-compatible technologies, RI sensors offer low-cost and large-scale fabrication capabilities while enabling the monolithic integration of electrical circuitry with optical sensors, resulting in compact, distributed, real-time, and remote sensing systems.

This thesis addresses four critical aspects of RI sensing:

1. Identifying the optimal platform and device for RI sensing.
2. Enhancing sensor selectivity to accurately identify medium compositions.
3. Improving sensor robustness against process and temperature variations.
4. Determining the optimal operating wavelength for surface sensing.

The work begins with a comprehensive comparison of various on-chip RI platforms and optical devices, including a detailed analysis of their performance parameters. Based on

this study, a compact and highly sensitive interferometric gas sensor using a slot-based loop-terminated Mach-Zehnder Interferometer (LT-MZI) design is proposed.

Subsequently, a micro-ring resonator design is introduced to address the issue of RI sensor selectivity. This design enables the simultaneous detection of both the real and imaginary parts of the medium's refractive index at multiple wavelengths, facilitating the determination of concentration composition in multi-element mediums. Artificial intelligence algorithms are employed to enhance detection accuracy, extend dynamic range, and mitigate noise in the measured spectrum.

To address robustness challenges, a system of four LT-MZIs utilizing wavelength splitting is developed. This system demonstrates significant improvements over conventional MZI-based designs in mitigating process and temperature-induced variations.

Finally, an extensive surface sensitivity analysis of silicon nitride waveguides is conducted over a broad wavelength range, from visible to mid-infrared. This study identifies the optimal wavelength for surface sensing and supports the development of a RI-based virus sensor and edge coupler designs for efficient light coupling from fibers.

This work is dedicated to my parents, Samir and Soheir, my sister Marietta, and my cousin Marionetta. Without your support and love, I would not be able to do anything.

# Acknowledgements

This research was partially funded by the Natural Sciences and Engineering Research Council (NSERC), and I gratefully acknowledge their support. I would like to express my deepest gratitude to Dr. Xun Li for granting me the opportunity to join his research group and for providing invaluable guidance as my Ph.D. supervisor. His mentorship has been instrumental throughout this journey.

The long and challenging path of a doctorate would not have been possible without the unwavering support of my incredible family. I extend my heartfelt thanks to my parents, Samir and Soheir, my sister, Marietta, and my cousin, Marionetta, for their boundless love, encouragement, and support. Your daily conversations and constant presence have been my anchor, and I could not have achieved this without you.

I am profoundly grateful to my committee members, Dr. Mohamed Bakr, Dr. Shiva Kumar, and Dr. Jamal Deen, for their invaluable guidance, encouragement, and constructive feedback, which greatly enriched my research.

I would also like to thank Dr. Mohamed Swillam for his close collaboration and unwavering support throughout my Master's and Ph.D. research. His contributions have been pivotal to the success of my work. Furthermore, I express my sincere gratitude to Dr. Diaa Khalil, whose supervision during my Master's studies helped shape my skills and prepared me for the challenges of the Ph.D. journey.

Finally, and most importantly, I thank God Almighty for granting me the strength, knowledge, and perseverance to undertake this research. Without His blessings, this achievement would not have been possible.

# Contents

Abstract ..... iv

Acknowledgements ..... vii

Contents ..... ix

List of Figures ..... xiv

List of Tables ..... xxii

Abbreviations ..... xxiv

1 Chapter 1: Introduction ..... 1

1.1 On-Chip Optical Refractive Index Sensing ..... 1

1.2 Objectives and Contributions ..... 5

1.3 On-chip optical RI Sensing fundamentals ..... 8

1.3.1 Optical waveguides ..... 8

1.3.2 Evanescent Field RI Sensing ..... 14

1.4 Literature Review ..... 19

1.4.1 Label-Free Optical Sensing ..... 19

1.4.2 Planar waveguide RI sensors for on-chip sensing ..... 28

1.4.3 Challenges and Prospects ..... 40

1.5 Summary of Thesis ..... 44

2 Chapter 2: Comparative Study of Photonic Material Platforms and Devices for On-Chip Sensing ..... 46
2.1 Introduction ..... 46
2.2 Photonic Material Platforms ..... 49
2.2.1 Comparative Study ..... 49
2.2.2 Discussion ..... 58
2.2.3 Other waveguide platforms ..... 61
2.3 Photonic Devices for RI Sensing ..... 66
2.3.1 Analysis ..... 68
2.3.2 Discussion ..... 70
2.4 Summary ..... 74
3 Chapter 3: Compact Gas Sensor Using Silicon-on-Insulator Loop-Terminated Mach–Zehnder Interferometer ..... 76
3.1 Introduction ..... 76
3.2 LT-MZI device ..... 78
3.3 Sensor Design ..... 82
3.3.1 LT-MZI Structure and Analysis ..... 82
3.3.2 Components Design ..... 84
3.3.3 Simulation Setup ..... 88

3.4 LT-MZI Sensor Simulation Results ..... 89
3.5 Temperature Variations Immune Design ..... 92
3.6 Experimental Characterization ..... 93
3.7 Summary ..... 96
4 Chapter 4: On-Chip Complex Refractive Index Detection at Multiple Wavelengths for Selective Sensing ..... 98
4.1 Introduction ..... 98
4.2 Complex Refractive Index Detection ..... 102
4.2.1 Methods ..... 102
4.2.2 MRR Design ..... 104
4.3 Results & Discussions ..... 107
4.3.1 Single element medium ..... 107
4.3.2 Multi-element medium ..... 111
4.4 Experimental Characterization ..... 113
4.5 Improving sensor's performance using deep learning ..... 117
4.5.1 Data generation ..... 117
4.5.2 CNN for noise removal ..... 117
4.5.3 DNN for concentration measurement ..... 120
4.6 Conclusion ..... 122

5 Chapter 5: On-Chip Sensing System Employing Wavelength Splitting for Noise Suppression ..... 124
5.1 Introduction ..... 124
5.2 Four LT-MZI Sensor ..... 126
5.2.1 MZI RI Sensor Limitations ..... 126
5.2.2 General Analysis and Design ..... 128
5.3 Liquid Sensor Design and Results ..... 132
5.4 Experimental Characterization ..... 140
5.5 Summary ..... 144
6 Chapter 6: Study of Silicon Nitride Waveguide Platform for On-Chip Virus Detection: Prospects for COVID-19 ..... 145
6.1 Introduction ..... 145
6.2 Virus Sensing Waveguide Analysis and Discussion ..... 148
6.3 Virus Sensors Design ..... 157
6.4 Edge Fiber Coupling of Designed Sensors ..... 163
6.5 Summary ..... 166
7 Chapter 7: Conclusions, Future Work and Publications ..... 168
7.1 Conclusions ..... 168
7.2 Future Work ..... 170

7.2.1 Enhancing Performance Through the Application of AI ..... 170

7.2.2 Consider other designs immune to fabrication and temperature variations 171

7.2.3 Experimental verification ..... 172

7.3 Publications ..... 173

7.3.1 Journal papers ..... 173

7.3.2 Conference papers ..... 174

APPENDIX A ..... 176

BIBLIOGRAPHY ..... 183

# List of Figures

Figure 1.1: Schematic of (a) optical fiber waveguide and (b) integrated planar waveguide. ..... 9

Figure 1.2: Slab waveguide (a) waveguide structure and (b) refractive index profile. .... 10

Figure 1.3: Electric field profile of a slab waveguide with V=4 (a) $TE_0$ mode (b) $TE_1$ mode and (c) $TE_2$ mode. ..... 13

Figure 1.4: TE (solid) and TM (dotted) dispersion curves of a slab waveguide. ..... 13

Figure 1.5: (a) TE mode and (b) TM mode electric field profile of SOI waveguide with w=500nm and h=220nm at λ=1550 nm. ..... 14

Figure 1.6: (a) Schematic of a sensing waveguide and (b) diagram showing the principle of operation of RI evanescent field sensing. ..... 15

Figure 1.7: (a) Schematic of a MZI sensor and its transmission spectrum (b) Schematic of a MRR sensor and its transmission spectrum. ..... 17

Figure 1.8: Schematic of: (a) Fiber MZI sensor. Figure adapted with permission from Reference [37], (b) Fiber MI sensor. Figure adapted with permission from Reference [37], (c) FBG sensor. Figure adapted with permission from Reference [37], (d) Kretschmann SPR sensor, (e) Fiber based SPR sensor. Figure adapted with permission from Reference [37] and (f) Plasmonic square resonator. ..... 26

Figure 1.9: Schematic of: (a) PS-BG sensor, (b) 1D-PhC cavity sensor and (c) Cascaded MRRs employing Vernier effect. ..... 34

Figure 1.10: Schematic of: (a) Slot waveguide, (b) Suspended waveguide and (c) SWG. ..... 36

Figure 2.1 shows $S_{wg,med}$ at different dimensions for the three different platforms..... 51

Figure 2.1: (a) Schematic of the waveguide platform. $S_{wg,med}$ of (b) Silicon, (c) silicon nitride and (d) silica platform versus normalized waveguide width at different waveguide thickness h at λ=1.55 μm. ..... 53

Figure 2.2: $S_{wg,T}$ of (a) Silicon, (b) silicon nitride and (c) silica platform versus waveguide width w at different waveguide thickness h at λ=1.55 μm. ..... 55

Figure 2.3: $S_{wg,w}$ of (a) Silicon, (b) silicon nitride and (c) silica platform versus waveguide width w at different waveguide thickness h at λ=1.55 μm. ..... 56

Figure 2.4: $S_{wg,ncore}$ of (a) Silicon, (b) silicon nitride and (c) silica platform versus waveguide width w at different waveguide thickness h at λ=1.55 μm. ..... 56

Figure 2.5: $S_{wg,med}$ of silicon strip waveguide and slot waveguide, (b) Schematic of MIM plasmonic waveguide, (c) $S_{wg,med}$ of MIM plasmonic waveguides (b) Propagation losses of plasmonic waveguides. ..... 65

Figure 3.1: (a) Schematic of the conventional MZI, (b) schematic of our proposed LT-MZI, and (c) transmission as a function of the phase difference between interferometer arms for both LT-MZI and C-MZI. ..... 80

Figure. 3.2: The H parameter as a function of the a1 coefficient for various a2 values. The transmission spectrum as a function of wavelength for the TE mode in an SOI waveguide with ΔL=20 μm and w=500 nm: (b) with a2=0.8 for different a1 values, (c) for different a1=a2 values, and (d) for different a1 values with a2=1-a1. ..... 82

Figure 3.3: (a) Three-dimensional schematic of the LT-MZI gas sensor; insets: cross sections of the strip (reference arm) and slot (sensing arm) waveguides; (b) top view of the

LT-MZI gas sensor showing the paths for the light beams with red and blue arrows, while marking the mode converters with a dotted-line box; (c) schematic of a typical LT-MZI interferometer transmission and the shift in the spectrum that occurs due to gas presence. ..... 83

Figure 3.4: (a) Waveguide sensitivity $S_{wg}$ of the quasi-TE mode of both the strip and the slot waveguides as a function of the silicon widths at $\lambda = 1.55$ μm. The fundamental quasi-TE mode of: (b) the slot waveguide with $W_{slot} = 270$ nm and (c) the strip waveguide with $W_{strip} = 500$ nm..... 86

Figure 3.5: (a) Schematic of the mode converter; (b) quasi-TE-mode profile of the mode converter as depicted from FDTD solver; (c) the insertion loss of the mode converter as a function of the length of its taper section at $\lambda = 1.55$ μm. ..... 87

Figure 3.6: (a) The output transmission spectra of the LT-MZI sensor with a length L of 50 μm at different refractive indices ranging from 1.000 to 1.009; (b) LT-MZI sensor transmission versus the medium's refractive index around $\lambda = 1.55$ μm with L = 50 μm; (c) LT-MZI sensor minimum wavelength versus the medium refractive index with L = 50 μm. ..... 91

Figure 3.7: (a) LT-MZI sensor FOM calculated at different medium index changes ($\Delta n$) with L = 50 μm; (b) LT-MZI sensor FOM versus interferometer arm length. ..... 92

Figure 3.8: (a) Slot-based LT-MZI gas sensor transmission spectrum at different temperatures and medium index and (b) Slot-based MRR gas sensor transmission spectrum at different temperatures and medium index ..... 93

Figure 3.9: (a) Silicon waveguide fabrication steps using EBL; (b) schematic of the measurement setup. ..... 95

Figure 3.10: (a) Layout of the fabricated LT-MZI with L = 50 μm; insets showing the SEM images of various components. Simulated and measured transmission spectra of the LT-MZI structure with oxide cladding: (b) L = 50 μm; (c) L = 150 μm. ..... 96

Figure 4.1: (a) $S_{wg}$ versus waveguide width w at λ=1.55 μm. (b) Schematic of the MRR sensor; inset: electric field intensity profile of the used SOI waveguide (w=320 nm) fundamental quasi-TE mode at λ=1.55 μm. ..... 107

Figure 4.2: Transmission spectra of the micro-ring resonator at reference and different sensing media. ..... 108

Figure 4.3: Actual (from FDE simulation) and detected (from our method) effective index change versus wavelength for different sensing media (a) Real part $\Delta n_{eff}$ and (b) Imaginary part $\Delta k_{eff}$. ..... 109

Figure 4.4: $S_{wg}$ of the fundamental quasi-TE mode versus wavelength of the used SOI waveguide with w=320 nm and its 4th degree polynomial fitting. ..... 109

Figure 4.5: Actual (from data in [130-131]) and detected (from our method) medium refractive index versus wavelength for different sensing media (a) Real part n and (b) Imaginary part k. ..... 110

Figure 4.6: (a) Layout of the fabricated SOI MRR and (b) SEM image of the point coupler between bus waveguide and ring cavity. ..... 114

Figure 4.7: (a) Measured transmission spectra of the different fabricated MRR designs and (b) zoomed-in view around 1588nm wavelength. ..... 116

Figure 4.8: Refractive index of silicon dioxide from references [25], [181], and as detected from the five different MRR designs using our method. ..... 116

Figure 4.9: Average error as a function of number of iterations. ..... 120

Figure 4.10: Loss versus training epoch. ..... 121

Figure 5.1: (a) Schematic of LT-MZI structure; (b) Typical LT-MZI transmission spectrum with $\alpha_{12}=\alpha_{22}=0.5$, $\alpha_{12}=\alpha_{22}=0.15$, $\alpha_{12}=0.15$ and $\alpha_{22}=0.85$..... 129

Figure 5.2: Schematic of the proposed four LT-MZI sensor. ..... 131

Figure 5.3: Transmission spectrum of LT-MZI 1 and LT-MZI 3 and their cascaded transmission with Lsens,1=500 μm, Lref,1=427.7 μm..... 134

Figure 5.4: (a) Transmission spectrum of the whole sensor at different medium index change $\Delta n_{med}$ with $\Delta n_{eff,other}=0$. (b) Wavelength splitting $\Delta\lambda$split versus medium index change $\Delta n_{med}$. ..... 135

Figure 5.5: (a) Transmission spectrum of the whole sensor at different phase noise $\Delta n_{eff,other}$ with $\Delta n_{med}=0.002$. (b) Transmission spectrum of the whole sensor at different combinations of phase noise and medium index. ..... 137

Figure 5.6: Transmission spectrum of the sensor at DC corners as well as the original design spectrum with $\Delta n_{med}=0.004$. ..... 139

Figure 5.7: (a) Layout and (b) SEM images of the fabricated reference LT-MZI design. ..... 141

Figure 5.8: Measured transmission spectra of the different fabricated LT-MZI designs in Table 5.4, with arrows indicating the wavelength splitting and dotted boxes indicating different operating wavelengths..... 143

Figure 5.9: (a) Splitting wavelength versus sensing arm length at different wavelengths determined from both experimental (solid) and simulation (dotted) data. (b) Sensitivity (experimental and simulation) of the four LT-MZI to sensing arm length (or phase) changes versus operating wavelength. ..... 143

Figure 6.1: (a) Schematic of surface functionalized $Si_3N_4$ on $SiO_2$ strip waveguide for virus detection with a layer representing virus attachment. Strip waveguide single-mode width versus waveguide thickness at different wavelengths of: (b) TE mode and (c) TM mode. ..... 150

Figure 6.2: Strip waveguide sensitivity ($S_{wg}$) versus waveguide width at different thicknesses at $\lambda$=450 nm of: (a) TE mode and (b) TM mode. (c) Strip waveguide sensitivity ($S_{wg}$) of both TE and TM modes versus the optimum waveguide width ($w_{opt}$) at $\lambda$=450 nm. ..... 152

Figure 6.3: Strip waveguide sensitivity ($S_{wg}$) versus waveguide aspect ratio (AR) for: (a) TE mode and (b) TM mode at different operating wavelengths. Optimum width (wopt) versus waveguide aspect ratio (AR) for: (c) TE mode and (d) TM mode at different operating wavelengths. ..... 154

Figure 6.4: (a) Slot waveguide sensitivity $S_{wg}$ versus waveguide width at the optimum wave-guide thickness at $\lambda$=1.55μm and $\lambda$=3.6μm with $w_{slot}$=200 nm. (b) Strip waveguide sensitivity $S_{wg}$ of TE mode (solid) and TM mode (dashed) versus virus size for different optimized designs. ..... 155

Figure 6.5: (a) Schematic of $Si_3N_4$ on $SiO_2$ strip waveguide with functionalization layer virus attachment layer and (b) $S_{wg}$ versus functionalization layer thickness for three different designs. ..... 156

Figure 6.6: Fitted $S_{wg}/\lambda$ (solid line) and $S_{wg}\times\lambda$ (dashed line) of strip waveguide versus wavelength at different AR with (*) representing the simulated results for: (a) TE mode and (b) TM mode. ..... 157

Figure 6.7: Schematic of MZI sensor configurations: (a) s-MZI, (b) a-MZI and (c) LT-MZI. ..... 162

Figure 6.8: (a) Fiber to strip waveguide coupling efficiency versus waveguide width at $\lambda$=450 nm and $\lambda$=650 nm with h=20 nm and h=40 nm, respectively. (b) Schematic of two-step fiber edge coupling. ..... 165

Figure 7.1: Layout of the fabricated SiN die. ..... 173

Figure A.1: Layout of the fabricated (a) slot-based LT-MZI sensor, (b) MRR sensor and (c) four LT-MZI sensor. ..... 177

Figure A.2: SEM images of fabricated devices (a) GC, (b) half ring DC, (c) MZI DC splitter, (d) MRR, (e) MMI, (f) straight DC. ..... 177

Figure A.3: (a) Schematic of the optical setup for experimental characterization. (b) Image of the real optical setup. ..... 179

Figure A.4: (a) Images of the devices on the SiN die from the captured from the camera. ..... 179

Figure A.5: Measured transmission spectrum of (a) GC and (b) LT-MZI. ..... 180

Figure A.6: Image of modified setup. ..... 181

Figure A.7: Measured transmission spectrum of (a) GC and (b) LT-MZI. ..... 182

Figure A.8: Image of fabricated microfluidic channel on top of silicon wafer. ..... 182

# List of Tables

Table 1.1: $S_{wg}$ achieved by different SOI waveguide structures (with h=220 nm) at λ=1.55 μm [71]..... 35

Table 1.2: Structure, dimensions and performance metrics of the different remarkable integrated optical RI sensors proposed recently in the literature. TM denotes TM mode, *i denotes iLoD and *S denotes simulated results..... 40

Table 2.1: The minimum $w_{min}$ and maximum $w_{max}$ waveguide widths (single mode range) at each waveguide thickness h for the silicon, silicon nitride and silica platform..... 52

Table 2.2: Comparison of experimentally measured sensitivity/LoD of RI sensors on different platforms. ..... 54

Table 2.3: The average losses and bend radius (footprint) of silicon, silicon nitride and silica material platforms as reported in published work. ..... 58

Table 2.4. The average values of the waveguide medium sensitivity $S_{wg,med}$, temperature sensitivity $S_{wg,T}$, width variations sensitivity $S_{wg,w}$, optical losses and footprint for the three photonic platforms included in this study. ..... 58

Table 2.5: The values of the different defined FOM for the three platforms. ..... 60

Table 2.6. Comparison of experimentally measured sensitivity, LoD and length/radius of MZI and MRR based RI sensors..... 73

Table 4.1: Average error in the detected effective index changes both real $\Delta n_{eff}$ and imaginary $\Delta k_{eff}$ for different sensing media..... 109

Table 4.2: Average error in the detected refractive index both real n and imaginary k for different sensing media. ..... 110

Table 4.3. Actual (input) and detected (using our method) concentration composition for different multi-element media as well as the percentage error. $c_W/e_W$, $c_M/e_M$, $c_E/e_E$ and $c_P/e_P$ are the water, methanol, ethanol and propanol concentrations/error, respectively..... 112

Table 4.4: The hyperparameters of the DNN..... 121

Table 4.5: The average RMSE of the different substances..... 122

Table 5.1: Comparison of the sensor performance parameters between our proposed four LT-MZI, single LT-MZI, and single MZI with the same waveguide structure and lengths. ..... 133

Table 5.2: Sensitivity and iLoD comparison of our proposed sensor design with other works..... 138

Table 5.3: Effect of fabrication process variations on DCs coefficients, the percentage change in the sensors' sensitivity, FWHM and FOM..... 140

Table 5.4: Fabricated LT-MZIs arm lengths..... 142

Table 6.1: FOM and $\Delta n_{min}$ of S-MZI sensors with $L_{MZI}$ = 1000 µm. ..... 158

Table 6.2: FOM and $\Delta n_{min}$ of S- MZI sensors with large feature size using TM mode at $L_{MZI}$ = 1000 µm..... 159

Table 6.3: $\Delta n_{min}$ and $r_{min}$ of S-MZI sensors with $L_{MZI}$ = 1000 µm. ..... 160

Table 6.4: S, $\Delta n_{min}$, and FOM of the TM mode for different MZI sensors configurations with wsens=270 nm, h=100 nm and $w_{ref}$ =300 nm at λ=450 nm and $L_{sens}$=1000µm. .... 162

Table 6.5: Comparison of our proposed fiber couplers and recently demonstrated ones, showing the output waveguide (w×h) in each case with (S) and (M) denoting simulated and measured results, respectively..... 165

# Abbreviations

| | |
|---|---|
| MZI | Mach-Zehnder interferometer |
| MRR | Micro-ring resonator |
| LT-MZI | Loop-terminated Mach-Zehnder interferometer |
| LoD | Limit of detection |
| iLoD | Intrinsic limit of detection |
| TE | Transverse-Electric |
| TM | Transverse-Magnetic |
| RI | Refractive index |
| FDE | Finite difference eigenmode |
| FDTD | Finite difference time domain |
| SOI | Silicon-on-Insulator |

# 1 Chapter 1: Introduction

## 1.1 On-Chip Optical Refractive Index Sensing

Chemical and biological detection and monitoring of various molecules is a critical task across numerous fields, including healthcare, medical diagnostics, the food industry, environmental monitoring, safety, and industrial process control. Effective sensing of gases, liquids, and viruses is essential for these applications.

Conventional detection methods are often hindered by several issues: low sensitivity, limited dynamic range, complex and lengthy sample preparation, and the necessity for bulky and expensive equipment. Consequently, there is a significant need for a detection method that is cost-effective, compact, simple, real-time, and accurate [1], [2], [3].

Optical sensors, in general, offer higher sensitivity, a broader dynamic range, and immunity to electromagnetic interference compared to their electrical counterparts [3], [4], [5], [6], [7]. Furthermore, integrated optical sensors based on complementary metal-oxide-semiconductor (CMOS) compatible technologies, such as silicon-on-insulator (SOI), provide low-cost and mass-scale fabrication capabilities. Additionally, electrical circuitry can be monolithically integrated with the optical sensor, facilitating the development of compact devices capable of distributed, real-time, and remote sensing [8].

There are mainly two techniques in integrated label-free optical sensing namely refractive index (RI) and absorption sensing [4], [9]. RI sensor detects the real part (n) of the sensing medium refractive index [3], [6], [7], [10], [11], [12], [13], [14], [15], while absorption

sensors detect the imaginary part (k) of the medium refractive index [16], [17], [18]. On-chip absorption based optical sensors suffer from poor sensitivity due to the small sample volume and hence weak light-matter interaction [18]. On the other hand, in RI sensors small changes in n over small distances can lead to large change in the propagating wave phase reaching very high sensitivities through different optical devices. For instance, Mach-Zehnder Interferometers (MZI) [10], [11], [12], [19] and micro-resonators [13], [14], [15], [20], [21]. This renders RI sensors suitable for small volume samples, as n is related to concentration not the total sample mass [9]. Hence, RI sensors are good candidate for integrated on-chip optical sensing.

The sensing performance of a RI sensor is determined mainly by its sensitivity and limit of detection (LoD) [3]. Sensitivity is the ratio between the change in the measured optical quantity, intensity or wavelength shift, and the change in the medium refractive index $\Delta n_{med}$. LoD is the smallest resolvable index change $\Delta n_{med,min}$ that can be detected by the sensor. To increase RI sensor accuracy and resolution we always want to increase sensitivity and decrease the LoD. While these are the main parameters that determine sensing performance, other important aspects contribute to the robustness of the sensor. The most important of these are the sensitivity of the sensor to fabrication process variations and temperature variations. These variations contribute to optical phase errors (noise), leading to intensity or wavelength shifts similar to those detected from changes in the medium index, resulting in errors that limit the sensor's accuracy and resolution. While the original design may achieve exceptional performance, the effect of such variations can significantly deteriorate this performance.

To optimize the design of RI sensor we need optimize three main things. First is the waveguide platform and dimensions, second is the optical device design and dimensions and the third thing is the operating wavelength.

Silicon on Insulator (SOI) platform is one of the most widely used integrated photonics platforms due to its CMOS compatibility, leveraging all the advantages of this mature technology. Consequently, many on-chip RI sensors have been implemented using this platform. Due to its high refractive index contrast (between core and clad), SOI platform can be highly sensitive to sensing medium. Another emerging photonics platform is silicon nitride, which, like the SOI platform, is CMOS compatible but offers several advantages over SOI. Silicon nitride exhibits lower propagation losses, higher fabrication tolerance, lower temperature sensitivity, and a wider transparency range [22], [23], [24]. Silica is another photonic material platform known for its low phase errors and compatibility with optical fibers; however, it suffers from very large footprints. There are other integrated photonic material platforms, such as those based on III-V materials like indium phosphide (InP). These materials are primarily used for optical communication applications due to their excellent electro-optical properties. However, they are significantly more expensive than other platforms and have a limited number of fabrication facilities. Thus, they are beyond the scope of this thesis.

The devices used for refractive index (RI) sensing can be classified into two main categories: interferometric-based devices and resonant-based devices. The most well-known devices in these categories are the Mach-Zehnder interferometer (MZI) and the micro-ring resonator, respectively. Interferometric-based sensors typically require a larger footprint compared to resonant-based sensors to achieve comparable performance; however, they exhibit higher immunity to fabrication process and temperature variations, making the design more reliable, as such variations can significantly degrade the sensor's performance.

Another important aspect is the operating wavelength. Optical communication is the origin of the photonics field; therefore, the near-infrared range, which is optimal for optical communication, is the wavelength range with the most development. A wide range of low-cost and high-performance sources, detectors, and other optical devices are available in the near-infrared range. Consequently, most proposed RI sensors operate within this range. However, this wavelength range is not necessarily the best for optical sensing. For instance, the optimal operating wavelength for absorption-based sensors for gas detection is the mid-infrared range (wavelengths greater than 2μm), as this is the range where gases have their absorption footprints. Similarly, for RI sensing, especially for surface sensing applications such as virus detection, the operating wavelength is a crucial aspect. Recently, significant work has been conducted in other wavelength ranges, leading to the development of many efficient and cost-effective devices at other wavelengths.

Despite the numerous advantages offered by on-chip RI sensors and the extensive efforts done in the optical design engineering, there are still remains a considerable amount of work to be done, and several drawbacks still need to be addressed.

According to our discussion, two primary questions remain inadequately covered: What is the best platform and device for RI sensing? And what is the optimal operating wavelength? Moreover, two major drawbacks in the optical design of RI sensors still need to be addressed. The first pertains to the robustness of the sensor. While numerous designs have been proposed in the literature aiming to enhance the sensing performance of RI sensors, aspects related to reliability and robustness are often overlooked. As mentioned, optical phase errors due to process and temperature variations can significantly degrade the sensor's performance. The second drawback is that RI sensors are not selective; that is, they cannot identify the specific substance or substances present in the medium. This limitation arises because RI sensors merely detect changes in the medium's index, which can result from various different substances.

## 1.2 Objectives and Contributions

The main objective of this thesis is to address the limitations in the optical design of on-chip refractive index (RI) sensors and to fill the gaps not addressed by previous research in this area. As discussed in the preceding section, it is essential to determine the optimal platform and device for on-chip RI sensing as well as the most suitable operating wavelength range for surface (virus) sensing applications. Additionally, this thesis aims to

tackle the issues of optical phase errors resulting from fabrication process and temperature variations, as well as the lack of selective and multi-element detection in RI sensors, while simultaneously improving their sensing performance.

This thesis primarily focuses on CMOS-compatible platforms for on-chip sensing, as our goal is to develop mass-scale, low-cost sensors capable of integration with electrical circuitry.

Hence, the main contributions of this thesis are as follows:

1. A comprehensive comparison between the different integrated photonics platforms and optical devices for RI sensing is introduced, where we determine the best platform and best device for on-chip RI sensing.

2. A compact and highly sensitive SOI interferometric-based design for RI gas sensing is introduced, utilizing a novel LT-MZI design together with a slot waveguide. This design is engineered to be immune temperature variations. Such design reduces the footprint of a conventional MZI interferometer, which is the main drawback of interferometric-based devices. While maintaining its main advantage over the resonant-based devices which is the high immunity to process and temperature variations.

4. A new approach for selective and multi-element medium detection is introduced. We use a simple SOI MRR to determine the complex refractive index of the sensing medium at multiple wavelengths, thus determining not only a single change index change ($\Delta n$) at a specific wavelength, but also the dispersion of the medium. Using this method, we also

developed artificial neural networks that are trained to remove electrical noise from measured spectrum and improve the detection accuracy and range.

5. A novel method for optical phase errors suppression using wavelength splitting is introduced. This wavelength splitting cannot be achieved using conventional MZIs. A system of four SOI based LT-MZIs are used to achieve wavelength splitting. This system also doubles the wavelength sensitivity, increase the resolution and detection range significantly.

3. The best operating wavelength range for RI virus sensing is determined. Unlike gas or liquid (bulk) sensing, surface sensing application such as virus sensing show high dependence on operating wavelength. Leveraging the wide transparency range of silicon nitride waveguide, a rigorous sensitivity analysis from visible to mid-infrared range is introduced together with optimum waveguide dimensions. Then, a virus sensor utilizing the optimum operating wavelength and dimensions are implemented along with edge couplers that couple light from fibers.

## 1.3 On-chip optical RI Sensing fundamentals

### 1.3.1 Optical waveguides

From Snell's law equation (1.1a) light incident from a medium with refractive index n1 at the interface with other medium with refractive index n2, with $n_1 > n_2$, at an angle θ will experience total internal reflection if $\theta > \theta_{crit}$ where $\theta_{crit}$ given by equation (1.1b)

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2) \qquad (1.1a)$$

$$\theta_{crit} = \sin^{-1}\left(\frac{n_2}{n_1}\right) \qquad (1.1b)$$

This phenomenon of total internal reflection is the basic principle of guiding lightwaves. Optical waveguide is a structure with a uniform cross-section (in x and y dimensions) along the propagation direction (z-dimension), with the cross-section formed of a high refractive index surrounded by a low refractive index, allowing to guide the light in the two dimensions (x and y) while propagating in the third dimension.

Optical fibers, Figure 1.1(a), guide lightwaves using a core with diameter $d_{core}$ inside a cladding with diameter $d_{clad}$ with $n_{core} > n_{clad}$. Similarly, integrated planar waveguides, Figure 1.1(b), are formed from a strip (core) with width *w* and thickness *h* above a substrate and capped with a clad, the strip refractive index is higher from both the substrate and the clad. Silicon-on Insulator (SOI), a well-known dielectric waveguide platform, use a silicon strip above a silicon dioxide substrate and covered with silicon dioxide clad, where $n_{Si}=3.457$ and $n_{SiO2}=1.447$ at $\lambda=1.55$ μm [25].

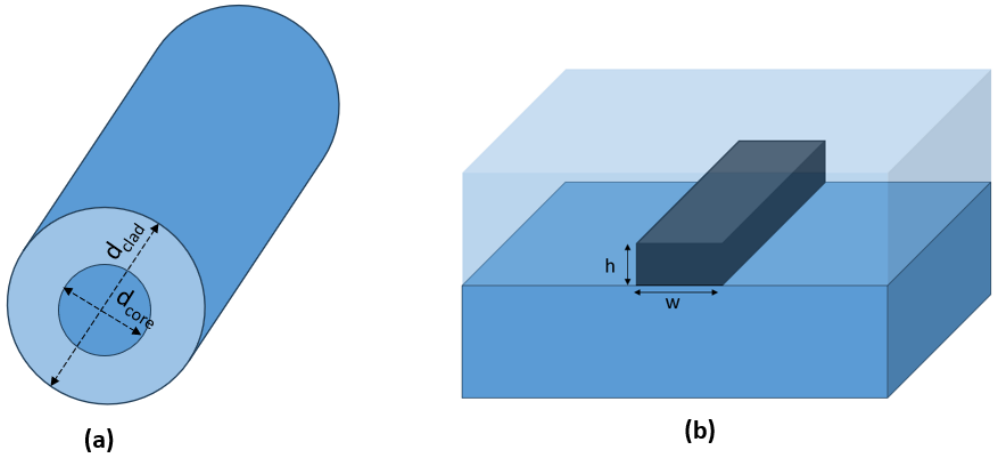



Figure 1.1: Schematic of (a) optical fiber waveguide and (b) integrated planar waveguide.

A fundamental concept in guided waves is the waveguide modes. Here, the waveguide mode we are interested in is defined as a certain field distribution in the transversal plane (cross-section) of the waveguide that preserve its field distribution across the waveguide propagation length and only exhibit a phase shift across the propagation direction. Thus, the electric *E* and magnetic *H* mode fields are defined as:

$$E(x,y,z) = E(x,y)*e^{-j\beta z} \qquad (1.2a)$$

$$H(x,y,z) = H(x,y)*e^{-j\beta z} \qquad (1.2b)$$

$$\text{with } \beta = 2\pi\, n_{eff} / \lambda \qquad (1.2c)$$

where $\beta$ the mode propagation constant, the rate of phase change in the propagation direction, and $n_{eff}$ the effective index of the mode.

To determine the mode field distribution, propagation constant and effective index, for a given waveguide structure, Maxwell's equation need to be solved. For a slab waveguide with thickness $2a$, Figure 1.2, guiding light in x-axis and propagation in z-axis, while assuming a uniform and infinite y-axis, $\partial E/\partial y$=0 and $\partial H/\partial y$=0. Assuming the field distribution of equation (1.2) Maxwell's equations shows two independent mode solutions described by:

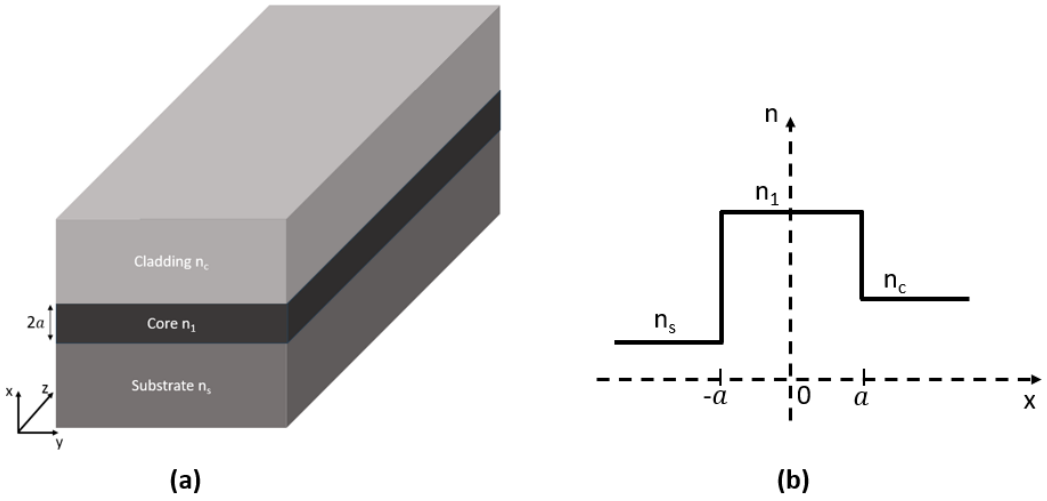



Figure 1.2: Slab waveguide (a) waveguide structure and (b) refractive index profile.

Equations (1.3) are representing a transverse-electric (TE) modes, as its electric field only have transversal components (perpendicular to the propagation direction), while equations (1.4) are representing a transverse-magnetic (TM) modes, as its magnetic field only have transversal components.

$$\frac{d^2E_y}{dx^2} + (k^2n^2 - \beta^2)E_y = 0 \qquad (1.3a)$$

$$H_x = -\frac{\beta}{\omega\mu_0} E_y \qquad (1.3b)$$

$$H_z = \frac{j}{\omega\mu_0} \frac{dE_y}{dx} \qquad (1.3c)$$

$$E_x = E_z = H_y = 0 \qquad (1.3d)$$

$$\frac{d}{dx}\left(\frac{1}{n^2}\frac{dH_y}{dx}\right) + \left(k^2 - \frac{\beta^2}{n^2}\right)H_y = 0 \qquad (1.4a)$$

$$E_x = \frac{\beta}{\omega\varepsilon_0 n^2} H_y \qquad (1.4b)$$

$$E_z = -\frac{j}{\omega\varepsilon_0 n^2} \frac{dH_y}{dx} \qquad (1.4c)$$

$$E_y = H_x = H_z = 0 \qquad (1.4d)$$

Solving equation (1.3) and (1.4) and applying the boundary conditions we reach the electromagnetic fields distribution and the propagation constants for the guided waveguide modes. The electric field $E_y$ distribution of the TE modes is given by equation (1.5a), with magnetic fields $H_x$ and $H_z$ given by equations (1.3b) and (1.3c). Similarly, the magnetic field $H_y$ distribution of the TM modes are given by equation (1.5a) by replacing $E_y$ with $H_y$, with electric fields $E_x$ and $E_z$ given by equations (1.4b) and (1.4c).

$$E_y = \begin{cases} A\cos(\kappa a - \varphi)e^{-\gamma(x-a)} & (x > a) \\ A\cos(\kappa x - \varphi) & (-a \le x \le a) \\ A\cos(\kappa a + \varphi)e^{\zeta(x+a)} & (x < -a) \end{cases} \qquad (1.5a)$$

$$\kappa = \sqrt{k^2 {n_1}^2 - \beta^2} \qquad (1.5c)$$

$$\gamma = \sqrt{\beta^2 - k^2 {n_c}^2} \qquad (1.5c)$$

$$\zeta = \sqrt{\beta^2 - k^2 {n_s}^2} \qquad (1.5c)$$

For guided modes $kn_c<\beta<kn_1$ hence, $n_s<n_{eff}<n_c$

The propagation constants of each mode are given by the dispersion relation of equation (1.6) and equation (1.7) for the TE and TM modes, respectively.

$$2\,V\sqrt{1-b} = m\pi + tan^{-1}\sqrt{\frac{b}{1-b}} + tan^{-1}\sqrt{\frac{b+\delta}{1-b}} \qquad (1.6a)$$

$$V = \frac{2\pi}{\lambda} a\sqrt{n_1^2 - n_s^2} \qquad (1.6b)$$

$$\delta = \frac{n_s^2 - n_c^2}{n_1^2 - n_s^2} \qquad (1.6c)$$

$$b = \frac{n_{eff}^2 - n_s^2}{n_1^2 - n_s^2} \qquad (1.6d)$$

$$2\,V\sqrt{1-b} = m\pi + tan^{-1}\frac{n_1^2}{n_s^2}\sqrt{\frac{b}{1-b}} + tan^{-1}\frac{n_1^2}{n_c^2}\sqrt{\frac{b+\delta}{1-b}} \qquad (1.7)$$

Where *m* the mode order, *b* the normalize propagation constant and *V* the normalized frequency, also called V-number.

Figure 1.3 shows the electric field distribution of the different TE modes supported by a slab waveguide with *V*=4 and $n_1$ = 3.4, $n_s$ = 3.2 and $n_c$ = 1 (δ = 6.6) and Figure 1.4 shows the corresponding TE and TM modes dispersion relation. As the waveguide thickness h increases or the wavelength decreases (V-number increases) as the waveguide can support more modes and modes are more confined in the core.

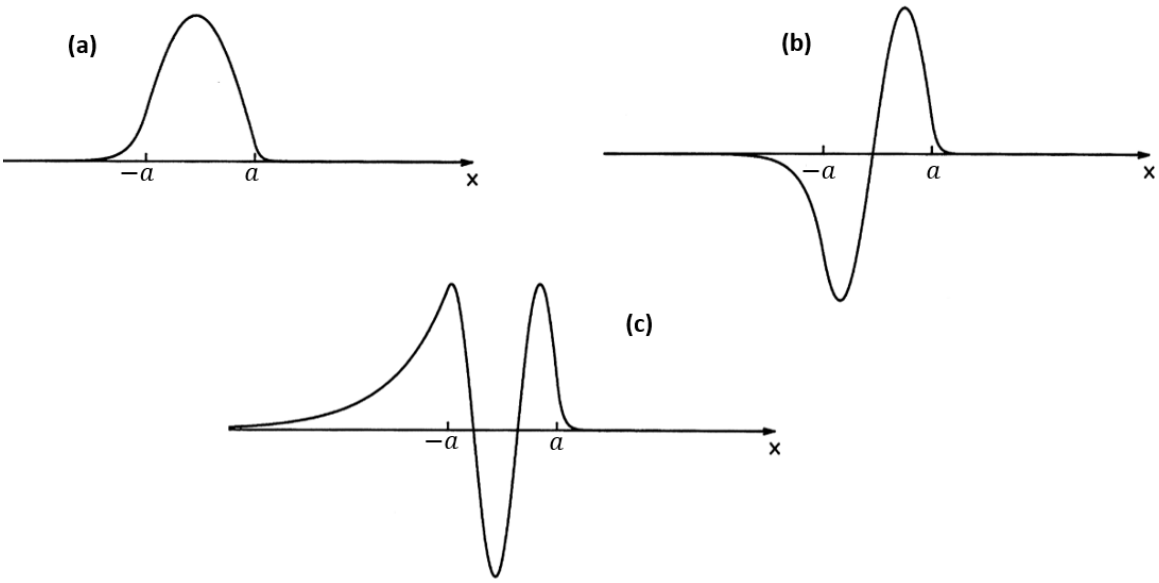



Figure 1.3: Electric field profile of a slab waveguide with V=4 (a) $TE_0$ mode (b) $TE_1$ mode and (c) $TE_2$ mode.

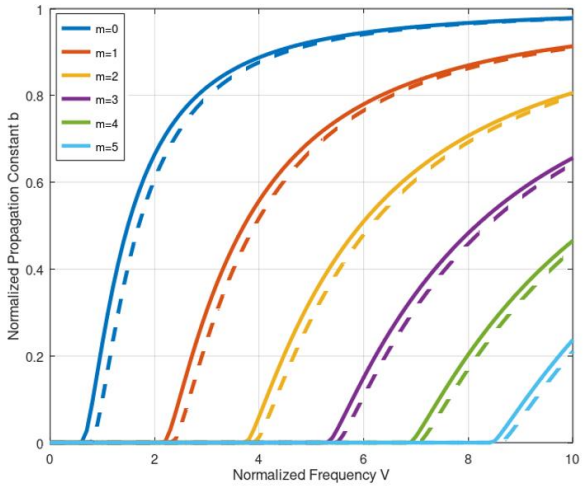



Figure 1.4: TE (solid) and TM (dotted) dispersion curves of a slab waveguide.

For a real dielectric waveguide, guiding the light in two dimensions, Figure 1.1(b), there is no general rigorous analytical closed form expression for the dispersion relation. There are

different methods proposed to analytically solve this problem for specific types of waveguides and under some assumptions, such as Marcatili, Kumar , and effective index method (EIM) [26], [27], [28].

Thus, numerical simulations such as finite difference eigenmode (FDE) solver or finite element method (FEM) [29] are usually used to determine the mode profiles and propagation constants of these waveguides. Figure 1.5 shows the TE and TM mode fields and effective indices of a typical SOI waveguide with w=500 nm and h=220 nm at 1550 nm wavelength, calculated using FDE solver [29].

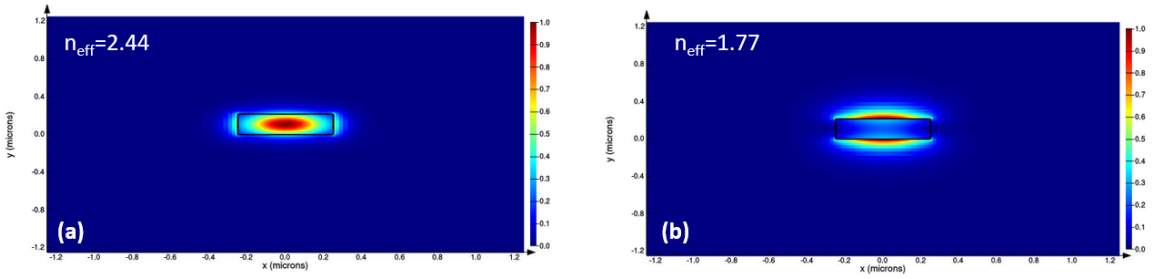



Figure 1.5: (a) TE mode and (b) TM mode electric field profile of SOI waveguide with w=500nm and h=220nm at λ=1550 nm.

### 1.3.2 Evanescent Field RI Sensing

As described in previous section a mode field propagating in a waveguide exhibits a tail in the clad, see Figure 1.3, called evanescent field. This field is interacting with the cladding medium. Evanescent field is the basis of on-chip label-free optical waveguide sensing. The cladding medium is replaced with the sensing medium so that the mode evanescent field interacts with it, Figure 1.6(a). The effective index, field phase change, of the propagating

mode is dependent on the wavelength, waveguide dimensions and waveguide materials refractive indices. Hence, changing the cladding refractive index by Δn will change the mode effective index $\Delta n_{eff}$ and accordingly the field phase Δφ. As phase change is not a detectable quantity several optical devices, such as interferometers and resonators, are used to convert this phase change to a measurable physical quantity, intensity or wavelength shift. The principle of operation of waveguide-based evanescent field refractive index (RI) sensor is shown in the diagram of Figure 1.6(b).

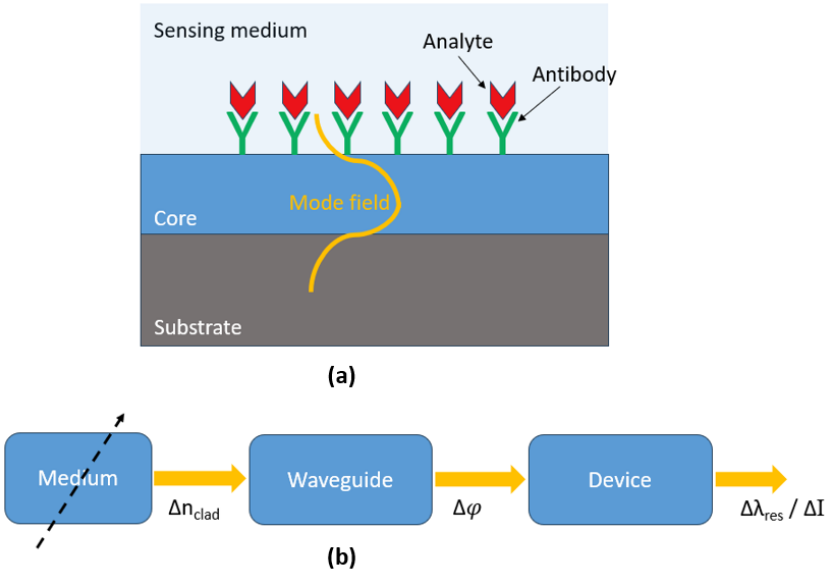



Figure 1.6: (a) Schematic of a sensing waveguide and (b) diagram showing the principle of operation of RI evanescent field sensing.

In interferometer-based sensors input beam splits into two lightwaves with different optical path lengths $\varphi_1$ and $\varphi_2$, then combined (interfere) at the output. According to the phase difference $\Delta\varphi = \varphi_2 - \varphi_1$, equation (1.8), at the output the interference of these two waves can be constructive or destructive thus affecting the intensity of the output beam. This

phase difference $\Delta\varphi$ and hence, the output intensity depends on the sensing medium index. Figure 1.7(a) shows a widely used interferometer in integrated optics called Mach-Zehnder interferometer (MZI) along with its response. In a resonator light is transmitted through a cavity. This cavity is formed by a waveguide section of length $L_{cav}$, between two partially transmitting (reflecting) interfaces with transmission coefficient t, such interface can be simply a coupler like in the case of micro-ring resonator (MRR). According to the cavity optical path length ($\beta L_{cav}$) only specific wavelengths, namely resonant wavelength $\lambda_{res}$, equation (1.9), will add constructively inside the cavity and hence this wavelength is filtered out from the output, resulting in a dip in the output intensity spectrum. In sensing this cavity (waveguide) is subjected to the sensing medium hence, its optical path length and accordingly $\lambda_{res}$ is dependent on the sensing medium index. Figure 1.7(b) shows a well-known resonator in integrated optics called MRR along with its Lorentzian response. A detailed analysis of these two structures is given in the next chapter.

$$\Delta\varphi = \varphi_2 - \varphi_1 = \beta_2 L_2 - \beta_1 L_1 = \frac{2\pi}{\lambda}\left(n_{eff2} L_2 - n_{eff1} L_1\right) \qquad (1.8)$$

$$\Delta\varphi = 2m\pi \rightarrow \beta\, L_{cav} = 2m\pi \rightarrow n_{eff} L_{cav} = m\lambda_{res} \qquad (1.9)$$

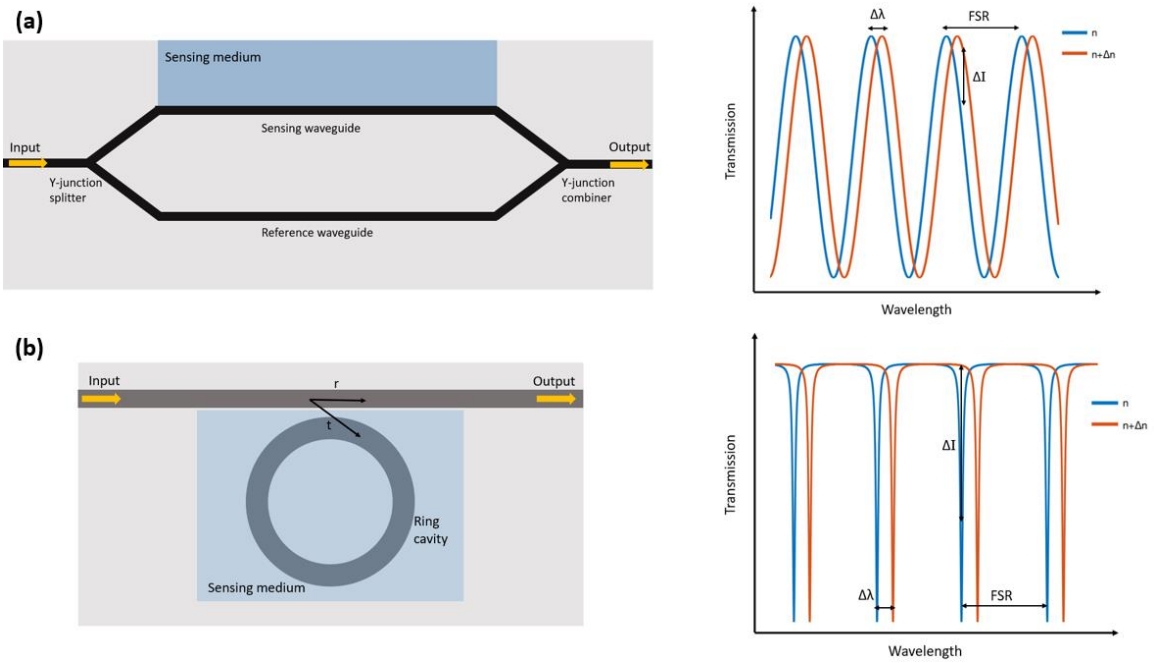



Figure 1.7: (a) Schematic of a MZI sensor and its transmission spectrum (b) Schematic of a MRR sensor and its transmission spectrum.

The sensitivity (S) of optical sensor is defined as ratio between the change in the wavelength (or intensity) and the change in the medium index, equation (1.10). The sensor's sensitivity can be divided into two terms, the device sensitivity ($S_D$) and the waveguide sensitivity ($S_{wg}$). Device sensitivity $S_D$ is defined as the ratio between the change in the resonant wavelength (or intensity) and the change in the waveguide mode effective index, $S_D = d\lambda_{res}/dn_{eff}$. $S_D$ is determined by the optical sensor configuration used and its dimensions such as Mach-Zhender interferometer and its arm's length [19], [30], [31] or ring resonators and its ring radius [32], [33], [34]. While waveguide sensitivity is defined as the ratio between the change in the mode effective index and the change in the medium index, $S_{wg} = dn_{eff}/dn_{med}$. $S_{wg}$ is determined by the sensing waveguide structure

such as strip, rib or slot waveguide and its dimensions (w and h). The overall sensitivity of the optical sensor is the product of both parameters, $S = S_D \times S_{wg}$. Hence, to maximize any RI sensor performance we should maximize both the device sensitivity and the waveguide.

$S_{wg}$ is proportional to the cladding confinement factor $\Gamma_{clad}$. $\Gamma_{clad}$ is the fraction of the mode optical power inside the waveguide cladding, equation (1.11). As $\Gamma_{clad}$ increases as larger part of the mode field is penetrating the cladding and hence stronger interaction with the medium and higher $S_{wg}$. Thus, thin, suspended and slot waveguides which have high $\Gamma_{clad}$ are used in many optical sensors designs, as will be discussed in the next section.

$$S = \frac{\partial \lambda_{res}}{\partial n_{med}} \; or \; \frac{\partial I}{\partial n_{med}} \qquad (1.10a)$$

$$S = \frac{\partial \lambda_{res}}{\partial n_{med}} = \frac{\partial \lambda_{res}}{\partial n_{eff}} \times \frac{\partial n_{eff}}{\partial n_{med}} = S_D \times S_{wg} \qquad (1.10b)$$

$$\text{or } S = \frac{\partial I}{\partial n_{med}} = \frac{\partial I}{\partial n_{eff}} \times \frac{\partial n_{eff}}{\partial n_{med}} = S_D \times S_{wg} \qquad (1.10c)$$

$$\Gamma_{clad} = \frac{\iint_{cladding} |E(x,y)|^2 \, dxdy}{\iint_{\infty} |E(x,y)|^2 \, dxdy} \qquad (1.11)$$

Another key performance metric is the limit of detection (LoD), which represents the smallest detectable change in index (concentration). The LoD is influenced not only by the design of the optical sensor but also by several other factors. These include the measurement system's characteristics, such as the source power level, detector resolution and responsivity, as well as optical and electrical noise. To separate these influences from the sensor's optical design, the intrinsic limit of detection (iLoD) is defined [3]. The iLoD

depends solely on the optical performance of the sensor and is, therefore, a useful metric for comparing different optical sensor designs. It does not account for any sources of noise.

The iLoD is defined as the ratio between the wavelength sensitivity (S) and the full width at half maximum (FWHM) bandwidth of the sensor's spectrum, equation (1.12). While higher sensitivity indicates better sensor performance, a higher FWHM makes it more challenging to differentiate resonance wavelength shifts or intensity shifts, thereby degrading the LoD. A figure of merit (FOM), defined as FOM = 1/iLoD, is commonly used in the literature to evaluate the performance of optical sensors [35].

$$iLoD = \Delta n_{med,min} = \frac{1}{FOM} = \frac{S}{FWHM} \qquad (1.12)$$

In addition to the discussed RI sensing, evanescent field can also be used to detect the imaginary part *k* of the medium refractive index namely absorption sensing. However, for integrated on-chip sensing and due to the small sample volume, absorption based evanescent field sensing suffers from poor performance. While RI sensing can be highly sensitive to small changes in *n* over small lengths.

## 1.4 Literature Review

### 1.4.1 Label-Free Optical Sensing

Chemical and biological detection and monitoring are increasingly important across many fields. Liquid sensing, for instance, water quality monitoring is crucial in healthcare, marine life, and agriculture. Monitoring various gases is essential for environmental

oversight, industrial process control, military applications, and safety. Detecting viruses and bacteria is vital for healthcare, medical diagnosis, and food quality, such as identifying Salmonella or E. coli in food or influenza in humans. Additionally, monitoring biochemical compounds like hemoglobin concentration and glucose levels in blood is necessary.

There are still significant challenges limiting the progress of chemical and biological detection, hindering the high quality of life aimed for in the 21st century. Conventional sensing methods for gases and liquids require bulky and expensive equipment for accurate detection, and most of them lack real time monitoring [2], [3]. Biochemical detection methods like enzyme-linked immunosorbent assay (ELISA) and polymerase chain reaction (PCR) are very sensitive and accurate. However, such methods need a whole laboratory with bulky and expensive lab equipment and they involve complex and lengthy sample preparation and lab procedures that must be done by skilled personnel [2], [3], [36]. An ideal detection method should be real-time, user-friendly, direct (label-free), and highly sensitive and selective, allowing for multiple simultaneous detections using compact devices at low cost [3], [4].

Several direct label-free detection approaches exist, including electrical, optical, and acoustic sensing devices. In general, optical techniques offer high sensitivity, wide dynamic range, metal-free operation, immunity to electromagnetic fields, and superior performance in aquatic and harsh conditions.

Label-free optical sensors can be classified into three main groups; planar waveguide sensors, optical fiber sensors and plasmonic sensors. Fiber based sensors was the first

developed domain of optical sensing driven by the great growth and advancement that happened in the optical fiber communication industry. Numerous optical fiber sensors were proposed using different types of optical fibers and different structures [37], [38]. Most designs utilize evanescent field sensing discussed above, where interferometers and resonators are mainly used to detect medium refractive index changes. In such sensors a part of the fiber cladding is partially or completely removed, in some design even a part of the core is removed, and replaced by the sensing medium so that the mode field is exposed and interacts with the medium. Hollow fiber with a suspended core, single-hole suspended twin-core fiber are other types of fiber where almost all cladding around the core is removed and replaced by the sensing medium thus increasing the sensitivity [38]. In a fiber based MZI light from input fiber is coupled into two different modes, a core-mode and a cladding-mode, then combined to an output fiber, as shown in Figure 1.8(a). The core mode is highly confined in the core region thus, not affected by sensing medium, while the cladding mode are not restricted to the core with field spreading through the cladding thus, sensitive to the surrounding medium. The output intensity is related to the optical phase difference of the two modes hence, which is related to the surrounding medium index. Another, MZI configuration use a twin-core fiber with on core surrounded by cladding as reference arm and the other one is subjected to the sensing medium.

A fiber Michelson interferometer (MI) based sensor was also proposed [37], see Figure 1.8(b). Similar to MZI, in MI based fiber sensor light split and propagate into two different modes (or cores) however, in MI the two modes (or cores) reflect back using a mirror, usually formed by a metal layer at the fiber end. Sagnac interferometer (SI) is another type

of interferometer-based fiber sensors, where input fiber is connected to a 3-dB coupler with the coupler outputs forming a loop. Light entering the coupler will split and propagate in opposite directions then combine and exit from the same coupler. This loop contains a high-birefringence (Hi-Bi) fiber which introduce a phase difference to the counter-propagating modes, which is related to the refractive index of the sensing medium surrounding the Hi-bi fiber.

Fabry-Perot (FP) resonator was also implemented for fiber sensing. In such sensor input light enters a fiber-based cavity with a single guided mode subjected to the sensing medium. This cavity is formed using two reflecting interfaces. Similar to any optical resonant device, the reflection (and transmission) spectrum will exhibit a resonant peak (and dip) at a wavelength $\lambda_{res}$ that is related to the optical path length, hence surrounded sensing medium, of this cavity.

Another type of optical fiber sensors is based on fiber Bragg gratings (FBGs), see Figure 1.8(c). FBGs are formed by modifying the fiber core refractive index periodically along the propagation direction with a period $\Lambda$, this is done using a UV light that modulates the photosensitive core refractive index. This periodic change results in multiple reflections from the different interfaces. At a specific wavelength, called Bragg wavelength $\lambda_B$, satisfying the Bragg condition $\lambda_B=2n_{eff}\Lambda$, these multiple reflections add constructively resulting in a dip in the transmission spectrum and a peak in the reflection spectrum. This peak location $\lambda_B$, is sensitive to the fiber mode effective index $n_{eff}$ and hence the surrounding medium.

MZI and SI based fiber sensors can reach ultra-high sensitivity of 11570 nm/RIU [39] and 13490 nm/RIU [40], respectively, and ultra-low detection limit in the order of nano-moles per liter detection limit. On the other hand, FP resonators and FBG sensor achieves moderately-high sensitivity, of around 1000 nm/RIU [41], [42], [43], [44], and low-sensitivity, less than 1000nm/RIU [45], [46], respectively, and able to reach a few micro-moles per liter detection limit [37].

Although plenty of optical fiber sensors have been proposed for chemical and biological sensing, they still suffer from serious disadvantages that limit their applications. They are difficult to miniaturize and integrate on a chip, having poor structural strength, sensitive to strain and vibrations. In addition, many designs need non-standard and complicated hence, expensive fabrication to achieve high sensing performance.

Plasmonic sensors utilize surface plasmon polariton (SPP) waves, which are guided waves that propagate along the interface between a metal with negative permittivity and an insulator. These waves are generated by the oscillations of free electrons charge density at the surface of the metal when excited by electromagnetic waves. Surface plasmons can confine light in tiny area, below the diffraction limit hence, it allows for significantly high field intensities that leads to high light-matter interactions, and accordingly, highly sensitive devices. There are three approaches employed in plasmonic sensing. Surface Plasmon Resonance (SPR) excitation, Localized Surface Plasmon Resonance (LSPR), and Propagating Surface Plasmon Polaritons (SPP). In SPR sensor, a prism is used to couple light to SPP waves on a metal film, this is known as Kretschmann configuration [47], see Figure 1.8(d). For an incident light of a specific wavelength will be reflected at all incident

angles except a specific angle that will be coupled to the SPP wave and hence, resulting in a null in the reflection. This angle is the one that satisfies the momentum matching condition, $k_{inc}(\theta)=\beta_{spp}(n_{med})$. This angle is highly sensitive to the refractive index of the medium present on the metal surface, which is replaced by the sensing medium $n_{med}$. To mitigate the drawback of mechanical motion similar structure can be used with a broadband light, where a specific wavelength is coupled to SPP wave. While SPR based sensors can achieved ultra-high sensitivity, theoretically reaching 13800 nm/RIU [48], and ulta-low iLoD of $10^{-6}$ RIU, its bulky size and sensitive alignment hinders its integration into a compact device with high-throughput. Fiber based SPR were proposed to mitigate the bulky prism based SPR, where part of the cladding or even the core is removed and a metal layer is deposited to allow for SPP wave excitation, Figure 1.8(e). Such sensors had a penalty in the sensitivity and iLoD compared to the bulk version with sensitivities ranging from 1000 nm/RIU to 10000 nm/RIU and iLoD ranging between $10^{-3}$ and $10^{-5}$ RIU [37].

LSPR sensors rely on SPP waves confined in metals nanostructures. These nanostructures exhibit sharp absorption and scattering peaks at specific wavelength in their spectrum. The location of this peaks is dependent on the nanostructure shape, size and surrounding medium refractive index. Hence, the location of these peaks can be used for chemical and biological sensing. Array of nanoparticles, nanoholes and nano-slits have all be implemented [49], [50], [51]. Such structures can be fabricated on-chip thus, replacing bulky SPR device. However, the measurement setup is still not robust and most designs exhibit a relatively low sensitivity and poor LoD, with a maximum of 560 nm/RIU achieved using nanoslits array [50].

Lastly, propagating SPPs are guide waves at a metal-dielectric interface. Metal-insulator-metal (MIM) and insulator-metal insulator (IMI) are two widely used plasmonic waveguides. Refractive index sensors based on an interferometer, such as plasmonic MZI, and various resonant devices, such as square and circular ring resonators, using plasmonic waveguides were also proposed [52]. Figure 1.8(f) shows a MIM plasmonic square resonator. While such sensors can achieve moderately high sensitivities easily above 1000 nm/RIU [53], [54], due to the highly confined field in the sensing medium, guiding using metal limits its performance and applications. The intensity of plasmonic modes is decaying exponentially with the propagation due to the presence of metal. Hence, sensors lengths must be kept small, which limits their iLoD to higher than $10^{-4}$ RIU for most designs, except when special effects such as fano-resonance are implemented [52]. Operating at wavelengths far from the surface plasmon wavelength will reduce the propagation loss but at the cost of lowering the sensitivity. Moreover, plasmonic modes profile is far from the gaussian like profile of lasers, optical fiber mode and standard silicon waveguide mode, due to its exponential decaying profile, this results in high coupling loss. Finally, plasmonic sensors involve the deposition and patterning of noble metals, like silver and gold hence, they are not compatible with standard fabrication processes, such as CMOS. Accordingly, similar to fiber-based sensors plasmonic sensors still difficult to miniaturize and integrate on a chip, and lack the ability of mass-scale production with low-cost.

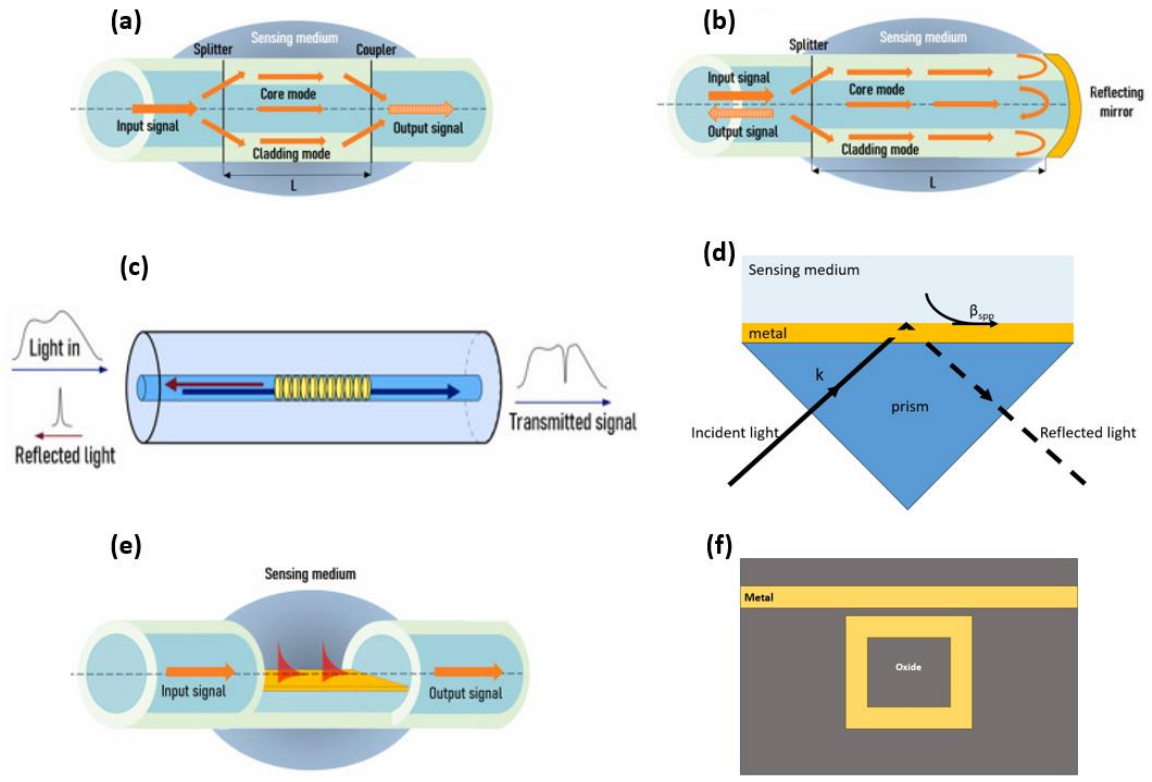



Figure 1.8: Schematic of: (a) Fiber MZI sensor. Figure adapted with permission from Reference [37], (b) Fiber MI sensor. Figure adapted with permission from Reference [37], (c) FBG sensor. Figure adapted with permission from Reference [37], (d) Kretschmann SPR sensor, (e) Fiber based SPR sensor. Figure adapted with permission from Reference [37] and (f) Plasmonic square resonator.

Alternatively, optical planar waveguide-based sensors, discussed in the next section, can be easily integrated on-chip. On-chip optical sensors can reduce production costs while adding mechanical stability and high integration levels, making real-time multiplexed recognition feasible. Silicon photonics integrated circuits (PICs) technology has rapidly developed over the last decades. Silicon-on-insulator (SOI) is the most widely used platform in this technology. The high refractive index difference between silicon and silicon dioxide, or other surrounding materials, allows for the implementation of compact,

miniaturized sensing devices. This also facilitates the production of multiple sensors on a single chip. Moreover, silicon photonics serve as outstanding transducers for quantitative, label-free chemical and biosensing, capable of directly responding to real-time interactions.

Another significant advantage is that SOI technology is compatible with complementary metal-oxide semiconductor (CMOS) processes, utilizing all the benefits of this well-established technology. This compatibility enables efficient, high-volume manufacturing of photonic circuitry with added robustness, reproducibility, low power consumption, and reduced production costs. More importantly, it allows for the co-integration of optical and electrical functions on the same chip, using monolithic integration.

Thus, silicon photonics is a promising technology for on-chip integrated optical sensing. Many silicon photonic sensing devices have been proposed over the last decades for chemical and biological detection applications. Finally, silicon photonics is an excellent candidate for integrating chemical, optical, microfluidic, and electronic functions into a single platform, ideal for lab-on-a-chip (LOC) devices. LOC devices miniaturize all functionalities, from sample preparation to signal delivery, and provide easily interpretable signals for non-skilled personnel. They may include sample preparation, fluid handling, analyte detection, readout, and signal processing. Although the LOC field is still emerging, significant progress has been made in recent years.

### 1.4.2 Planar waveguide RI sensors for on-chip sensing

In this section we review the work done in the literature on integrated planar waveguide sensors for chemical and biological detection. There is a plethora of work done in this field and many interesting and useful reviews trying to analyze, summarize and conclude the great work done [3], [55], [56], [57]. We here mention the different sensing mechanism and devices used for RI evanescent field sensing and present the latest and most remarkable work. We also discuss the different ways proposed to enhance the performance of conventional waveguide-based optical sensors. Finally, we mention the different implemented chemical and biological applications and the achieved LoDs. We focus on the CMOS compatible silicon photonics platform while mentioning few works using silicon nitride and silica platforms too.

Planar waveguide evanescent field optical sensors can be classified into two main groups according to the sensing mechanism as; interference-based sensors and resonant-based sensors. Interference based sensors include MZIs, bimodal and directional coupler (DC) sensors, while resonant based sensors include MRR, phase-shifted Bragg grating and photonics crystals.

In a MZI light splits to two different waveguides (or arms) each supporting a single mode. One waveguide is surrounded by the sensing medium with mode effective index $n_{eff,sens}$ named sensing waveguide, the other waveguide is not subjected to the sensing medium having mode effective index $n_{eff,ref}$ name reference waveguide. When sensing medium changes the effective index $n_{eff,sens}$ changes while effective index $n_{eff,sens}$ is fixed. The

intensity seen at the output port is dependent on the optical phase difference $\Delta\varphi = \varphi_{sens} - \varphi_{ref}$ between the two arms, see equation (1.8), in a $\cos^2$ function. Thus, the phase difference $\Delta\varphi$ and hence, the output intensity depends in the sensing medium index. MZI can have different architectures; the two waveguide arms can have the same length but different waveguide structures, or the same waveguide structure but different lengths. Also, the splitting and combination of the light in the input and output can be using y-junction, directional couplers (DCs) or multi-mode interferometers (MMIs). MZI using y-junctions have only one output while DCs based MZI have two complementary outputs which enhances the accuracy of the sensor. However, DC are very sensitive to wavelength and fabrication changes. MMIs are broadband and fabrication tolerant splitters/combiners and can be used for both one and two outputs MZI. MZI sensors can achieve ultra-high sensitivity by designing the two waveguide arms to have close optical path lengths. However, such high sensitivity doesn't necessary reflect to a low detection limit, as the FWHM is also large in this case, see equation (1.12). Various MZI sensors were proposed using different waveguide designs and platforms reaching theoretical LoD as low as $2.74\times10^{-6}$ RIU using large cross-section (2.5 μm × 10 μm) SOI rib waveguide [58] with 10 mm long sensing arm length. Compat submicron SOI waveguide (450 nm × 260 nm) was also utilized in a MZI with length of only 2 mm folded in just 150 μm diameter spiral shape and integrated with microfluidic channel have reached a measured detection limit of just 10 fg of streptavidin protein [59].

In a bimodal interferometer light input from a single mode waveguide couples into two different modes of a two-mode waveguide then couple again into a single mode output

waveguide after propagating the same distance L. The two modes of the bimodal section have different effective indices hence, experience different phase shift for the same length. Bimodal sensor structure is more compact when compared to MZI sensors however, unlike MZI based sensor here both modes have overlap with the sensing medium, this limits the sensitivity of such sensor, and because of this the sensor should be sufficiently long, usually longer than 5 mm, in order to achieve reasonable LoD comparable to that of a MZI sensor. A LoD as low as $2.5\times10^{-7}$ RIU was experimentally demonstrated using rib SOI waveguide (3 μm × 350 nm) with length as large as 30 mm [60] Also note that, a precise etching step of only 2 nm is needed to define the rib waveguide.

DC is another interferometer-based sensor. A DC is formed by two single mode waveguides, waveguide A and waveguide B, separated by a distance *s*. Similar to bimodal device, this system of the two waveguides supports two modes namely, even and odd mode. With waveguide B subjected to the sensing medium, input light form waveguide A can couple to waveguide B only at a specific wavelength that satisfy the phase matching condition, i.e. $\beta_A = \beta_B$, resulting in a dip in the output spectrum of waveguide A. Changing the medium index will shift the dip wavelength. While such sensor can achieve ultra-high sensitivity, theoretically up to 7000 nm/RIU [61], it is very sensitive to fabrication errors and also need specific waveguide thicknesses not supported by standard fabrication processes.

MRR shown in Figure 1.7(b), is one of the most widely used resonant-based device in integrated photonics. In a MRR light entering the bus waveguide will couple to the ring-cavity (with coefficient t) and interfere after each round trip. As mentioned in section 1.1.2,

by exposing the ring-cavity to the surrounding medium the resonant wavelength $\lambda_{res}$ (at which light is added constructively in the cavity) at the bus waveguide output spectrum will be sensitive to the medium index.

Unlike interferometers in which light propagates only once through the device, in a resonant based devices light circulates multiple times inside the cavity, increasing the effective length of the sensor and allowing for higher interaction between light and the surrounding medium. Thus, MRR sensors are more compact when compared to interferometer-based sensors and are more suitable for multiplexed sensing. The effective length is proportional the resonator quality factor Q. Thus, the resonator Q-factor is very important as this is the main parameter that determines the sensor LoD. Q-factor describes how fine this resonance is and hence, it is inversely proportional to the FWHM bandwidth of the resonance, $Q=\lambda/FWHM$. For higher Q-factor the cavity losses should be decreased. Also, larger cavity lengths will result in higher Q-factors. Silica based MRR achieved LoD of $1.8\times10^{-5}$ RIU with 60 μm radius [62]. Compat SOI based (500 nm × 200 nm) MRR of only 15 μm radius was experimentally demonstrated. While having low sensitivity of only 163 nm/RIU this MRR achieves a significantly low LoD of $7.6\times10^{-7}$ [63].

Micro-disc resonator sensors utilizing the whispering gallery (WGM) mode were also proposed. Unlike MRR a WGM in micro-disk resonator only interacts with the outer surface of the disk, not both inner and outer surface of the ring in MRR, thus reducing the losses due to side-wall roughness and hence can achieve higher Q-factors. Also, using WGM micro-disc resonators can have even smaller footprint than MRRs. However, they also exhibit lower sensitivity compared to MRR sensors due to weaker interaction with

outer (sensing) medium. An ultra-compact SOI Micro-disk with 200 nm silicon device layer and radius of only 3 μm is proposed in [64] achieving a low sensitivity of only 26 nm/RIU and iLoD of $1.8\times10^{-3}$ using fundamental TE mode. This result was enhanced but with larger radius of 10 μm using TM mode achieving 142 nm/RIU and iLoD of $6.8\times10^{-4}$ [64]. Silicon nitride based micro-disc sensor was also proposed with radius of 15 μm achieving a LoD of $10^{-4}$ [65].

Phase-shifted Bragg grating (PS-BG) is another type of resonance-based sensors. In PS-BG sensor, Figure 1.9(a), Bragg gratings are used as partially reflecting mirrors on the two sides of a waveguide cavity. Hence, a resonant peak appears in the output waveguide transmission inside the Bragg stop band. Waveguide-based Bragg gratings are formed by periodically changing the waveguide width or thickness by a $\Delta w$ and $\Delta h$, respectively. SOI based PS-BG sensor was theoretically designed and optimized in different works [66], [67] reaching sensitivity of 182 nm/RIU [67], and LoD as low as $5\times10^{-5}$ RIU with only 13 μm device length [68].

In photonic crystal (PhC) sensors a cavity is formed by introducing a (line) defect in a PhC structure, Figure 1.9(b). PhC structure is a structure with a refractive index modulated periodically in the space. When the period with which the refractive index changes is comparable to the lightwave wavelength this structure blocks a band of wavelengths, called photonic band gap, to propagate inside the PhC structure. When a defect in this periodically changing refractive index is formed a mode can propagate along this defect forming a cavity, resulting in a sharp resonant peak in the output transmission spectrum within the photonic band gap. Similar to all resonant structures, the wavelength of this peak is

dependent on the optical path length of the cavity and hence, in PhC sensor this cavity is immersed in our sensing medium. The light is strongly confined in the small cavity leading to compact devices with high light-matter interaction allowing for high sensitivity. However, similar to PS-BG, PhC based sensors require very precise fabrication and also very sensitive to any small deviations or imperfections in its periodic profile. They also exhibit optical losses due to scattering and absorption within the PhC structure which limits its LoD in most cases. PhC structure can be one-dimensional 1D, two-dimensional 2D or three-dimensional (3D) according to the refractive index periodicity profile. Note that, PS-BG can be considered as 1D-PhC cavity sensor. 1D and 2D PhC based cavities are the ones usually implemented for RI sensing. A SOI based 2D-PhC sensor with a defect hole cavity was proposed in achieving LoD of $2\times10^{-3}$ with a footprint of only 10 $\mu m^2$ [69]. Later a 1D-PhC with a 390 nm line defect cavity was presented [70], with a sensitivity of 130 nm/RIU and expected LoD of $7\times10^{-5}$ RIU.

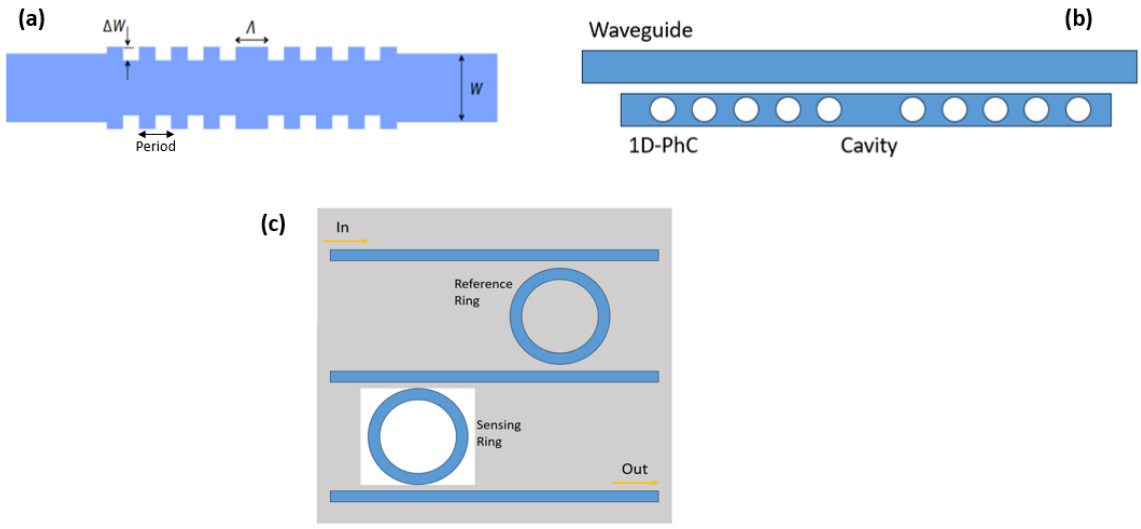



Figure 1.9: Schematic of: (a) PS-BG sensor, (b) 1D-PhC cavity sensor and (c) Cascaded MRRs employing Vernier effect.

There are several techniques that were used to enhance the sensing performance of the aforementioned designs. Enhancement can be on the structure of the sensing waveguide or on the device design or both. As mentioned in section 1.1.2, a waveguide structure that increases the overlap between the propagating mode field and the sensing medium will directly increase the waveguide sensitivity ($S_{wg}$) hence, the overall sensitivity of the sensor. One way to increase the mode overlap with the medium is by decreasing the waveguide core dimensions, width and/or thickness. This will push the mode field out of the core and increase the evanescent field. Note that, in most cases where the substrate index is higher than the cladding (sensing medium) index, there is optimum dimensions to maximize the sensitivity, because shrinking the waveguide dimensions [71] further will push the mode field more to the substrate and not the cladding, reducing the overlap with the medium again [71]. Another way is to use TM mode rather than TE mode, again because TM mode

is less confined to the core having higher overlap with the clad. Slot waveguide proposed in [72], is another highly sensitive waveguide structure. The waveguide is formed by two strip waveguides of small width and separated by a small gap G, Figure 1.10(a). In this case, the evanescent fields of the two strip waveguides are coupled to form a high field intensity in the slot thus, enhancing the mode overlap with the medium. However, to achieve considerable enhancement the gap G should be small enough < 200 nm which makes the fabrication challenging using standard lithography process. Also, it is hard and even impossible for certain molecules to fill such small gap. One of the reasons limiting the sensitivity of conventional waveguides is the substrate evanescent field. Thus, suspended waveguide, Figure 1.10(b), where the substrate is removed and the core is covered all around by the sensing medium significantly enhances the waveguide sensitivity. However, the fabrication of such waveguide is challenging too and it is lacking the robustness. It is important to note that, the mode in all the above-mentioned waveguide structures is more susceptible to losses as they are less confined to the core compared to the conventional waveguide. They also need larger bend radius which will increase the footprint of the sensor.

Table 1.1: $S_{wg}$ achieved by different SOI waveguide structures (with h=220 nm) at λ=1.55 μm [71].

| | Standard w=500nm | Standard w=500nm TM mode | Small width w=280nm | Slot waveguide G=100nm | Suspended w=390nm | SWG w=450 nm |
|---|---|---|---|---|---|---|
| $S_{wg}$ | 0.15 | 0.52 | 0.73 | 0.86 | 1.22 | 0.9 |

Another way to increase the mode overlap is to replace the conventional strip waveguide with subwavelength grating waveguide (SWG), Figure 1.10(c). SWG is formed by

alternating the waveguide core and cladding periodically with a period Λ. For a period much smaller than the wavelength, $\Lambda << \lambda/2n_{eff}$, light propagates through this structure as a uniform waveguide with new core effective index without any perturbation (scattering or diffraction). With cladding replaced by the sensing medium a high overlap between propagating wave and sensing medium is achieved enhancing the sensitivity of the waveguide. But again, such waveguides are very challenging in fabrication due to the small features and suffer from high losses. Table 1.1 shows the waveguide sensitivity $S_{wg}$ of the different SOI waveguide structures.

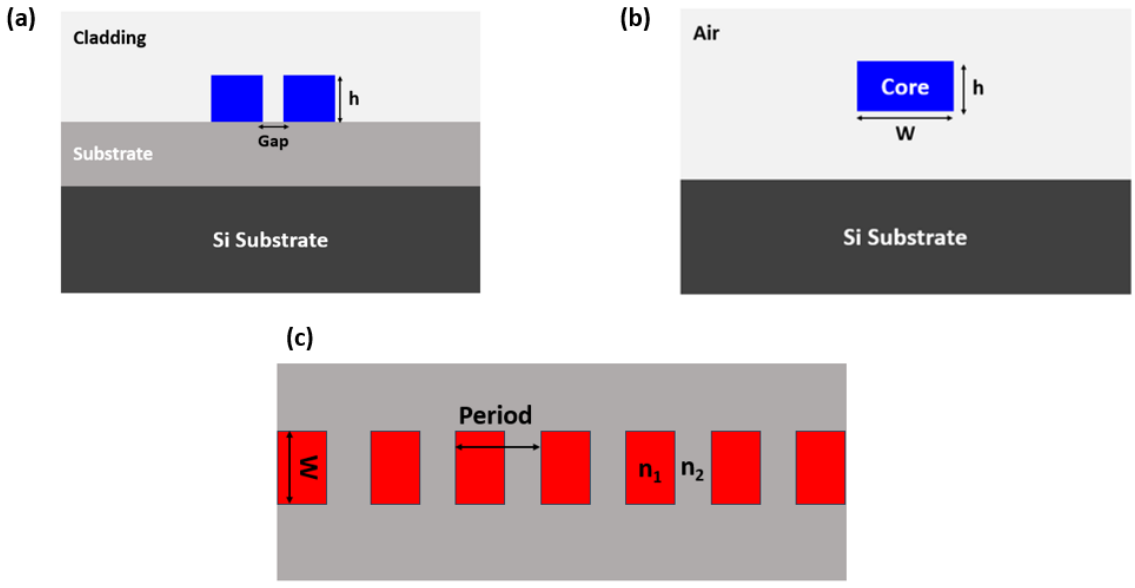



Figure 1.10: Schematic of: (a) Slot waveguide, (b) Suspended waveguide and (c) SWG.

Accordingly, a lot of work utilizing the aforementioned waveguide structures instead of the conventional fundamental TE mode of a strip or rib waveguide were proposed in the literature. Slot waveguide based MZI, MRR, PS-BG and PhC sensors were all proposed

[31], [73], [74], [75], [76]. In addition, suspended MZI, MRR and micro-disc sensors were implemented [20], [77]. SWG MRR sensor were proposed and many designs using the TM mode rather than the TE mode were also presented [78], [79], [80], [81]. Table 1.2 summarize all the previously mentioned sensors along with the design configuration and achieved results. It also contains the designs utilizing different waveguide structures for comparison. From Table 1.2 we can see that, designs implementing non-conventional (strip) waveguides have enhanced the sensitivity of the sensors as expected. However, it is not necessary accompanied by enhancement in the achieved LoD. As mentioned, these enhancement techniques rely on increasing the mode overlap with the surrounding medium by pushing the mode field out of the high index core. This will also increase the field overlap with the waveguide surface resulting in higher propagation losses due to roughness [82], [83].

On the other hand, many works have proposed different device designs and configurations to enhance one or more aspect of planar waveguide RI sensors. Here we mention two of these designs namely Vernier-effect based sensor and ring-assisted MZI (RAMZI) sensor. In a Vernier based sensor two resonators (cavities) with close but not identical FSRs, $FRS_1$ and $FSR_2$, are cascaded, see Figure 1.9(c). The output spectrum of the cascaded device is formed of major and minor peaks with a new larger FSR (determined from the major peaks). The major peaks happen when the resonance wavelength of both devices coincide with each other $\lambda_{res,1}=\lambda_{res,2}$. In Vernier based sensor one of the resonators is not subjected to the sensing medium, reference resonator, and the other one is subjected to the sensing medium, sensing resonator. When the refractive index of the medium changes, the

resonance wavelengths of the sensing resonator will shift. If this shift is large enough, the major peak wavelength of the cascaded structure will jump by $FSR_1$. This consequently enhance the wavelength sensitivity significantly compared to single resonator. The enhancement is M times, with $M=FSR_1/(FSR_1-FSR_2)$. The minimum index change of the sensing medium that results in major peak shift is given by equation (1.13). Note that, the resonant shift in this case is discrete (multiples of $FSR_1$) not continuous which can limit the actually LoD due to quantization.

A sensor using two cascaded MRRs employing the Vernier effect were proposed achieving a sensitivity of 2169 nm/RIU and LoD $8.3\times10^{-6}$ RIU [84]. Higher sensitivities of 24300 nm/RIU, $10^5$ nm/RIU and $4.6\times10^5$ nm/RIU were achieved using TM mode, slot and suspended waveguide, respectively [80], [85], [86] achieving LoD as low as $4.8\times10^{-6}$ RIU [86]. Also, MZI cascaded with ring resonator reference was proposed [87] achieving a sensitivity of 21500 nm/RIU that is able to easily detect just a 1 ng/ml immunoglobulin G (IgG).

$$\Delta n_{med,min} = \frac{n_{eff}}{S_{wg}} \frac{FSR_1 - FSR_2}{\lambda} \qquad (1.13)$$

Although Vernier based sensors have achieved ultra-high sensitivity, their LoD don't enhance with the same factor and they exhibit a LoD comparable to what have been achieved by conventional MZI and MRR based sensors, see Table 1.2. Also, Vernier based sensors are not robust enough when considering a real product. They are very sensitive to fabrication and temperature variations, and different other sources of noise.

Finally ring assisted MZI was also proposed [88], where a ring is inserted in a MZI configuration. RAMZI expand the detection range of conventional MRR sensors by around 5 times. It also offers large resonance extinction ratio, more than 7 dB higher than conventional MRR, which should lead to a higher LoDs [88]. Another major advantage of RAMZI devices over conventional MRR is that this device is more tolerant to fabrication errors making such design more robust. However, the LoD that can be achieved using this device is still not verified experimentally yet.

While there are plenty of designs and ideas trying to enhance the sensing performance of integrated on-chip optical sensors, as we have seen in this section, many of them are challenging in fabrication, due to the small features (below 100 nm), like slot-based waveguides, Bragg gratings, SWGs and PhC sensors. Such designs are difficult to implement using standard lithography process and needs lot of development and optimization in the fabrication process, usually an electron beam lithography (E-Beam) is used for such designs which is not cost-effective. Also, other designs, for instance vernier based designs, are highly sensitive to fabrication process and other sources of variations. So, while the original design can achieve a really good performance such variations can deteriorate it significantly. Hence, a simple design using conventional waveguides and techniques would be more robust and suitable for a real product. Consequently, among the different planar waveguide RI sensors proposed, designs utilizing strip or rib waveguide in a MZI or MRR which have achieved one of the highest LoDs, see Table 1.2, are preferred and actually implemented in real products [82], [89], [90]. A detailed comparison between the MZI and MRR based sensors are given in the next chapter.

Table 1.2: Structure, dimensions and performance metrics of the different remarkable integrated optical RI sensors proposed recently in the literature. TM denotes TM mode, *i denotes iLoD and *S denotes simulated results.

| Sensor Configuration | Waveguide Structure (W×H) | Footprint | Sensitivity nm/RIU | LoD | Ref |
|---|---|---|---|---|---|
| **MZI** | SOI (10 µm × 2.5 µm) | L = 10 mm | | $2.74\times10^{-6}$ *S | [49] |
| | SOI (450 nm × 260 nm) | L = 3 mm | | 10 fg streptavidin | [50] |
| | SiN (H=250 nm) TM | L = 15 mm | | $7\times10^{-6}$ RIU | [72] |
| | Slot SiN (gap=200 nm) | L = 7 mm | | $5.4\times10^{-6}$ RIU | [24] |
| | Suspended SOI (450 nm x 220 nm) | L = 10 mm | 740 | $4\times10^{-5}$ RIU | [68] |
| | Vernier SOI (500 nm × 250 nm) | L = 5 mm | 21500 | 1ng/ml IgG | [78] |
| **Bimodal** | SOI (3 µm × 350 nm) | L = 30 mm | | $2.5\times10^{-7}$ RIU | [51] |
| **DC** | SOI | | 7000 *S | $9.3\times10^{-4}$ RIU *i,*S | [52] |
| **MRR** | Silica | R = 60 µm | | $1.8\times10^{-5}$ RIU | [53] |
| | SOI based (500 nm × 200 nm) | R = 15 µm | 163 | $7.6\times10^{-7}$ RIU | [54] |
| | Slot SOI (H=220 nm gap=104 nm) | 13 µm x 10 µm | 298 | $4.2\times10^{-5}$ RIU<br>$1.6\times10^{-2}$ RIU *I | [65] |
| | Slot SiN | R = 70 µm | 246 | $7\times10^{-6}$ RIU | [67] |
| | Thin SOI (900 nm × 150 nm) *TM | R = 40 µm | 270 | $1.2\times10^{-3}$ RIU *i | [70] |
| | SWG SOI (500 nm × 220 nm) | R = 30 µm | 490 | $2\times10^{-6}$ RIU<br>$5.5\times10^{-4}$ RIU *I | [69] |
| | Vernier SOI (450 nm × 220 nm) | $R_1$= 128.5 µm<br>$R_2$= 129.5 µm | 2169 | $8.3\times10^{-6}$ | [75] |
| | Vernier SOI (450 nm × 260 nm) *TM | $R_1$ = 128.5 µm<br>$R_2$ = 129.5 µm | 24300 | $7.2\times10^{-4}$ | [71] |
| | Vernier Suspended SOI (500 nm × 220 nm) | $R_1$ = 48.5 µm<br>$R_2$ = 50 µm | $4.6\times10^{5}$ | $4.8\times10^{-6}$ | [77] |
| **Micro-disc** | SOI (H=200 nm) | R = 3 µm | 26 | $1.8\times10^{-3}$ *I | [55] |
| | SOI (H=200 nm) *TM | R = 10 µm | 142 | $6.8\times10^{-4}$ *I | [55] |
| | SiN | R = 15 µm | | $10^{-4}$ | [56] |
| | Suspended SOI (H=340 nm) *TM | R = 0.8 µm | 130 | $8\times10^{-4}$ RIU<br>$1.2\times10^{-1}$ RIU *i | [20] |
| **PhC** | 2D-PhC SOI (H=260 nm) | 10 µm² | | $2\times10^{-3}$ | [60] |
| | 1D-PhC SOI (450 nm × 250 nm) | L = 390 nm | 130 | $7\times10^{-5}$ | [61] |
| | Slot SOI (H=220 nm gap=80 nm) | 20x2 µm² | 851 *S | $5\times10^{-5}$ *i,*S | [66] |
| **Bragg** | SOI | L = 13 µm | 90 *S | $5\times10^{-5}$ *i,*S | [59] |
| | Slot SOI (H=220 nm gap=150 nm) | L = 132 µm | 340 | $3\times10^{-4}$ | [64] |

### 1.4.3 Challenges and Prospects

Unlike labeled based detection which require complex sample preparation including selection and attachment of label molecules (for example fluorescent tag) in addition to,

slow label-based detection. Label-free detection, where native medium properties such as the RI are detected, eliminate sophisticated labeling processes allowing for simple and real-time detection. Consequently, label-free detection is the way to achieve a LoC device for convenient, cheap, fast and accurate detection of different molecules which will have a huge impact on our future quality of life.

In the last decades different technologies for chemical and biological sensing have been proposed and developed with lots of designs and devices. For instance, semiconductor-based transducers, microelectrodes and carbon nanotube-based sensors and optical based sensors. Among all the different technologies and techniques integrated planar waveguide-based photonics sensors are the most promising achieving with performance surpassing other technologies achieving low detection limits (as shown from our review). In addition, they are integrated on a small chip hence highly compact and suitable for multiple chemicals detection at the same time using array of sensors. Moreover, using CMOS compatible platforms make them robust and allow for mass-scale fabrication which reduce the cost significantly which paves the way for fully disposable sensor.

Although a plethora of integrated waveguide-based RI sensors were implemented there are still some challenges that slowed the development of a commercialized and complete single chip product. The challenges are both on the optical design level and the integration/ packaging level.

Despite numerous optical designs achieving ultra-high sensitivity and low detection limits, the robustness and reproducibility of these sensors under fabrication variations remain largely unaddressed in nearly all the proposed works.

Also, the immunity of the sensor to optical phase noise such as temperature variations is another challenge. In addition, specific sensing is another obstacle in waveguide-based RI sensing. In RI sensing the sensor detect a certain refractive index change in the medium but cannot specify the exact substance that resulted in this change.

Robustness and reproducibility are covered in to a large extent when using CMOS compatible technology such as silicon photonics however, the optical design itself need also to be immune to fabrication variations. Specific sensing can be addressed by absorption-based sensing however, as mentioned such method shows poor performance when dealing with small volume of on-chip samples. Biofunctionalization of the waveguide surface to be selective to a specific molecule has been widely used to address this issue. A lot of work has been done for biofunctionalization of different waveguides and mainly silicon waveguides [91].

The aforementioned challenges in the optical design of the sensor will be addressed to a large extent in this thesis.

On the other hand, the integration of the different functions on the same chip is another major obstacle. Different units need to be incorporated on the same platform which are: photonic sensor, source, detector, microfluidics and processing electronics.

Thanks to the significant advancements in silicon photonics for optical communication, various approaches have been demonstrated to efficiently integrate III/V lasers onto silicon dies, including direct mounting, heterogeneous integration, and monolithic integration. Heterogeneous integration is scalable and cost-effective hence the most promising approach [92], [93]. Similarly, photodetectors (PDs) on silicon have been extensively developed for optical communication, with Ge on Si PDs achieving high efficiency in the near-infrared wavelength range with responsivity approaching one and only few nano-ampere dark current.

Extensive work has been done for microfluidic systems integration with the optical sensor for sample handling and manipulation. Polydimethylsiloxane (PDMS), negative tone photoresist SU-8, glass epoxy and other materials have been employed for on-chip microfluidic integration. In addition, an emerging technology named digital microfluidics, where microliter or pico-liter droplets are generated and transported in miniaturized chambers, was also considered [94], [95].

The integration of the different aforementioned units can be monolithic or hybrid. In monolithic integration all the functions are implemented on the same chip. While in hybrid integration different functions can be implemented on different chips and co-packaged together. Hybrid integration is easier to implement and also more economic, as the optoelectronic functions can be reusable in this case and only the cartridge of the optical sensor and microfluidics is replaced.

There are already a few commercial products utilizing silicon photonics biosensors [5]. Additionally, numerous large companies and governments are investing in the research and development of on-chip silicon photonic biosensors, recognizing the significant and promising role this technology can play in our society. Given the extensive research and efforts in this field, it is certain that lab-on-chip devices based on integrated optical sensors will soon become a reality, significantly enhancing our quality of life and making a substantial positive impact on society.

## 1.5 Summary of Thesis

This thesis is organized as follows: In the rest of this chapter the fundamentals of integrated optics and basis of RI sensing are presented. A detailed literature review for the different on-chip optical sensing techniques is also presented. Then, we focus on the recently proposed designs for on-chip RI sensing and discuss the current state and challenges.

Next, a comparison between the different on-chip RI platforms as well as the different RI optical devices are discussed in chapter 2. A rigorous analysis for the different performance parameters for RI sensing for the different platforms and devices are included in this chapter. In chapter 3, a compact and highly sensitivity interferometric based SOI gas sensor is proposed using slot-based loop-terminated MZI (LT-MZI) design. This design is also engineered to compensate for any temperature variations.

In chapter 4, we propose a method for selective on-chip RI sensing by detecting both the real and the imaginary parts of the medium refractive index at multiple wavelengths thus,

determining the dispersion of the detected substance. Using this method, we can also determine the concentration composition of multi-element medium. This method was then empowered significantly using artificial intelligence algorithms to further improve the detection accuracy and detection range and more importantly remove the electrical noise from the measured spectrum.

Chapter 5 propose a novel method that use wavelength splitting for optical phase noise suppression utilizing a system of four LT-MZIs. The proposed sensing system also doubles the sensitivity and increase the detection range and significantly increase the limit of detection LoD compared to conventional MZI based designs.

Chapter 6 includes a study of the silicon nitride platform for on-chip RI virus sensing. A rigorous sensitivity analysis for the silicon nitride waveguide spanning a wide wavelength range from visible to mid-infrared is presented to determine the best operating wavelength for virus sensing. Then, based on this study we design a RI based virus sensors together with edge couplers designs for light coupling from standard fibers.

Finally, chapter 7 concludes the thesis and discusses the future work that can further assure and improve our proposed RI sensors designs.

# 2 Chapter 2: Comparative Study of Photonic Material Platforms and Devices for On-Chip Sensing

## 2.1 Introduction

RI sensors performance is determined mainly by its sensitivity and limit of detection (LoD) [3]. While these are the main parameters that determine the sensing performance, there are other important aspects that determine the effectiveness, reliability and yield of the sensor. For instance, the sensitivity of the sensor to process and temperature variations. While the actual design may achieve exceptional performance, the effect of such variations could deteriorate this performance significantly. Moreover, the sensor footprint is another important aspect for low-cost on chip sensors. Hence, when evaluating on-chip sensor all these parameters should be taken into consideration. The mentioned parameters and hence the performance of on-chip RI optical sensors is determined by the design of the optical devices as well as the platform used to implement this design. In most cases the same design can be implemented on the different platforms. Hence, choosing the suitable platform for a specific application become of significant importance.

Today there are many integrated photonic material platforms available. Silicon, silicon nitride, silica and III-V materials-based platforms, such as indium phosphide (InP), are the

main technologies for integrated photonic circuits [8], [96], [97], [98]. Silicon and silicon nitride are widespread photonic platforms due to their compatibility with the mature CMOS technology. Silicon photonics is well-known for many decades now however, silicon nitride is recently emerging as it offers lower losses, better tolerance to process and temperature variations and wider operating wavelength range compared to silicon [22], [23], [24], [99]. CMOS compatibility offer cost-effective and monolithic solution for sensors where both optical and electrical devices can be implemented on single chip which result in a more robust sensor and reduce packaging cost. While III-V based technologies are still not widely used for integrated photonics. Their cost is much higher when compared to other platforms with very limited number of foundries. Silica based platforms on the other hand, are also a cost-effective platform that can provide acceptable performance, has fiber-compatible mode field size and small phase error [100]. However, its chip size is in the order of centimeters [100].

In this chapter we present a comparative study for the different photonic material platforms that can be used for on-chip RI sensing. As mentioned previously, there are plenty of structures/technologies used for optical RI sensing such as plasmonic and photonic crystals. The context of this study is mass-scale and low-cost RI sensors hence, we focus on the standard, dielectric waveguide based, integrated photonics fabrication process, which is mature and can be easily fabricated in many foundries with high yield. Our study includes silicon, silicon nitride and silica platforms. While most of literature works focus on enhancing the sensors' sensitivity and LoD, other parameters such as process and temperature variations affecting the sensor reliability and effectiveness are usually

overlooked. In this study we compare the different platforms, using dielectric strip waveguide, according to five different aspects that determine the detection performance of the sensor as well as its yield, reliability and cost-effectiveness. The aspects are the waveguide sensitivity to medium index change, the fabrication tolerance, the temperature sensitivity, optical losses and footprint.

A rigorous analysis based on finite difference eigen-mode (FDE) solver [29] is used to determine and compare the first three aspects for the different platforms. While a literature survey is used to get the values of optical losses and footprints. We then define a figure of merit to assess the different platforms and find the most-suitable one for RI sensing application.

Next, other waveguide types employing different guiding mechanisms are also investigated namely, slot waveguide and plasmonic waveguide. We compare the different waveguide types and we show that the simple dielectric strip waveguide is the most-suitable platform for high sensing performance, cost-effective and robust integration.

At the end, we compare interferometer-based and resonator-based RI designs in terms of device sensitivity, limit of detection, detection range and immunity to phase noise. An analysis for the different parameters is presented for both devices. As we saw in the literature review section 1.5.2, while more complex designs can achieve better performance, they are usually not robust enough and come with their own drawbacks. Thus, here we focus on simple MZI and MRR based sensors, while a discussion and results about grating devices and also shown.

This chapter is organized as follows; section 2.2 presents the comparison between the different photonic material platforms for RI sensing, including a rigorous modal analysis and literature survey. Different waveguide types are also discussed in this section. Next, section 2.3 presents the comparison between the main optical RI sensing devices. Finally, section 2.4 has the summary of the chapter.

## 2.2 Photonic Material Platforms

### 2.2.1 Comparative Study

In this section we compare the different integrated photonic material platforms with respect to five different aspects. The waveguide sensitivity to medium (analyte) index change, the fabrication tolerance, the temperature sensitivity, optical losses and footprint. These five aspects determine the performance, yield (robustness) and cost-effectiveness of the RI optical sensor. The waveguide sensitivity to medium index change is the main parameter for our RI sensing application as it determines both the overall sensor sensitivity and the LoD. As mentioned in section 1.4, to maximize the sensor's sensitivity $S_{sensor}$ one should select the platform that maximizes the waveguide sensitivity to medium index change $S_{wg,med}$, equation (2.1). Also, as will be seen in the next section, iLoD is also proportional to the waveguide sensitivity $S_{wg,med}$ for both interferometric and resonant (cavity-based) devices.

$$S_{wg,med} = \frac{\partial n_{eff}}{\partial n_{med}} \qquad (2.1)$$

The fabrication tolerance determines how much this platform is insensitive to fabrication process variations. Here we include process variations in fabricated waveguide width and core index. This is so important in RI sensors as just few nanometers change in waveguide dimensions or small changes in fabricated core index can lead to significant change in the optical field phase. This will introduce an error in the detected analyte index limiting sensors' accuracy, LoD and accordingly the fabrication yield. Temperature fluctuations is the main source of noise in optical devices. Similar to process variations changes in temperature will degrade sensors' accuracy and LoD.

On the other hand, optical losses of the sensor will determine the specifications of the source and detector needed for this sensor. Higher losses will require more powerful source and/or detector which can significantly increase the price of the sensor. Finally, in the context of mass-scale low-cost optical sensors, a significant part of the sensor's price depends on its footprint. The footprint will determine how many sensors can be fabricated on a single wafer. Hence, the sensor's price is proportional to wafer price over the footprint.

The first four parameters can be assessed for the different platforms using numerical modal analysis. Change in medium refractive index, temperature, core refractive index and dimensions will result in change in the mode effective index $\Delta n_{eff}$ which will accordingly change the detected optical signal. So we also define the waveguide sensitivity to temperature variations $S_{wg,T}$, the waveguide sensitivity to core index variations $S_{wg,core}$, and the waveguide sensitivity to width variations $S_{wg,w}$, as:

$$S_{wg,T} = \frac{\partial n_{eff}}{\partial T} \quad (2.2), \quad S_{wg,core} = \frac{\partial n_{eff}}{\partial n_{core}} \quad (2.3) \quad S_{wg,w} = \frac{\partial n_{eff}}{\partial w} \quad (2.4)$$

Here we use finite difference eigenmode (FDE) solver [29] to calculate the $S_{wg,med}$, $S_{wg,T}$, $S_{wg,core}$ and $S_{wg,w}$ of a strip waveguide structure for the different platforms. The substrate used in all platforms is silicon dioxide and the waveguide core is covered with water cladding, see Figure 2.1(a). The simulation setup settings used in the FDE solver are mentioned in the next chapter, section 3.3.3. In RI sensing, $S_{wg,med}$ represents the signal we want to detect while the other three parameters represent the noise (error). For each platform we sweep the dimensions, width w and thickness h, and calculate these four parameters for the fundamental quasi-transverse electric (TE) mode. In general, quasi-TE mode can achieve higher $S_{wg,med}$ compared to quasi-transverse magnetic (TM), as the TM mode field has larger overlap with the substrate. Note that, the dimensions used in this analysis always satisfy the single mode operation. The operating wavelength used in the analysis is the near-infrared at $\lambda$=1.55 μm. For the silica platform the index contrast $\Delta$ between core and cladding is 5.5%. The range of thicknesses used for each platform is different as platform with lower core index, like Silica, needs higher thickness to be able to support optical mode.

Figure 2.1 shows $S_{wg,med}$ at different dimensions for the three different platforms. The waveguide width (single mode) range used at each thickness h for the different platforms is mentioned in Table 2.1. Note that, figure shows $S_{wg,med}$ versus normalized width, i.e. $w/w_{max}$. As the waveguide thickness decreases as the width increases to be able to support mode. Note that, the minimum waveguide width included in the analysis is 200 nm. FDE results show that as $\Delta n=n_{core}-n_{clad}$ increase as we can achieve higher $S_{wg,med}=dn_{eff,sens}/dn_{med}$.

Table 2.1: The minimum $w_{min}$ and maximum $w_{max}$ waveguide widths (single mode range) at each waveguide thickness h for the silicon, silicon nitride and silica platform.

| | | | | | | |
|---|---|---|---|---|---|---|
| Silicon | | h=80 nm | h=220 nm | h=400 nm | h=500 nm | |
| | $w_{min}$ (nm) | 400 | 250 | 200 | 200 | |
| | $w_{max}$ (nm) | 950 | 550 | 350 | 300 | |
| Silicon Nitride | | h=220 nm | h=400 nm | h=500 nm | h=750 nm | h=1000 nm |
| | $w_{min}$ (nm) | 750 | 500 | 450 | 400 | 350 |
| | $w_{max}$ (nm) | 2000 | 1300 | 1200 | 1000 | 700 |
| Silica | | h=0.75 μm | h=1 μm | h=3 μm | h=5 μm | |
| | $w_{min}$ (nm) | 2650 | 1850 | 1150 | 1100 | |
| | $w_{max}$ (nm) | 6350 | 4500 | 1950 | 1300 | |

We can see that silicon platform $S_{wg,med}$ can reach up to 1.2 while silica maximum $S_{wg,med}$ is only around 0.2. We then calculate the average $S_{wg,med}$ for the different dimensions. The average $S_{wg,med}$ for the silicon, silicon nitride and silica platform are 0.53, 0.23 and 0.09 respectively. Hence, silicon is 5.6 times more sensitive to medium (analyte) change than Silica and 2.3 times higher than SiN.

Table 2.2 shows published and experimentally measured sensing performance (sensitivity and LoD) of different MZI-based and MRR-based RI sensors for comparison. The results shown in the table aligns with our analysis showing that silicon platform achieves the highest sensitivity among the three platforms while silica show the lowest sensitivity. Note that, these are just a sample from a numerous of works [3], [52], [55], [56], [57] reported

in this field. Much higher sensitivities have also been achieved but with much complex designs for instance, cascaded MRRs. However, here we just selected simple MZI or MRR sensors for the sake of comparison. Furthermore, it is important to note that, the reported sensitivity includes the device sensitivity which depends on the device design not only the platform. In [6] the authors also compared silicon to other platforms based on the different sensor structures and the show the same conclusion.

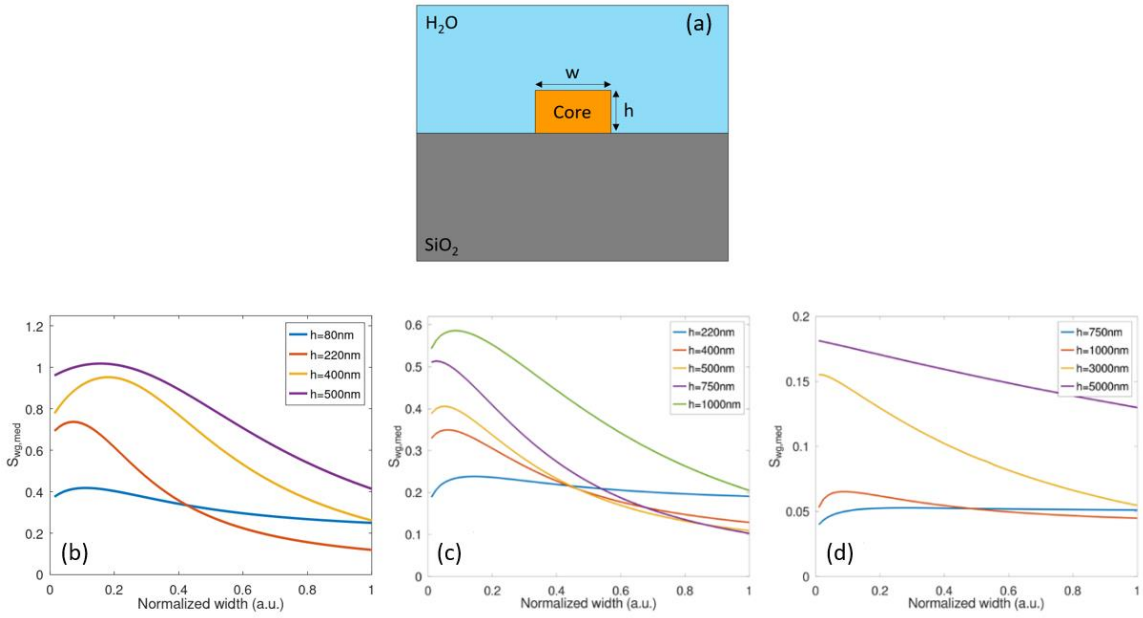



Figure 2.1: (a) Schematic of the waveguide platform. $S_{wg,med}$ of (b) Silicon, (c) silicon nitride and (d) silica platform versus normalized waveguide width at different waveguide thickness h at λ=1.55 μm.

For the temperature sensitivity $S_{wg,T}$ we use the thermo-optical (TO) coefficient ($\partial n/\partial T$) of the different materials [106], [107] and calculate it as:

$$S_{wg,T} = \frac{\partial n_{eff}}{\partial T} = \left.\frac{\partial n}{\partial T}\right|_{cld} * \Gamma_{cld} + \left.\frac{\partial n}{\partial T}\right|_{core} * \Gamma_{core} + \left.\frac{\partial n}{\partial T}\right|_{sub} * \Gamma_{sub} \qquad (2.5)$$

where Γ the confinement factor, which is the fraction of the optical power inside a certain medium [108]. Here cladding medium is water (liquid), substrate medium is silicon dioxide and the core is silicon, silicon nitride and silica, respectively.

Table 2.2: Comparison of experimentally measured sensitivity/LoD of RI sensors on different platforms.

| Platform | Sensitivity | LoD | Ref. |
|---|---|---|---|
| Silica | 75 | 4e-4 | [101] |
| | 142.5 | >2e-3 | [102] |
| Silicon Nitride | 91 | - | [103] |
| | 91.8 | - | [104] |
| | 240 | - | [105] |
| | 246 | - | [76] |
| Silicon | 490nm/RIU | 1e-4 | [57] |
| | 476 nm/RIU | 1e-5 | [13] |
| | 740 nm/RIU | >4e-5 | [77] |

For each platform $\Gamma_{cld}$, $\Gamma_{core}$, $\Gamma_{sub}$ is determined for different dimensions using modal analysis. Figure 2.2 shows $S_{wg,T}$ of the three platforms. When we calculate the average over the different waveguide dimensions we find that, the average $S_{wg,T}$ for the silicon, silicon nitride and silica platform are 8.8e-5 RIU/K, 1.1e-5 RIU/K and 0.6e-5 RIU/K. Silica platform has the lowest change to temperature fluctuations 15 times lower than Si and around 2 times lower than SiN. This is expected as silicon have much higher thermo-optical

coefficient $1.8\text{x}10^{-4}$ $K^{-1}$ than silica and silicon nitride, $2.45\text{x}10^{-5}$ $K^{-1}$ and $8.5\text{x}10^{-6}$ $K^{-1}$, respectively [106], [107]. However, note that some points in Figure 2.2 have zero $S_{wg,T}$. This means that at certain dimensions the sensitivity to temperature variations can be minimized (ideally be zero). This is because the TO coefficient of the water $-10^{-4}$ $K^{-1}$ is negative and the other materials have a positive TO coefficient. So, for a certain dimension the effect of temperature in the mode effective index can be compensated.

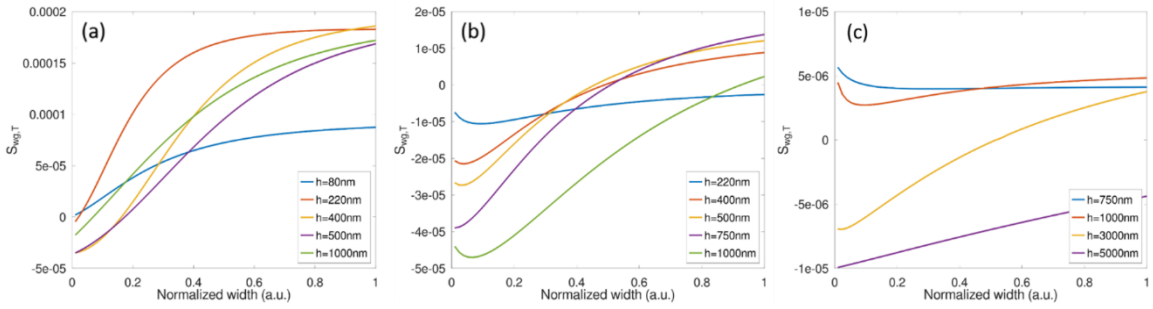



Figure 2.2: $S_{wg,T}$ of (a) Silicon, (b) silicon nitride and (c) silica platform versus waveguide width w at different waveguide thickness h at $\lambda$=1.55 µm.

Figure 2.3 shows the $S_{wg,w}$ of the different platforms. It can be seen that, silica platform is the most tolerant to width variations which is expected as it has small $\Delta n=n_{core}-n_{clad}$ which results in relatively large waveguide dimensions and hence more tolerant to fabrication variations. FDE results show that, the average $S_{wg,w}$ for the silicon, silicon nitride and silica platform are 49e-4 RIU/nm, 3e-4 RIU/nm and 0.4e-4 RIU/nm, respectively. Hence, silica platform is the 123 times more tolerant than silicon and 8 times more than SiN. Also, SiN is 16 times more tolerant than Si. Figure 2.4 illustrates the $S_{wg,core}$ across the three

platforms. Again, we can see that, silica shows the lowest change in $n_{eff}$ due to core index variations. The average $S_{wg,core}$ for the silicon, silicon nitride and silica platform are 0.39 , 0.36 and 0.28, respectively.

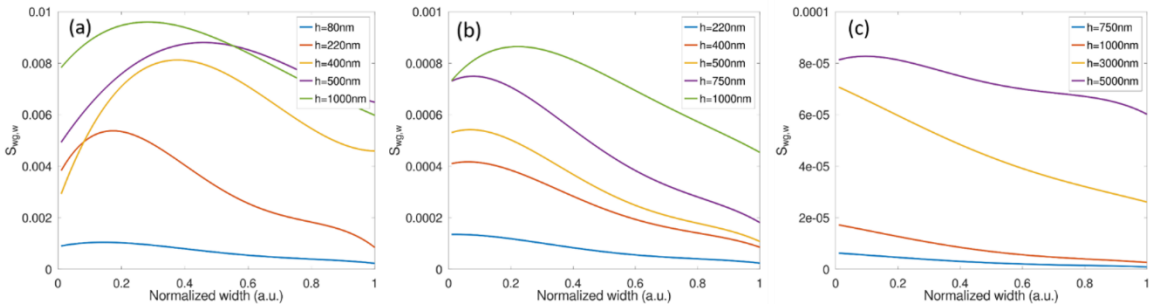



Figure 2.3: $S_{wg,w}$ of (a) Silicon, (b) silicon nitride and (c) silica platform versus waveguide width w at different waveguide thickness h at λ=1.55 μm.

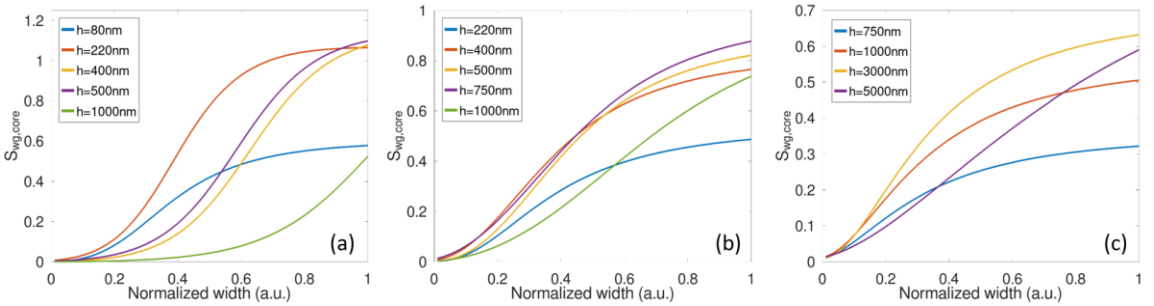



Figure 2.4: $S_{wg,ncore}$ of (a) Silicon, (b) silicon nitride and (c) silica platform versus waveguide width w at different waveguide thickness h at λ=1.55 μm.

For the losses and footprint of the different photonic platforms we gather different data reported from previous published work. The losses can be divided into two parts, the propagation loss and the coupling loss. The propagation losses are due to the quality (defects) and the absorption of the waveguide materials as well as the quality of the surfaces (surface roughness). Higher index contrast between core and clad (like in silicon) results

in higher propagation loss due to higher scattering from the surface roughness. The coupling loss is the loss due to the coupling of the light source to the sensor chip. In most cases a fiber is used to couple the light source to the chip. For the SOI platform the waveguide dimensions are sub-micron which results in high coupling loss due to large mode mismatch between fiber and silicon waveguide. However, the devices are compact hence the propagation losses are low. While, silica waveguides have low mismatch with fiber hence low coupling loss. However, silica-based devices are of millimeter lengths hence large propagation loss. On the other hand, optical circuit footprint is determined by two parameters the phase change per unit length and the waveguide bend size. Both depend mainly on the core-clad index difference. The phase change of the EM field after propagating distance L is $\Delta\varphi=2\pi n_{eff}L/\lambda$, where $n_{eff}$ the mode effective index. The effective index is proportional to the core index. Hence, waveguide like silica with low core index will need a larger length compared to silicon which has larger core index. Another thing that determines the footprint is the waveguide bends size. Platforms like silica with small core-clad index difference need large bend radius (in millimeter range) to route the optical signal with minimal losses. The waveguide bend radius is the parameter that dominates the circuit footprint. Because while the phase change per unit length can be two times higher when comparing silicon to silica. The bend radius can change three orders of magnitude. Hence, here we compare the footprint using the bend radius of the different platforms. Table 2.3 below shows the average losses and footprint of the different platforms and the references used to calculate them. Results show that silicon platform exhibit around 27 times higher losses than silica while the footprint is more than three orders of magnitude

lower in silicon. As mentioned before, this will be reflected into significant reduction in the cost of the silicon-based sensors.

Table 2.3: The average losses and bend radius (footprint) of silicon, silicon nitride and silica material platforms as reported in published work.

| | Silicon | Silicon Nitride | Silica |
|---|---|---|---|
| Propagation Loss (dB/cm) | 1.88 | 0.92 | 0.07 |
| Coupling Loss (dB) | 2.74 | 1.82 | 0.56 |
| Bend Radius (μm) | 4.35 | 233 | 6056 |
| References | [8], [109], [110], [111], [112], [113], [114], [115], [116] | [110], [111], [117], [118], [119], [120], [121] | [122], [123], [124], [125], [126], [127] |

### 2.2.2 Discussion

Table 2.4 concludes the results we presented so far for the three platforms. As we mentioned silicon is the best choice for compact and high medium sensitivity while silica is the best choice for low temperature sensitivity and fabrication tolerant sensor design with silicon nitride always being a compromise of both.

Table 2.4. The average values of the waveguide medium sensitivity $S_{wg,med}$, temperature sensitivity $S_{wg,T}$, width variations sensitivity $S_{wg,w}$, optical losses and footprint for the three photonic platforms included in this study.

| | Silicon | Silicon Nitride | Silica |
|---|---|---|---|
| $S_{wg,med}$ | 0.53 | 0.23 | 0.09 |

| | | | |
|---|---|---|---|
| $S_{wg,T}$ (RIU/K) | 8.8e-5 | 1.1e-5 | 0.6e-5 |
| $S_{wg,core}$ | 0.39 | 0.36 | 0.28 |
| $S_{wg,w}$ (RIU/nm) | 49e-4 | 3e-4 | 0.4e-4 |
| Propagation Loss (dB/cm) | 1.88 | 0.92 | 0.07 |
| Coupling Loss (dB) | 2.74 | 1.82 | 0.56 |
| Footprint (µm) | 4.35 | 233 | 6056 |

To assess these three platforms and choose the best candidate for on-chip RI sensing and Lab-on-Chip (LoC) application, we define a figure of merit (FOM). This FOM takes into account the six different aspects we discussed earlier; the platform sensitivity to medium change, sensitivity to temperature, fabrication tolerance (waveguide width and core index variations), optical losses and footprint. Our FOM is defined as:

$$FOM_1 = \frac{S_{\mathrm{wg,med}}}{S_{\mathrm{wg,T}} * S_{\mathrm{wg,w}} * S_{\mathrm{wg,core}} * \mathrm{L} * \mathrm{FP}} \qquad (2.6)$$

where L the losses and FP the footprint

Note that, as mentioned the losses term L have two components the propagation loss which is proportional to the length hence, it is multiplied by the footprint and the coupling loss.

Replacing the results of Table 2.4 in equation (2.6) we get that the $FOM_1$ for silicon, silicon nitride and silica are 53046, 3811 and 520, respectively. When included the different aspect that affect the performance and robustness of RI sensor, we find that silicon platform outperforms the other platforms, with 14 times higher FOM than silicon nitride and more than 100 times higher than silica. FOM may be defined in many other different ways, by changing the weights of each parameter depending on what is more significant for specific application. For example, for RI sensing $S_{wg,med}$ is the main parameter as it affects both the sensor's sensitivity and LoD and hence determines its sensing performance, so we can give it more weight by squaring it, see $FOM_2$ equation (2.7). Also, one may give less weight to fabrication variations if we are using a process that have a very good control on the fabrication variations, see $FOM_3$ equation (2.8). We can also give lower weight to the device footprint, $FOM_4$ equation (2.3.9). Table 2.5 includes the results of the different defined FOM shown in equation (2.6-2.9).

Table 2.5: The values of the different defined FOM for the three platforms.

| | $FOM_1$ | $FOM_2$ | $FOM_3$ | $FOM_4$ |
|---|---|---|---|---|
| Silicon | 53046 | 28114 | 2319 | 110636 |
| Silicon Nitride | 3811 | 877 | 39 | 58172 |
| Silica | 520 | 47 | 2 | 40490 |

$$FOM_2 = \frac{{S_{\mathrm{wg,med}}}^2}{S_{\mathrm{wg,T}} * S_{\mathrm{wg,w}} * S_{\mathrm{wg,core}} * \mathrm{L} * \mathrm{FP}} \qquad (2.7)$$

$$FOM_3 = \frac{S_{\mathrm{wg,med}}}{S_{\mathrm{wg,T}} * \sqrt{S_{\mathrm{wg,w}} * S_{\mathrm{wg,core}}} * \mathrm{L} * \mathrm{FP}} \qquad (2.8)$$

$$FOM_4 = \frac{{S_{\mathrm{wg,med}}}^2}{S_{\mathrm{wg,T}} * S_{\mathrm{wg,w}} * S_{\mathrm{wg,core}} * \mathrm{L} * \sqrt{\mathrm{FP}}} \qquad (2.9)$$

From the table we can see that for the four different definitions of the FOM silicon platform is always achieving the highest FOM. Silicon significantly outperforms other platforms when less weight is given to process variations or more weight is given to medium sensitivity. However, it becomes comparable to other platforms when less weight is given to footprint. This is mainly because of the very high sensitivity and extremely small FP due to the large index contrast. Thus, while silicon is more susceptible to variations than the other two platforms it is also much more sensitive to medium (analyte) variations and extremely compact leading to lower propagation losses even if the loss per unit length is large.

### 2.2.3 Other waveguide platforms

The previous discussed waveguide platforms are based on dielectric strip waveguide structure, Figure 2.1(a). As discussed in section 1.3.1, in dielectric strip waveguide, light is guided based on the total internal reflection mechanism. Waveguide mode is confined inside the waveguide core (gaussian-like profile of fundamental mode) and exponentially decaying out of the core (evanescent field). There are other waveguide types that have different guiding mechanism and which can be used as sensing waveguide. Among them slot waveguide and plasmonic waveguides are the most known.

Slot waveguide is formed by two strip waveguides of small width and separated by a small gap [72], see Figure 1.10(a). In this case, and due to the boundary condition of the TE polarized light at the high index contrast edges, the evanescent fields of the fundamental TE modes of the two strip waveguides are coupled to form a high field intensity in the slot. This, increases the mode filed overlap with the medium compared to standard strip waveguides, resulting in higher waveguide sensitivity. Figure 2.5(a) shows the sensitivity of silicon slot waveguide, with 100 nm and 60 nm gap width, compared to a strip waveguide. We can see that, slot waveguide can achieve slightly higher $S_{wg,med}$ than strip waveguide. However, this high sensitivity doesn't translate to better sensing performance, lower LoDs [82] [3], see Table 1.2. This is because this higher field overlap with the medium also, increases the overlap of the field with the roughness at the waveguide side-walls, leading to higher propagation losses of 3-7 dB/cm, for SOI with 220nm, which is more than 50 times higher than standard strip or rib waveguides [128], [129].

In addition, slot waveguides need much larger bend radii than conventional strip waveguide to minimize the bend losses, ranging from tens of micrometer to few hundreds for the SOI platform [130]. Also, a mode converter from standard strip waveguide to slot waveguide is required, which will add to the footprint and losses of the sensor.

At last, as mentioned in section 1.4.1 such waveguides are challenging to fabricate using standard lithography process due to the small feature size (slot width of less than 100 nm) and are more sensitive to fabrication variations [131], [132].

Plasmonic waveguides supporting the so-called surface plasmon polaritons (SPPs) waves, are another well-known waveguide platform, especially for sensing [133]. TM polarized electromagnetic (EM) waves can be confined at the metal-insulator interface due to its interaction with the free electrons at the metal surface, this wave is called surface plasmon polariton (SPP). SPP wave decays evanescently at both the metal and the insulator (dielectric). For this wave to be bounded, its propagation constant β must be larger than the wavenumber at the dielectric, $k_d = \sqrt{\beta^2 - k_0^2 \varepsilon_d}$ ($n_{eff}$>$n_d$), with $\varepsilon_d$ the permittivity of the dielectric [133]. The dispersion relation of the metal-insulator (MI) waveguide shows that, as the frequency increases as the propagation constant of the SPP mode increases, reaching a maximum at the surface plasmon frequency $\omega_{sp}$. This frequency $\omega_{sp}$ forms the upper bound of the bounded SPP modes. The surface plasmon frequency $\omega_{sp}$ is related to the metal permittivity and hence to the plasma frequency $\omega_p$, for the MI waveguide $\omega_{sp} = \omega_p/\sqrt{1+\varepsilon_d}$ [133]. Due to its nature surface plasmons are able to confine light below the diffraction limit and can also lead to high field enhancement [133], [134] which result in high-light matter interaction. Owing to these characteristics plasmonic waveguides can achieve high sensitivities.

However, as the EM wave is propagating by the metal, plasmonic waveguides experience significantly high losses, hence can only propagate over a distance of few hundreds or even tens of microns [134]. There is always a trade-off between the sensitivity a plasmonic waveguide can achieve and its propagation length, defined as the distance covered by the mode till its power reach 1/e its initial power. Increasing the sensitivity means higher field intensity at the metal-insulator interface thus, higher losses and lower propagation lengths.

There are different types of plasmonic waveguides for instance, metal-insulator-metal (MIM), insulator-metal-insulator (IMI) and hybrid plasmonic waveguides [135]. Figure 2.5(c) shows the $S_{wg,med}$ of MIM plasmonic waveguide using gold as metal, with 100 nm and 60 nm gap width, at two different wavelengths (infrared and visible). We can see that, plasmonic waveguide can achieve higher $S_{wg,med}$ than dielectric waveguide reaching a value of 1.6 which is around two times higher than what can be achieved my dielectric waveguides. However, as can be seen from Figure 2.5(d), the propagation losses are extremely high reaching 14 dB/μm. From results we can see the trade-off between the sensitivity and losses. Lower gap width shows higher sensitivity however, at the expense of higher propagation losses. Also, lower wavelength will lead to a higher $S_{wg,med}$ but at the expense of higher losses. A more detailed sensitivity and loss analysis of different plasmonic waveguides at different wavelengths is presented in [134], reaching the same conclusion and trends.

Consequently, while plasmonic waveguides can achieve higher sensitivities compared to dielectric waveguides, the achieved LoD are poorer due to the high losses and small propagation lengths. In [52] a review study of the different MIM plasmonic sensors is presented showing poor iLoD of larger than 3e-3. Same conclusion is also outlined by [136].

In [137] authors compared dielectric and plasmonic waveguides in a MZI configuration for sensing. Their results show that dielectric MZI achieves 18 times and 110 times lower LoD for bulk and surface sensing, respectively, compared to plasmonic waveguide biosensors.

It is demonstrated that these conclusions are also valid when a resonant-based devices (like ring resonator) are used.

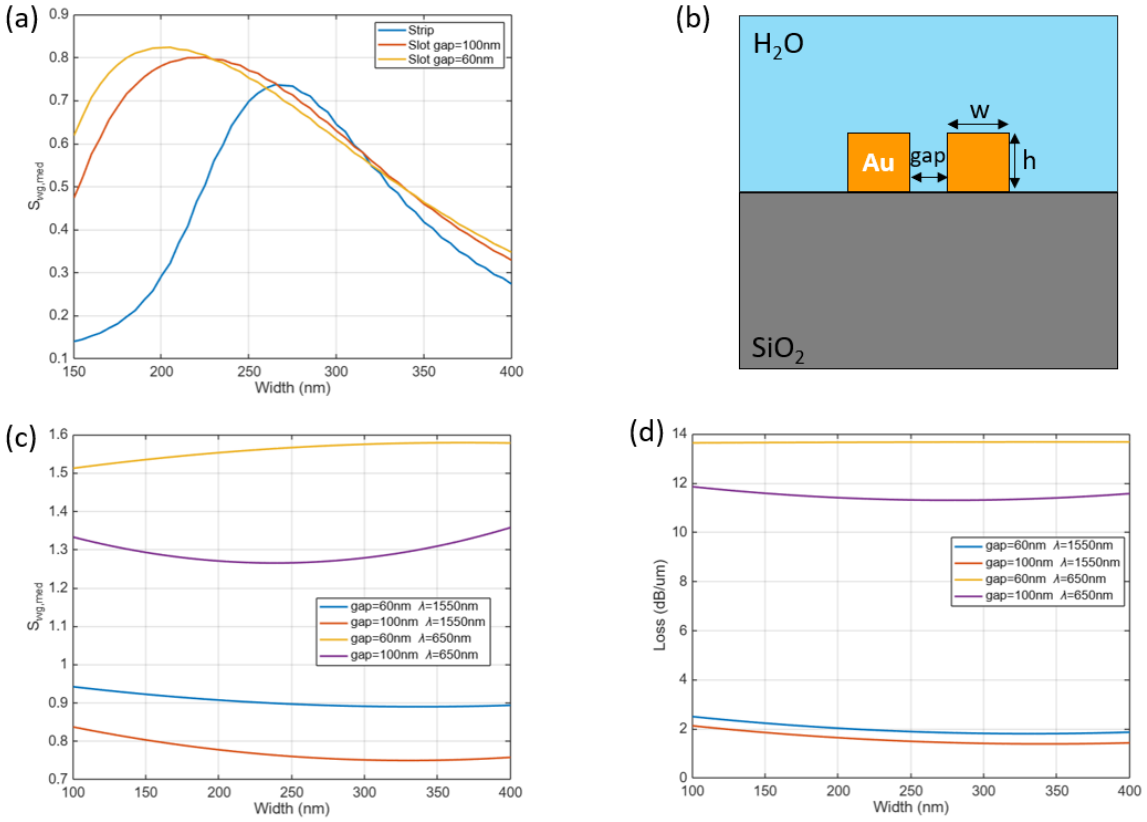



Figure 2.5: $S_{wg,med}$ of silicon strip waveguide and slot waveguide, (b) Schematic of MIM plasmonic waveguide, (c) $S_{wg,med}$ of MIM plasmonic waveguides (b) Propagation losses of plasmonic waveguides.

In addition, similar to slot waveguide, SPP mode of the plasmonic waveguide is not compatible with standard strip waveguide, fiber or laser modes (which are gaussian likely shaped). Hence, additional loss is added to couple the light to and from the plasmonic waveguides, around 4 dB [137] to more than 10 dB in some cases [138], [139].

Finally, plasmonic waveguides require the deposition and etching of silver or gold metal layer, which are not part of standard CMOS fabrication process. Hence, plasmonic waveguide-based sensors will lack the mass-scale and cost-effective solution.

## 2.3 Photonic Devices for RI Sensing

As we can see from the literature review in section 1.5, there is a plethora of designs either interferometric based or resonator based which can further enhance the performance of each one of them. However, we here compare between the standard interferometer-based and resonator-based RI sensor. This comparison depends on the fundamental difference between resonant-based (Lorentzian response) and interferometric-based (cosine response) devices which will still be valid even for a different (such as Phase-Shifted Bragg grating or Young interferometer) or more complex designs.

Another type of integrated optical devices is Bragg grating. Waveguide-based Bragg gratings are formed by periodically changing the waveguide width (or height) by a $\Delta w$ ($\Delta h$). This periodic change results in multiple reflections from the different interfaces. At Bragg wavelength $\lambda_B$, these multiple reflections add constructively resulting in a stop-band response in the transmission spectrum and a pass-band in the reflection spectrum. This reflection band can be used in sensing however, to achieve high sensitivity this band should be narrow enough (fine resonance). For standard integrated photonic platforms and due to large $\Delta n$ between core and cladding this reflection band is broad, resulting in a poor sensing performance. For fine response (high quality factor) the grating corrugation width $\Delta w$ need

to be very small (just few nanometers), which is extremely challenging to achieve even with state-of-the-art photolithography technology [140], [141], [142]. In this case, the Bragg gratings performance are very sensitive to small fabrication imperfections, such as silicon thickness variations, optical lithography smoothing, and quantization errors due to the mask grid size [141]. Also, Bragg gratings should be inserted in an interferometer to be able to extract the reflection spectrum. Hence, such device cannot be used solely for on-chip sensing. However, Bragg gratings are used to construct an on-chip Fabry-Perot (cavity) resonator, where gratings work as a partially reflecting mirrors on the two sides of a waveguide cavity. This structure exhibits the same Lorentzian response of resonant-based devices and many works have implemented it for on-chip RI sensing, see PS-BG sensors in section 1.4.2. Yet the fabrication of waveguide Bragg gratings is highly challenging and still not reliable enough. Almost all work in literature use electron-beam lithography for the fabrication of grating-based devices. Thus, we don't focus on the analysis of Bragg gratings in this study, as our study already includes resonant-based sensor, the MRR, which is easily fabricated with standard photolithography process.

Subwavelength grating is another type of gratings, where the waveguide core and cladding alternate periodically with a period $\Lambda$ much smaller than the wavelength. SWG are used to enhance the sensitivity of waveguide, see section 1.5.2. But again, such waveguides are very challenging in fabrication due to the small features and suffer from high losses.

### 2.3.1 Analysis

In this section we present an analysis for the MZI and micro-ring resonator (MRR) RI sensors. The main performance parameters of RI sensing are derived for both devices which are then used to compare them. The transmission of MZI with 3dB couplers and all-pass micro-ring resonator are [143], [144]:

$$T_{MZI} = \cos^2\left(\frac{\Delta\phi}{2}\right) \quad (2.10a) \text{ and } \quad T_{MRR} = \frac{a^2 - 2ra\cos\phi + r^2}{1 - 2ra\cos\phi + (ra)^2} \quad (2.10b)$$

In MZI $\Delta\phi=\beta_{sens}L_{sens}-\beta_{ref}L_{ref}$ is the phase difference between the MZI sensing and reference arm where $\beta_{sens}$, $L_{sens}$ and $\beta_{ref}$, $L_{ref}$ are the propagation constant, the length of the sensing and reference arm, respectively. For the MRR $\phi=\beta L$, $L=2\pi R$ and $\alpha$ are the round-trip phase shift, length and amplitude transmission respectively, where $\beta$ the propagation constant of the circulating mode, r the self-coupling coefficient and $\alpha_2=e^{-\alpha L}$ with $\alpha$ the power loss coefficient.

The resonance (peak) wavelengths of these devices are [143], [144]:

$$\lambda_{res,MZI} = \frac{1}{q}\left(n_{eff,sens}L_{sens} - n_{eff,ref}L_{ref}\right) \quad (2.11a) \text{ and } \quad \lambda_{res,MRR} = \frac{n_{eff,sens}\,L}{q} \quad (2.11b)$$

Accordingly, the free spectral range (FSR), wavelength sensitivity $S_{med}=d\lambda_{res}/dn_{med}$ and the full-width at half maximum (FWHM) bandwidth can be derived as [143], [144]:

$$FSR_{MZI} = \frac{\lambda^2}{(n_{g,sens}\,L_{sens} - n_{g,ref}\,L_{ref})} \quad (2.12a) \text{ and } \quad FSR_{MRR} = \frac{\lambda^2}{n_{g,sens}\,L} \quad (2.12b)$$

$$S_{med,MZI} = \frac{\lambda_{res}}{(n_{g,sens}\,L_{sens} - n_{g,ref}\,L_{ref})} L_{sens}\,S_{sens,med} \quad (2.13a)$$

$$S_{med,MRR} = \frac{\lambda_{res}}{n_{g,sens}} S_{sens,med} \quad (2.13b)$$

$$FWHM_{MZI} = \frac{\sqrt{2}\,FSR_{MZI}}{\pi} \quad (2.14a) \text{ and } \quad FWHM_{MRR} = FSR_{MRR}\,\frac{1-ra}{\pi\sqrt{ra}} \quad (2.14b)$$

where $S_{sens,med}=dn_{eff,sens}/dn_{med}$ is the sensing waveguide sensitivity to the medium index change.

We can also derive the sensitivity of the devices to any parameter x, $S_x=d\lambda_{res}/dx$ as:

$$S_{x,MZI} = \frac{\lambda_{res}}{(n_{g,sens}\,L_{sens} - n_{g,ref}\,L_{ref})}\,(S_{sens,x}L_{sens} - S_{ref,x}L_{ref}) \quad (2.15a)$$

$$S_{x,MRR} = \frac{\lambda_{res}}{n_{g,sens}} S_{sens,x} \quad (2.15b)$$

with $S_{sens,x}=dn_{eff,sens}/dx$ and $S_{ref,x}=dn_{eff,ref}/dx$ the waveguide sensitivity to changes in the parameter x of the sensing and the reference waveguide, respectively. This parameter can represent the noise (error) in our detected signal hence, we need to minimize it. The main noise sources in integrated RI sensors are the temperature and process variations. These variations will shift the resonance wavelength which will lead to an error in the detected medium index, limiting the accuracy and LoD of the sensor. We can then determine the ratio between the wavelength sensitivity to medium index change and the wavelength sensitivity to any other change (error), which represent the signal to noise (or error) ratio in the RI sensing application.

$$SNR_{MZI} = \frac{S_{med,MZI}}{S_{x,MZI}} = \frac{S_{sens,med}}{S_{sens,x}}\,\frac{1}{1-\frac{S_{ref,x}}{S_{sens,x}}\frac{L_{ref}}{L_{sens}}} \quad (2.16a)$$

and $$SNR_{MRR} = \frac{S_{med,MRR}}{S_{x,MRR}} = \frac{S_{sens,med}}{S_{sens,x}} \quad (2.16b)$$

The figure of merit (FOM) and minimum detectable refractive index ($\Delta n_{min}$), also known as intrinsic limit of detection (iLOD) [3] are given by:

$$FOM_{MZI} = \frac{1}{\Delta n_{min}} = \frac{S}{FWHM} = \frac{\pi S_{sens,med} L_{sens}}{\lambda \sqrt{2}} \quad (2.17a)$$

$$FOM_{MRR} = \frac{1}{\Delta n_{min}} = \frac{S}{FWHM} = \frac{\pi S_{sens,med} L_{sens}}{\lambda} \frac{\sqrt{ra}}{1-ra} \quad (2.17b)$$

The maximum detectable refractive index $\Delta n_{max}$ correspond to resonance wavelength shift equal to the FSR, hence:

$$S_{med} = \frac{\partial \lambda_{res}}{\partial n_{med}} \xrightarrow{\Delta \lambda_{res,max}=FSR} \Delta n_{max} = \frac{FSR}{S_{med}} \quad (2.18)$$

Now we can define the detection range of both MZI and MRR sensors as:

$$DR_{MZI} = \frac{\Delta n_{max}}{\Delta n_{min}} = \frac{FSR}{FWHM} = \frac{\pi}{\sqrt{2}} \quad (2.19a)$$

$$DR_{MRR} = \frac{\Delta n_{max}}{\Delta n_{min}} = \frac{FSR}{FWHM} = \frac{\pi\sqrt{ra}}{1-ra} \quad (2.19b)$$

### 2.3.2 Discussion

Based on this analysis we now compare the MZI and MRR with respect to the main features of RI sensing derived here. First, analysis show that MRR can achieve much higher FOM (hence low LoD) than MZI. From equation (2.17) we can see that, for the same sensing length ($L_{sens}$) and sensing waveguide ($S_{sens,med}$), $FOM_{MRR}$ is larger than $FOM_{MZI}$ by the

factor $\sqrt{ra}/(1-ra)$, owing to the resonators nature where the electromagnetic field circulates multiple round trips inside the cavity which results in Lorentzian like spectrum. For critically coupled MRR (r=α) this factor will depend on the round-trip losses. Typical values for α are between 0.88 and 0.98. Hence, $FOM_{MRR}$ is 4 to 25 times higher than $FOM_{MZI}$. Also note that, MRR is more compact than MZI even for the same Lsens due to its ring shape. Next, regarding the detection range equation (2.19), we can see that, for both devices if we want to reduce $\Delta n_{min}$, $\Delta n_{max}$ will also reduce as in both devices the FWHM is proportional to the FSR. The maximum detectable index change in MZI is only 2.2 times its minimum detectable value which is a very limited range. However, for critically coupled ring resonators (r=α) the detection range depends on the round-trip cavity losses. For typical values of α the detection range of MRR-based sensor is much larger than that of MZI-based sensor. For example, a MRR with α of only 0.9 will have around 7 times higher DR than MZI. While for a MRR with better quality, α can be as high as 0.98. In this case the $DR_{MRR}$ is more than 30 times higher than the $DR_{MZI}$. On the other hand, for certain waveguide platform ($S_{sens,med}$) and operating wavelength the wavelength sensitivity $S_{med,MRR}$ of the MRR sensor is constant and cannot be designed, equation (2.13) and you can only design the $FOM_{MRR}$ using the radius of the ring R. While for the MZI-based sensor the wavelength sensitivity $S_{med,MZI}$ and $FOM_{MZI}$ can be engineered independently through Lsens and Lref. Hence, MZI sensors can achieve high sensitivities however, at large lengths. Finally, from equation (2.16) we can see that MZI can maximize the signal to noise ratio by designing the interferometer arms. We can design sensing and reference waveguide dimensions together with their lengths to minimize the detection error to

temperature and process variations. However, in MRR this ratio is cannot actually be designed and is almost fixed for a given platform. This is extremely important as temperature and fabrication variations will have significant effect on the sensor design and can totally degrade its performance.

Table 2.6 below presents the published and experimentally demonstrated sensing performance (sensitivity and LoD) of various MZI-based and MRR-based RI sensors for comparison, as well as Bragg gratings sensors. From these results we can see that MZI based sensors can achieve much higher (wavelength) sensitivity compared to MRR designs. Also, we can see that the LoD achieved is comparable or slightly higher in the MZI case, 10 times higher on average. However, MZI based sensors have significantly larger length (footprint) compared to the MRR design, more than 50 times higher on average. These experimental verified results are well-aligned with our analysis and discussion and furtherly confirming our conclusions. On the other hand, sensors utilizing Bragg gratings can reach ultra-compact footprint, making them suitable for multiplexed sensing. However, they show limited sensitivity and LoD in addition to, its challenging fabrication as discussed abover.

Hence, MRR sensor is the best choice when a mature technology with very low process variations is used and with platform that has low temperature dependence such as silicon nitride or silica (discussed in the previous section). In this case MRR will achieve better LoD than MZI and with compact size too. However, for a technology with non-negligible process variations and with platform like the silicon which has high temperature

dependence MRR sensor design will not be able to compensate for such changes and hence MZI will be the device of choice in this case as discussed.

Table 2.6. Comparison of experimentally measured sensitivity, LoD and length/radius of MZI and MRR based RI sensors.

| Device | Sensitivity | LoD | Device Length/Radius | Ref. |
|---|---|---|---|---|
| Suspended MZI | 740 nm/RIU | 4e-5 RIU* | L=10 mm | [77] |
| Slot MZI | 1730x2π | 1.3e-5 RIU | L=7 mm | [145] |
| Wire MZI | 460x2π | 3e-5 RIU | L=2 mm | [59] |
| Rib MZI | 1450x2π | 7e-6 RIU | L=15 mm | [81] |
| Bimodal MZI | 2000 nm/RIU | 1e-7 RIU | L=10 mm | [146] |
| MZI | - | 9.2e-7 RIU | L=4.5 mm | [147] |
| MZI | - | 2ng/ml | - | [148] |
| Wire MRR | 70 nm/RIU | 7.1e-5 RIU | R=5 µm | [149] |
| MRR | 163 nm/RIU | 7.6e-7 RIU | R=15 µm | [63] |
| Slot MRR | 476 nm/RIU | 2e-6 RIU | R=30 µm | [76] |
| Slot MRR | 246 nm/RIU | 5e-6 RIU | R=70 | [76] |
| Slot MRR | 298 nm/RIU | 4.2e-5 RIU | 13 µm x 10 µm | [74] |
| Thin MRR | 270 nm/RIU | - | R=40 µm | [150] |
| Thin MRR | 133 nm/RIU | 5e-4 RIU* | R=30 µm | [14] |

| | | | | |
|---|---|---|---|---|
| Suspended MRR | 130 nm/RIU | 8e-4 RIU | R=0.8 μm | [20] |
| Cascaded MRR | | 83.5 fg/ml | R=125/128 μm | [151] |
| Slot Bragg | 340 nm/RIU | 3e-4 RIU* | L=132 μm | [73] |
| Rib Bragg | 34 nm/RIU | 1e-4 RIU | L=173 μm | [66] |
| Bragg | 90 nm/RIU | 5e-5 RIU | L=13 μm | [68] |

* Indicating calculated intrinsic LoD.

## 2.4 Summary

In this chapter we have presented a comparative study of different photonic material platforms and devices for on-chip RI sensing. We first compared the different material platforms namely, silicon, silicon nitride and silica with respect to five different parameters. The parameters are: the waveguide sensitivity to medium index change, fabrication tolerance, temperature sensitivity, optical losses and footprint. These parameters will determine not only the sensing performance, but also the robustness and cost-effectiveness of the sensor. A rigorous modal analysis that spans the waveguide dimensions was used to determine the waveguide sensitivity to medium index change, the effect of the process variations and temperature variations for the three different platforms. For the process variations we determined the waveguide sensitivity to core index changes and the waveguide sensitivity to waveguide width changes. Then a figure of merit is defined that includes all the parameters to compare the different platforms. Our study

showed than silicon is the best platform for RI sensing. Despite being sensitive to temperature and fabrication variations, its high sensitivity to surrounding medium and extremely compact footprint make silicon the platform of choice for RI sensing. Next, we presented a detailed analysis for MZI and MRR based RI sensors. The sensitivity, iLoD, signal to noise ratio and detection range were derived for both devices. Our analysis shows that MRR sensors shows higher iLoD and higher detection range compared to MZI ones. However, unlike MRR, MZI based sensors can be easily designed to maximize wavelength sensitivity and signal to noise ratio. Finally, the best choice between the two sensing devices is discussed. Our analysis and conclusions presented in the study of this chapter are also supported by experimental data from different published work.

# 3 Chapter 3: Compact Gas Sensor Using Silicon-on-Insulator Loop-Terminated Mach–Zehnder Interferometer

## 3.1 Introduction

Gas sensing is rapidly growing in importance due to its essential role in various fields [16]. The demand for integrated on-chip photonics gas sensors is rising and is expected to further increase.

As we have concluded in chapter 2, sensors based on microcavities are compact; however, they suffer from high sensitivity to any optical phase noise such as temperature fluctuations or fabrication variations. On the other hand, interferometers, while less sensitive to such fluctuations, need a large footprint to achieve high performance.

Increasing the percentage of the mode field that passes through the sensing medium will increase the change in $n_{eff}$, hence resulting in higher sensitivity. The percentage of the mode field depends on the waveguide shape, its refractive indices, and its dimensions [152], [153], [154], [155]. A slot waveguide has the capability of strongly confining and guiding light in nanoscale voids (low RI) between two high-index regions [72], [155]. As a result, slot waveguides allow for higher field interaction with the analyte compared to other waveguides. Therefore, compact and highly sensitive sensors can be realized.

In this chapter, we propose a SOI based loop-terminated Mach–Zehnder interferometer (LT-MZI) design. In LT-MZI, the conventional MZI is cascaded with a Sagnac loop [156] at its end, which reduces the interferometer length by a factor of two. Combining this LT-MZI structure with a slot waveguide in the sensing arm we have achieved a highly compact and sensitive RI gas sensor based on interferometric design. Also note that, unlike liquid sensing media gases can easily fill the slot waveguide gap.

In [157], the LT-MZI structure is proposed for RI sensing using optical fibers. In this study, the authors achieved a sensitivity as high as 10,000 nm/RIU with an 11 mm arm length. In [158], the structure is used as a tunable comb filter through phase modulating the Sagnac loop exhibiting ultra-fast wavelength tuning, also based on fibers. The authors in [159] implemented a similar structure as they utilized two Sagnac-loop reflectors to form a Fabry–Perot resonator for a bandwidth tunable filter using an SOI strip waveguide, achieving a tuning efficiency of 0.019 nm/mW. Furthermore, in [160], two loop mirrors are used to implement a Michelson interferometer modulator.

Here, the loop-terminated MZI structure based on SOI platform is proposed, and the analysis and different characteristics of this device are presented. Next, finite-difference eigenmode (FDE) [29] and finite-difference time-domain (FDTD) solvers [161] were used to design and optimize, and determine the performance of our LT-MZI gas sensor in the near-infrared (NIR) region. The results showed that our sensor can achieve high performance with a small footprint. In addition, the same structure, but with oxide cladding rather than air, was fabricated and experimentally characterized to further ensure the robustness of our proposed design. This chapter is structured as follows: Section 3.2 covers

the design, analysis, and main features of the LT-MZI design. In section 3.3 the SOI LT-MZI sensor is proposed and the optimization and design of its various components is presented. Section 3.4 presents and discusses the FDTD simulation results of the complete LT-MZI sensor. Finally, Section 3.5 provides a summary of the chapter.

## 3.2 LT-MZI device

The proposed loop-terminated MZI (LT-MZI) is shown in Figure 3.1(a). Similar to conventional MZI, Figure 3.1(b), input light is first split into the two interferometer arms using a DC. Then a second DC is used to combine the light beams from the two arms. Unlike MZI in LT-MZI, a Sagnac loop at the end of the second DC is used to reflect back the electromagnetic (EM) waves to the interferometer arms again. The output is then collected from the reflected port of the first DC, as shown in Figure 3.1(a). This doubles the optical path length, giving it twice the physical length compared to the conventional MZI.

The transmission of the conventional MZI and LT-MZI can be derived as:

$$T_{MZI} = a_1^2 a_2^2 + b_1^2 b_2^2 + 2a_1 a_2 b_1 b_2 \, cos(\Delta\phi) \qquad (3.1)$$

$$T_{LT-MZI} = A\left[\frac{H}{4} + cos(\Delta\phi)\right]^2 \qquad (3.2a)$$

with $A = 16a_1^2 a_2^2 b_1^2 b_2^2$ (3.2b)

and $H = -\frac{a_1 a_2}{b_1 b_2} - \frac{b_1 b_2}{a_1 a_2} + \frac{a_1 b_2}{a_2 b_1} + \frac{a_2 b_1}{a_1 b_2}$ (3.2c)

where $\alpha_1$, $b_1$ are the first DC forward and cross coupling coefficients, respectively, and $\alpha_2$, $b_2$ are the second DC forward and cross coupling coefficients, respectively. $\Delta\phi$ the phase difference between the interferometer arms.

We can then derive the resonance wavelength ($\lambda_{res}$), from the constructive interference condition, as:

$$\Delta\phi = q\pi => \lambda_{res,LT-MZI} = \frac{2}{m}(n_{eff,2}L_2 - n_{eff,1}L_1) \qquad (3.3a)$$

$$\frac{\Delta\phi}{2} = q\pi => \lambda_{res,MZI} = \frac{1}{m}(n_{eff,2}L_2 - n_{eff,1}L_1) \qquad (3.3b)$$

with m an integer number, $L_1$ and $L_2$ are the arm lengths, and $n_{eff,1}$ and $n_{eff,2}$ are the effective indices of the reference and sensing arms, respectively.

Then the free spectral range (FSR), wavelength spacing between two consecutive resonances, as:

with $FSR_{LT-MZI} = \frac{\lambda^2}{2(n_{g,2}L_2 - n_{g,1}L_1)}$ (3.4a)

and $FSR_{MZI} = \frac{\lambda^2}{(n_{g,2}L_2 - n_{g,1}L_1)}$ (3.4b)

We can also derive the full width half maximum (FWHM), which is the wavelength width $\Delta\lambda$ with transmission greater than half the maximum transmission ($T_{max}/2$), from transmission equations (3.1-3.2) using $T(\lambda_{res} + \frac{\Delta\lambda}{2}) = \frac{T_{max}}{2}$. For a 3-dB DCs, $a_1 = a_2 = b_1 = b_2 = \frac{1}{\sqrt{2}}$, we have H=0, and FWHM is:

$$FWHM_{LT-MZI} \approx \frac{FSR_{LT-MZI}}{\pi} \quad (3.5a)$$

$$FWHM_{MZI} \approx \frac{FSR_{MZI}}{\pi} \quad (3.5b)$$

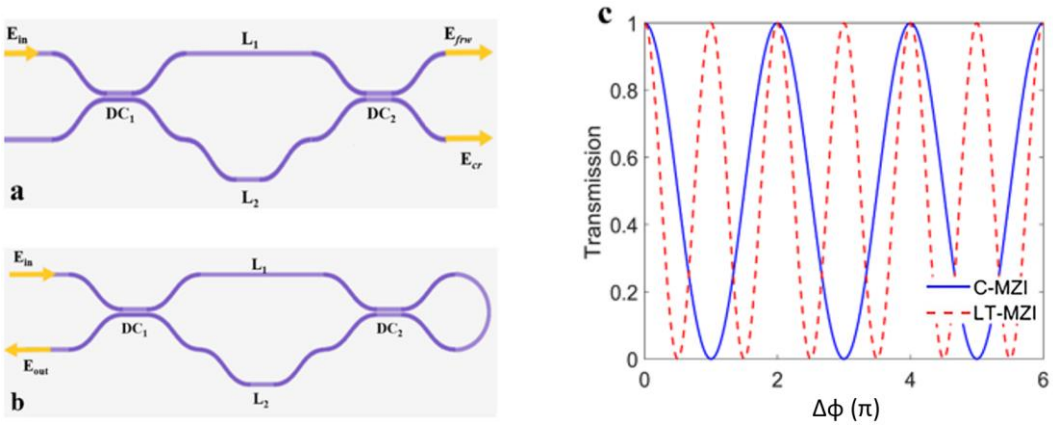



Figure 3.1: (a) Schematic of the conventional MZI, (b) schematic of our proposed LT-MZI, and (c) transmission as a function of the phase difference between interferometer arms for both LT-MZI and C-MZI.

Figure 3.1(c) compares the effect of the phase changes in the output transmission for both MZI and LT-MZI with 3-dB directional couplers ($\alpha_1=b_1=\alpha_2=b_2=1/\sqrt{2}$). Figure shows that LT-MZI have two times higher rate of change than MZI. This makes LT-MZI have the same FSR of the a conventional MZI with only half its length. Accordingly, for the same path lengths the FWHM of the LT-MZI is half that of MZI. Getting lower FWHM with the same physical length of the interferometer is useful in many applications such modulators and sensors.

LT-MZI exhibit another unique feature which is not present on other types of interferometers like conventional MZI or Michelson interferometer. LT-MZI transmission

spectrum can have different forms by designing the coupling coefficients of its DCs. Altering the DCs coefficient modifies the H parameter value, which in turn changes the transmission spectrum, as described in equations (3.1) and (3.2). Figure 3.2(a) illustrates the H values for different $a_1$ and $a_2$ values with $b_1^2=1-a_1^2$ and $b_2^2=1-a_2^2$, indicating lossless DCs. The H parameter is positive when $a_1>0.5$ and $a_2<0.5$ or vice versa, negative if both a1 and a2 are either greater than 0.5 or less than 0.5, and zero when a1=a2=0.5. Figure 3.2(b) depicts the transmission spectrum of the LT-MZI in an SOI waveguide with w=500nm and $\Delta L=20\mu m$ as a1 varies while a2 remains fixed at 0.8. This shows that adjusting the DC1 coefficient inversely alters the transmission intensity of even and odd resonances, allowing to switch from a spectrum with only odd resonances to one with only even resonances

There are two special cases worth mentioning. First, when the DCs coefficients change such that a1=a2, as shown in Figure 3.2(c), even resonances always peak at T=1, while the intensity of odd resonances can be varied from 0 to 1 by designing the DCs coefficients.

Second, when the DCs coefficients are adjusted so that a1+a2=1, as seen in Figure 3.2(d), odd resonances peak at T=1, and the intensity of even resonances can be modulated. This behavior is not observed in conventional MZI or the Michelson interferometer [159], where changing the DCs coefficients only uniformly affects the transmission intensity of the entire spectrum.

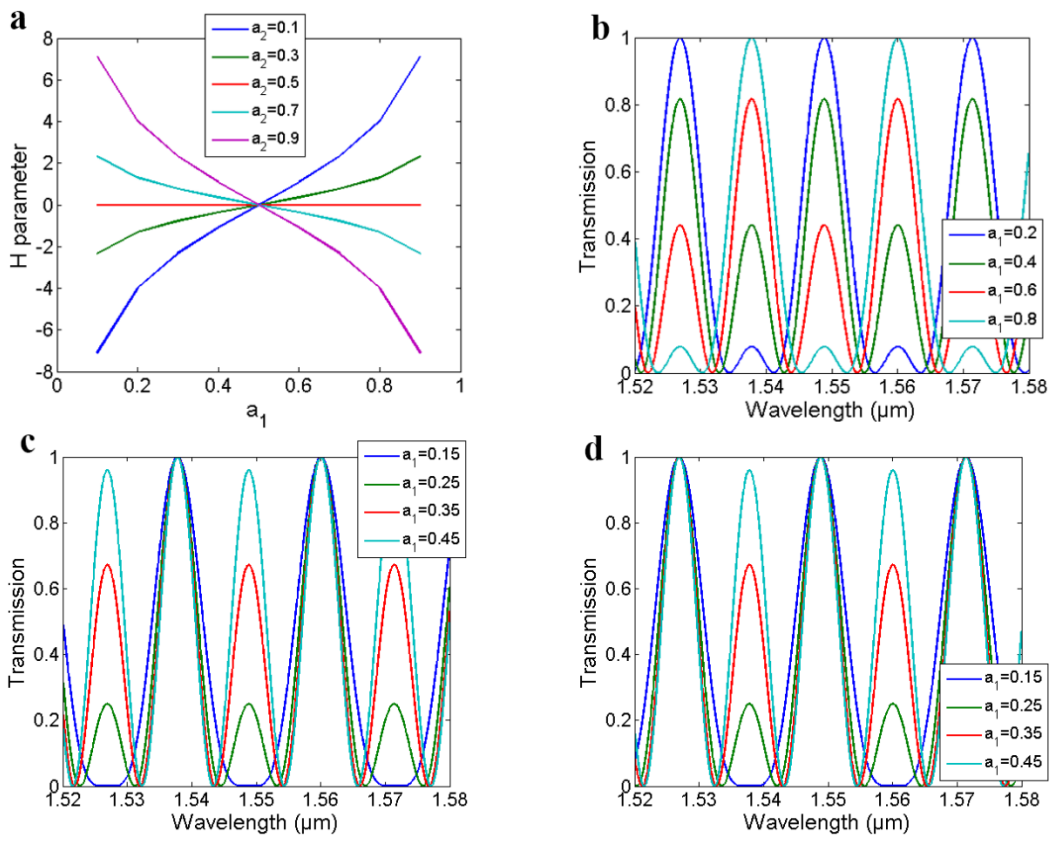



Figure. 3.2: The H parameter as a function of the a1 coefficient for various a2 values. The transmission spectrum as a function of wavelength for the TE mode in an SOI waveguide with ΔL=20 μm and w=500 nm: (b) with a2=0.8 for different a1 values, (c) for different a1=a2 values, and (d) for different a1 values with a2=1-a1.

## 3.3 Sensor Design

### 3.3.1 LT-MZI Structure and Analysis

The full structure of our LT-MZI gas sensor and its principle of operation are shown in Figure 3.3. The structure is based on the standard SOI technology with a 220 nm silicon device layer working at the 1.55 μm wavelength, with the whole sensor subjected to air cladding. A slot waveguide was used as the sensing arm to maximize the interaction with

the gaseous medium. On the other hand, a wide strip waveguide was used as the reference arm, as well as for the other parts of the sensor, so that there was minimal interaction with the sensing medium.

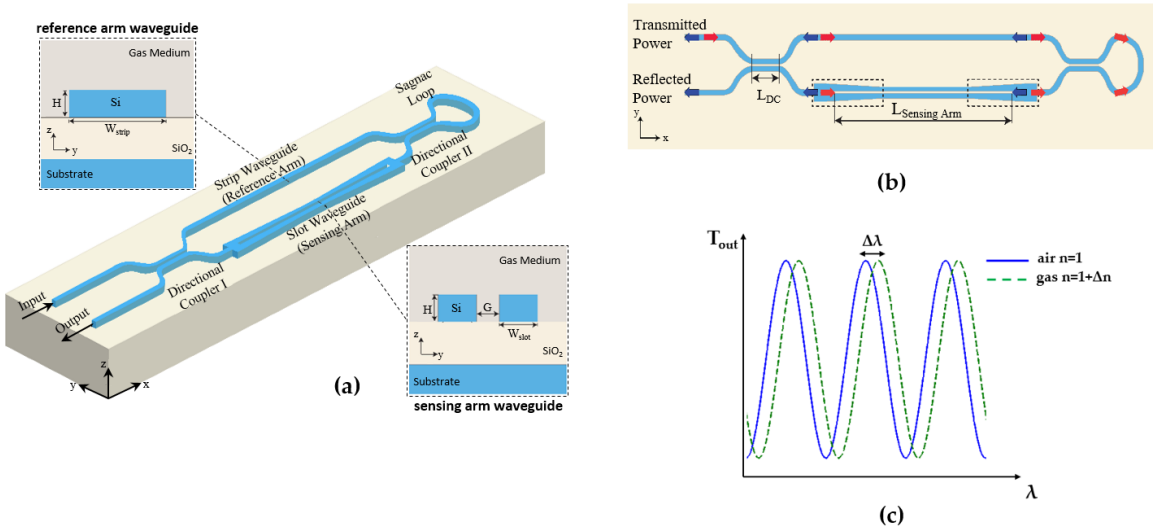



Figure 3.3: (a) Three-dimensional schematic of the LT-MZI gas sensor; insets: cross sections of the strip (reference arm) and slot (sensing arm) waveguides; (b) top view of the LT-MZI gas sensor showing the paths for the light beams with red and blue arrows, while marking the mode converters with a dotted-line box; (c) schematic of a typical LT-MZI interferometer transmission and the shift in the spectrum that occurs due to gas presence.

Note that a sensor with high sensitivity is not enough for good detection, as the FWHM of the sensor should also be minimized. Thus, a figure of merit (FOM) is defined for such sensors as the ratio between the wavelength sensitivity $S_\lambda$ and the FWHM, as discussed in previous chapters. The FOM determines the intrinsic limit of detection iLoD [35] of an RI sensor, i.e., the minimum detectable refractive index change iLoD = $\Delta n_{med,min}$, as follows:

$$FOM = \frac{\partial\lambda/\partial n_{med}}{FWHM} = \frac{S_\lambda}{FWHM} = \frac{1}{iLoD} \qquad (3.6)$$

where $S_\lambda$ is the wavelength sensitivity and $n_{med}$ is the refractive index of the sensing medium.

From the resonance wavelength, equation (3.3), the wavelength sensitivity of a LT-MZI and conventional MZI can be derived as:

$$S_{\lambda,LT-MZI} = \frac{\lambda_{res}}{(n_{g,sens}L_{sens} - n_{g,ref}L_{ref})} L_{sens}S_{wg} \qquad (3.7a)$$

$$S_{\lambda,MZI} = \frac{\lambda_{res}}{(n_{g,sens}L_{sens} - n_{g,ref}L_{ref})} L_{sens}S_{wg} \qquad (3.7b)$$

where Lref and Lsens are the arm lengths, and $n_{eff,ref}$ and $n_{eff,sens}$ are the effective indices of the reference and sensing arms, respectively. $S_{wg}=dn_{eff}/dn_{med}$ is the sensing arm waveguide sensitivity. So, both LT-MZI and MZI configuration have the same wavelength sensitivity $S_\lambda$ for the same waveguide structure and same dimensions. Using FWHM from equation (3.5) the FOM can be determined as:

$$FOM_{LT-MZI} = \frac{S_\lambda}{FWHM} = \frac{2\pi S_{wg}L_{sens}}{\lambda} = 2FOM_{MZI} \qquad (3.8)$$

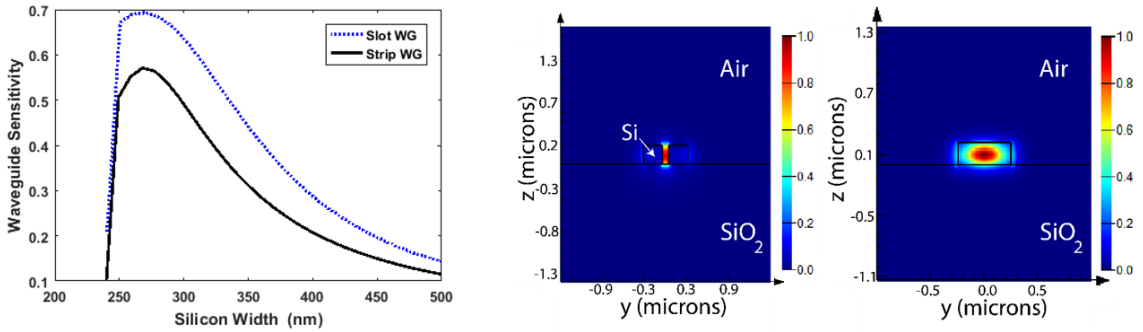
FOM of LT-MZI is double that of conventional MZI with both proportional to sensing arm length $L_{sens}$ and its waveguide sensitivity $S_{wg}$.

### 3.3.2 Components Design

A DC with a gap $G_{DC}$ of 100 nm and a length $L_{DC}$ of 12.1 µm was designed using 3D-FDTD to work as a 3 dB coupler at 1.55 µm. Next, we optimized the slot waveguide with a 100 nm slot gap width Gslot such that we maximized the waveguide sensitivity $S_{wg}$. $S_{wg}$ was determined at $\lambda = 1.55$ µm using the FDE solver. A change in the air cladding medium

index $\Delta n_{med}$ of 0.001 was introduced and the corresponding change in the mode effective index $\Delta n_{eff}$ was calculated using the FDE solver from which the $S_{wg}$ value was determined.

Figure 3.4(a) shows the $S_{wg}$ of the quasi-transverse electric (TE) mode for both the slot and strip waveguide versus the silicon waveguide width. The results show that there is an optimum width for both waveguides that maximizes $S_{wg}$. This is expected because, at small silicon widths a significant part of the evanescent field is confined in the substrate, resulting in a low $S_{wg}$, while for large silicon widths most of the field will be confined inside the silicon. Hence, there is an optimum width where the field in the cladding is maximized. According to the results shown in Figure 3.4(a), we chose the silicon width of the slot waveguide to be $W_{slot}$ = 270 nm, which gave the highest $S_{wg}$ value of 0.7. On the other hand, wide strip waveguides allow for better confinement, and hence lower sensitivity, and lower propagation and bend loss. Therefore, we chose $W_{strip}$ = 500 nm, which was the largest possible waveguide width, while maintaining single-mode operation, resulting in an $S_{wg}$ as low as 0.1. For this design, there is no need for capped oxide deposition and subsequent lithography and etching of a sensing area.




(a) (b) (c)

Figure 3.4: (a) Waveguide sensitivity $S_{wg}$ of the quasi-TE mode of both the strip and the slot waveguides as a function of the silicon widths at $\lambda = 1.55$ μm. The fundamental quasi-TE mode of: (b) the slot waveguide with $W_{slot} = 270$ nm and (c) the strip waveguide with $W_{strip} = 500$ nm.

The mode mismatch between the slot and strip waveguides (see Figure 3.4(b) and Figure 3.4(c)) makes the coupling challenging. To tackle this problem, a mode converter was designed. Various designs have been realized for such a converter [128], [162], [163], [164]. The work in [164] was adapted for our LT-MZI gas sensor. A conversion efficiency higher than 95% for both the quasi-TE and quasi-TM modes was experimentally demonstrated in this study. The converter also works over a wide bandwidth range from 1.475 μm to 1.6 μm, where our sensor is operating, and it shows high tolerance to fabrication errors. The mode converter is composed of two sections: a multi-mode interferometer (MMI) and a slot taper, see Figure 3.5(a). As illustrated in the field profile of the converter shown in Figure 3.5(b), the MMI is used to split the input light into two identical beams, while the taper section gradually pushes the field to the slot. The dimensions of the mode converter as indicated in Figure 3.5(a) were optimized using 3D-FDTD simulations, to reduce the insertion loss (IL) and to improve its overall performance. The IL is calculated as:

$$IL = 10 \times \log(T_{out}) \qquad (3.9)$$

where Tout=Pout/Pin is the output transmission, Pout is the output power transmitted through the slot waveguide quasi-TE mode, and Pin is the input power launched to the strip

waveguide fundamental quasi-TE mode. Tout is easily calculated using the power monitor of the FDTD solver.

The simulation results show that the IL is highly dependent on the taper length, $L_{Taper}$. As demonstrated in Figure 3.5(c), we found that for an IL lower than 0.05 dB, the length of the taper section must be greater than 3 µm. Based on this conclusion, the dimensions of the optimized mode converter were designed to be WMMI = 1.1 µm and LMMI = 1.2 µm, with $L_{Taper}$ equal to 5 µm. Finally, to ensure negligible losses, the Sagnac-loop bend radius R, was 5 µm [116]. It is important to note that perfectly-matched-layer (PML) boundary conditions and a minimum of 10 meshes in every dimension in the object's cross section were used in all simulations, with both the FDE and FDTD solvers. In addition, for the FDTD simulations, a spectral window of 100 nm from 1.5 µm to 1.6 µm was simulated.

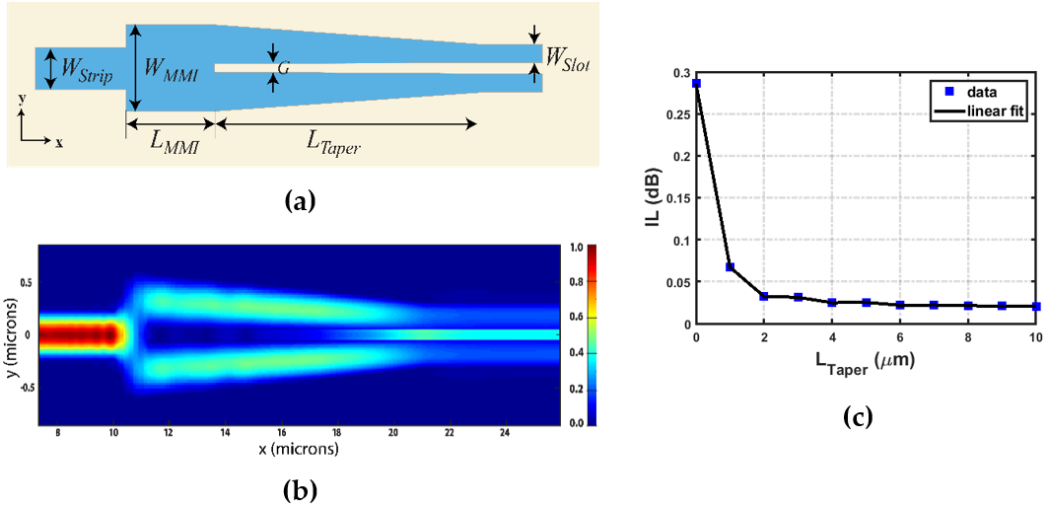



Figure 3.5: (a) Schematic of the mode converter; (b) quasi-TE-mode profile of the mode converter as depicted from FDTD solver; (c) the insertion loss of the mode converter as a function of the length of its taper section at λ = 1.55 µm.

### 3.3.3 Simulation Setup

In this section we mention details about the simulation setup used in FDTD and FDE simulations. Ansys-Lumerical is the software used for both FDTD and FDE simulations. The same settings are used in all the numerical results presented throughout this thesis. Both FDTD and FDE solvers use cartesian style mesh, cuboid for 3D-FDTD and rectangular for FDE. For the FDTD a non-uniform mesh is automatically generated by the software with a scale from 1 to 8 with 8 being the highest accuracy (but also highest computational cost). The meshing algorithm takes into account many factors, including source wavelength, material properties and the structure geometry. In most cases, auto non-uniform mesh with accuracy of 2 will be enough, corresponding to a target of 10 mesh points per wavelength. In our work we use auto non-uniform mesh with accuracy of 3 (14 points per wavelength).

For FDTD we use perfect matched layer (PML) boundary conditions at all sides. Ideally PML boundaries are used to absorb all incident field without creating any back-reflections. The quality of the PML boundary (how much of the incident field is absorbed and not reflected) is dependent on several parameters. Standard PML settings, shown in Table 3.1, works for most of the cases which we are using here.

For FDE a uniform mesh size of 80 nm is used for both x and y direction over the whole simulation region. Then, a finer mesh is used around the waveguide core with size 10 nm in both x and y sides. The boundary conditions used in FDE modal analysis is Metal (computationally efficient).

The simulation region should be wide enough in order to ensure that the mode filed is almost zeros at the boundaries, so that the mode is accurately described and to prevent losses due to the metal boundaries. We choose the simulation region such that mode field at the boundaries is less than 20 dB of its peak value. Here, a 5 μm for both the x span and the y span is used for the simulation region.

Table 3.1: FDTD standard PML boundary conditions settings.

| | |
|---|---|
| Layers | 8 |
| Kappa | 2 |
| Sigma | 1 |
| Polynomial | 3 |
| Alpha | 0 |
| Alpha polynomial | 1 |
| Min layers | 8 |
| Max layers | 64 |

## 3.4 LT-MZI Sensor Simulation Results

To determine the performance of the whole sensor, i.e., obtain the LT-MZI output transmission spectrum, the Lumerical INTERCONNECT [165] photonic circuit simulator was used. In this tool, the FDTD simulation results of the different components are used to obtain the overall LT-MZI sensor response via the scattering matrix method. This allows for fast and accurate simulation of the whole LT-MZI sensor for different lengths and

different medium indices. Note that, the refractive index of gases is typically very close to 1 with $\Delta n = n_{gas} - 1$ less than $10^{-3}$. However, this refractive index can vary significantly depending on the pressure, reaching a value of $\Delta n$ of the order of $10^{-3}$ [166]. Therefore, we varied the cladding medium index over a wide range from 1 to 1.009. In Figure 3.6(a), the output transmission spectra of our sensor with L = 50 μm for different surrounding medium indices are shown. The effective index dispersion $n_{eff}(\lambda)$ from 1.5 μm to 1.6 μm was calculated for the strip and slot waveguide modes for different cladding indices using the FDE solver, and these were then used in INTERCONNECT to determine the corresponding output transmission spectrum.

There are two main sensing schemes for the determination of the refractive index of the sensing medium: intensity interrogation and wavelength interrogation. In the intensity interrogation scheme, the medium's refractive index is determined by the change in the output intensity (transmission) at a specific wavelength. Figure 3.6(b) depicts the change in the output transmission at $\lambda$ = 1.55 μm due to the change in the refractive index of the medium for our LT-MZI sensor with L = 50 μm. In the wavelength interrogation scheme, the medium's refractive index is determined by the shift in the resonance wavelength $d\lambda_{res}$, see Figure 3.6(a). Figure 3.6(c) shows the shift in the resonance wavelength due to the change in the refractive index of the medium for our LT-MZI sensor with L = 50 μm.

Next, the FOM of our LT-MZI sensor design was calculated from the transmission spectra obtained by the INTERCONNECT simulations, where the wavelength sensitivity S and the FWHM were first calculated. In Figure 3.7(a), the dependence of the FOM on the medium index change $\Delta n_{med}$ is determined. It can be seen that, a lower $\Delta n_{med}$ results in a

higher FOM; hence, our design is suitable for gas sensing where the refractive index is very close to 1. The results also show that our LT-MZI gas sensor achieves a high wavelength sensitivity that reaches 1070 nm/RIU around λ = 1.55 µm with only 150 µm arm's length and a FOM as high as 280.8 $RIU^{-1}$. The FOM is further increased by increasing the interferometer's arm length with a linear dependence, as expected from equation (3.8), see Figure 3.7(b). Results show that, FOM reaches over 2500 $RIU^{-1}$ with L=1.5 mm, corresponding to an iLoD as low as 3.9e-4.

Finally, we point out that the design of the LT-MZI gas sensor can be applied using even longer sensing arms than the ones used in this study, to achieve higher values of sensitivity and the FOM.

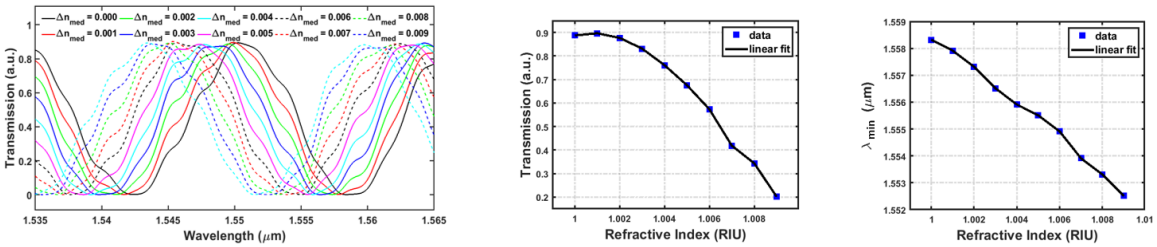



Figure 3.6: (a) The output transmission spectra of the LT-MZI sensor with a length L of 50 µm at different refractive indices ranging from 1.000 to 1.009; (b) LT-MZI sensor transmission versus the medium's refractive index around λ = 1.55 µm with L = 50 µm; (c) LT-MZI sensor minimum wavelength versus the medium refractive index with L = 50 µm.

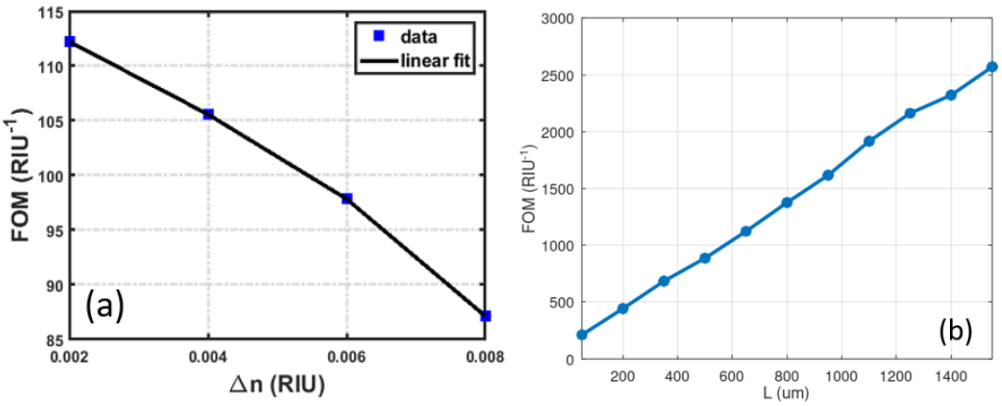



Figure 3.7: (a) LT-MZI sensor FOM calculated at different medium index changes (Δn) with L = 50 μm; (b) LT-MZI sensor FOM versus interferometer arm length.

## 3.5 Temperature Variations Immune Design

As mentioned in chapter 2, MZI based sensors can be designed to compensate for a certain type of variations, by engineering its arms lengths, see equation (2.15a). To compensate for temperature variations, we need:

$$\frac{S_{ref,T}}{S_{sens,T}} \frac{L_{ref}}{L_{sens}} = 1 \qquad (3.10)$$

where $S_{ref,T}$ and $S_{sens,T}$ are the sensitivity of the strip (reference) waveguide and slot (sensing) waveguide to temperature changes, respectively. Performing FDE simulations we get that $S_{ref,T} = 2.08 \times S_{sens,T}$. Hence, to be immune to temperature variations we design our LT-MZI sensor with $L_{sens} = 2.08 \times L_{ref}$. Figure 3.8(a) below, shows the transmission of our slot-based LT-MZI gas sensor with $L_{sens} = 100$ μm and $L_{ref} = 48$ μm. In this case we can see that, the sensor spectrum only shifts when the medium index is changed but it is immune to the temperature variations. To compare the same slot waveguide was also used

in a MRR configuration with $L_{sens} = 2\pi R$ also of a 100 μm. As discussed in chapter 2, we see that, temperature changes cannot be compensated in a MRR design. Calculating the SNR according to equation (2.16) we find that our LT-MZI sensor in this case has 27 times higher SNR than a MRR.

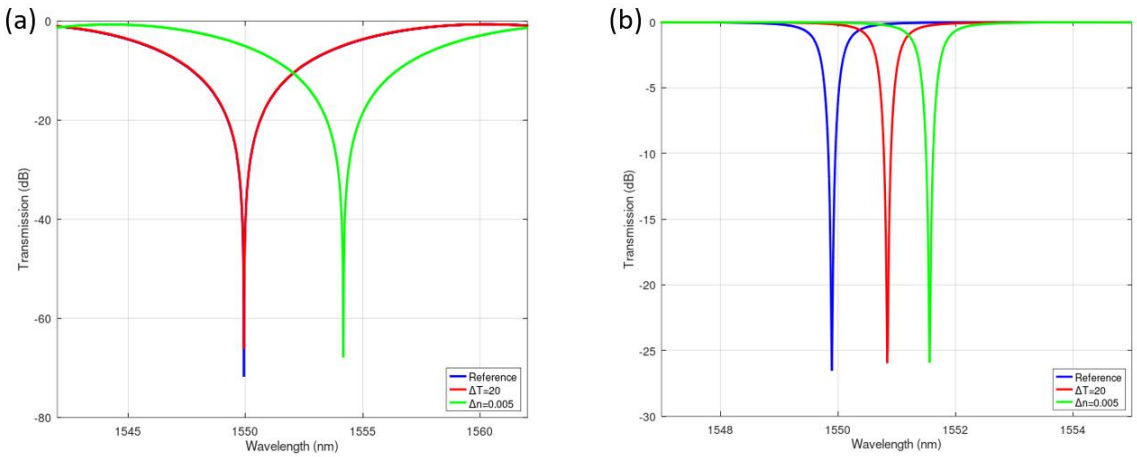



Figure 3.8: (a) Slot-based LT-MZI gas sensor transmission spectrum at different temperatures and medium index and (b) Slot-based MRR gas sensor transmission spectrum at different temperatures and medium index

## 3.6 Experimental Characterization

The same LT-MZI structure as in Figure 3.3(a) but with oxide cladding (limitation of the MPW run, cannot etch oxide cladding) rather than air, denoted the passive LT-MZI, was fabricated and experimentally characterized. However, no gas sensing measurement could be performed using this device as it is not subjected to air, and accordingly the device's sensing performance cannot be determined. Our goal in this fabrication run was to ensure the reliability of our proposed LT-MZI design. This was achieved by comparing the measured response of the fabricated devices (LT-MZI capped with oxide cladding) with

their corresponding numerical simulations. Here, two LT-MZIs capped with oxide with arm lengths L of 50 μm and 150 μm were simulated, fabricated, and characterized.

The fabrication of the structures was achieved using a multi-project wafer (MPW) on a 200 mm SOI wafer with a 220 nm silicon device layer at Applied Nanotools [167], using electron beam lithography (EBL). Figure 3.9(a) shows the waveguide fabrication steps. EBL with a negative resist and inductively coupled plasma (ICP) etching (using chlorine gas) were used for silicon patterning. Then, plasma-enhanced chemical vapor deposition (PECVD) was used to deposit around 2 μm of oxide cladding. The device characterization was performed by the MiNa group at the University of British Columbia [168] and at Maple Leaf Photonics [169]. The measurement setup, shown in Figure 3.9(b), included a tunable laser from 1500 nm to 1600 nm wavelength with 4 pm resolution, a polarization-maintaining fiber (PMF) array with four fibers for the i/o optical signal, and a multi-channel photodetector. Grating couplers provided by the fab PDK were used to couple the optical signal from/to the fiber array on each device on the chip. The chip was placed on an automated stage with 2D movement. The stage was linked to a computer with software that automated the measurements by controlling the movement of the stage using the GDS file of the chip to trace and measure the different optical devices on the chip. A temperature controller was also used to adjust the temperature of the stage. Using this setup, the transmission spectra of the different devices could be measured in a fast and accurate manner.

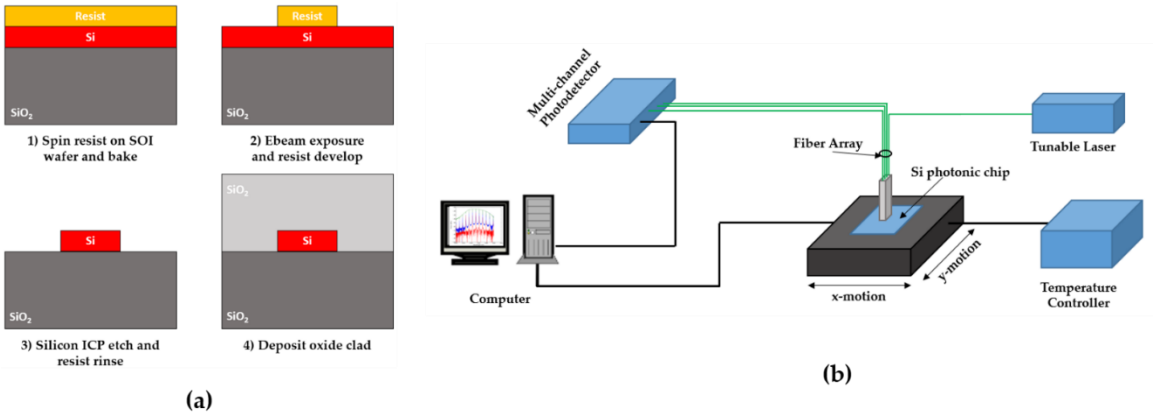



Figure 3.9: (a) Silicon waveguide fabrication steps using EBL; (b) schematic of the measurement setup.

The layout of the fabricated LT-MZI and SEM images of its various components are shown in Figure 3.10(a). The measured transmission spectra of the LT-MZIs with L = 50 µm and L = 150 µm are shown in Figure 3.10(b) and Figure 3.10(c), respectively, along with the corresponding INTERCONNECT simulation results. It can be seen that there is a good match in the device response between the simulation and the measurement results. The FSR calculated from the measurements was 11.24 nm and 4.19 nm with 1.1% and 2.18% deviation from the simulation results for the 50 µm and 150 µm devices, respectively. These results further ensured that the numerical simulation results we obtained previously could be trusted.

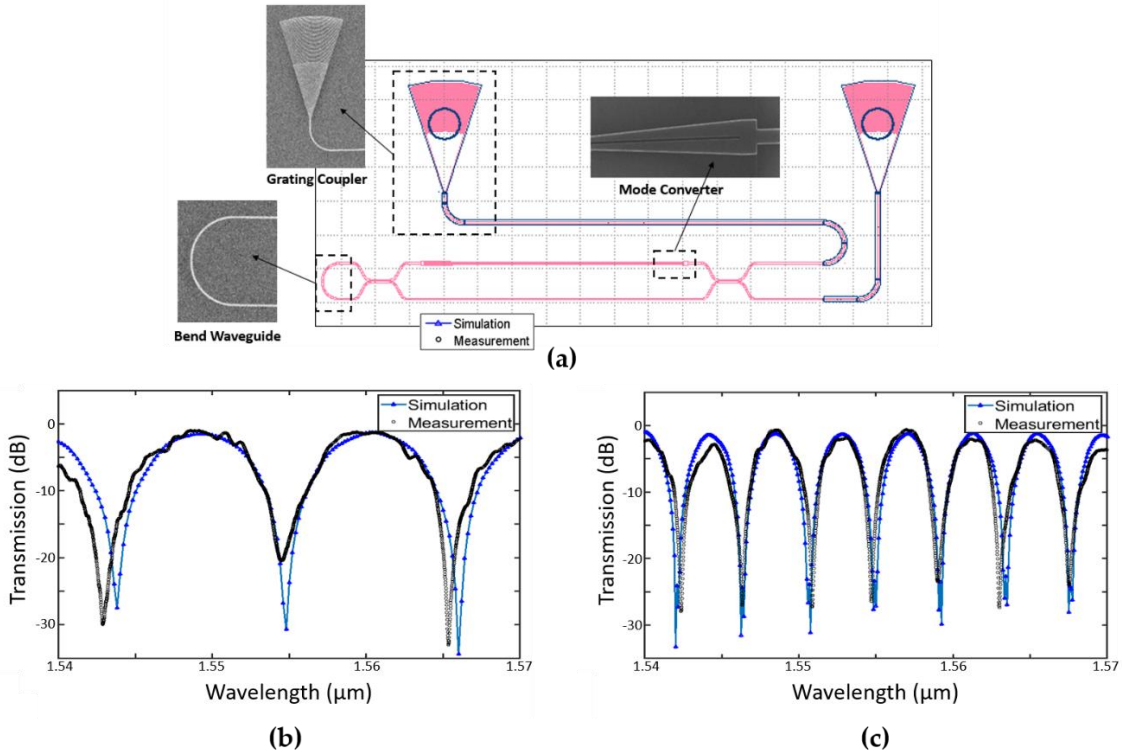



Figure 3.10: (a) Layout of the fabricated LT-MZI with L = 50 μm; insets showing the SEM images of various components. Simulated and measured transmission spectra of the LT-MZI structure with oxide cladding: (b) L = 50 μm; (c) L = 150 μm.

## 3.7 Summary

We presented a compact SOI-based RI gas sensor using a MZI interferometer cascaded with a Sagnac loop forming a loop-terminated MZI. The LT-MZI structure analysis, features were presented and compared with the conventional MZI. The LT-MZI sensor achieved the same performance as a conventional MZI with only half its length. The different sensor components were designed and optimized using FDE and FDTD solvers. A slot waveguide was designed to achieve maximum waveguide sensitivity of 0.7,

compared to a maximum of 0.56 for strip waveguide, which consequently maximizes the overall device sensitivity. A strip-to-slot waveguide mode converter was optimized to minimize the insertion loss between the two waveguides to less than 0.05 dB with only 6.2μm length. The simulation results based on the FDTD solver showed that our design could achieve a high sensitivity of 1070 nm/RIU and a high FOM of 280.8 $RIU^{-1}$ using a length of only 150 μm. Increasing the length of the sensing arm further increase the FOM almost linearly, reaching more than 2500 $RIU^{-1}$ with L=1.5 mm, which corresponds to an iLoD as small as 3.9e-4. The same LT-MZI structure but with oxide cladding was fabricated in MPW run and experimentally characterized, showing a good agreement with the theoretical simulations. Thus, our proposed LT-MZI gas sensor is a good candidate for efficient and compact on-chip sensors with low-cost and mass-scale fabrication, due to the CMOS-compatible SOI platform.

.

# 4 Chapter 4: On-Chip Complex Refractive Index Detection at Multiple Wavelengths for Selective Sensing

## 4.1 Introduction

Most optical RI sensors use wavelength or intensity shifts around single wavelength for refractive index detection. Such sensors are not selective, i.e. they determine only the value of the refractive index n but cannot determine the substance that resulted in this n. At a certain wavelength, different substances with different concentrations can result in the same n. Moreover, in the case of multiple element medium the sensor will determine the resultant n and not the elements composition of this medium. Thus, selectivity is one of the main issues regarding RI sensors. One widely used techniques to solve this problem in bio-sensing is surface functionalization [170], [171], [172], [173], [174], [175]. In surface functionalization the surface of the sensing waveguide is coated with specific molecules called binder or capture molecules and immobilized through certain process. This immobilized molecules selectively capture the analyte molecules to be detected from the whole sample. For instance, in [171] a MZI influenza virus sensor is proposed and experimentally demonstrated. Although this technique is selective and widely used in bio-sensing however, it involves many lengthy processes to functionalize the sensor surface with the capture molecules. In addition, a washing step after sensor exposure to the sample

is needed to make sure that only the analyte of interest is present in the sensor medium. Moreover, this surface functionalization process may be challenging or even not possible on some materials. Also, this technique is used mainly for biological samples. Furthermore, for multi-element detection multiple output signals need to be detected which complicates the sensor circuitry.

On the other hand, absorption sensors which detects the imaginary part k of the medium refractive index can be selective [16], [17], [18], [176]. The absorption spectrum of the sample is measured which then used to specify the different substances present in the sample through their absorption fingerprints [16]. However, the sensitivity of this technique depends on the sample volume because larger interaction lengths between the light and sample will result in higher light absorption hence higher sensitivity [18]. Consequently, integrated waveguide-based absorption sensors with micrometer dimensions suffer from poor sensitivity. To overcome this problem high quality factor (Q) micro-cavities were proposed to enhance the interaction length between the mode field and the analyte [93], [94], [95]. In high Q cavities the light circulates multiple times in the cavity and hence the effective optical path length can be increased significantly from micrometer to millimeter range [93], [94], [95]. In [95] authors used micro-ring resonators with Q > 100,000 to enhance the interaction length and hence sensitivity for on-chip absorption spectroscopy. The absorption spectrum of N-methylaniline was successfully obtained in this work. In [177] authors were detecting the complex refractive index of the sensing medium by measuring changes in the resonance wavelength and the resonance linewidth of a photonic crystal micro-cavity from which they detect the complex refractive

index of the medium. Using this technique, they were able to determine the concentration composition of a ternary mixture.

In this chapter we present a method for on-chip complex refractive index sensor for selective sensing without the need of surface functionalization. Using our proposed method both real n and imaginary k parts of the medium refractive index are determined at multiple wavelengths over certain wavelength range. This is done simply using a single micro-ring resonator (MRR) with multiple resonances along the wavelength range of operation where both n and k are determined at each resonance. In this case the medium complex refractive index dependence with wavelength, i.e. medium dispersion, is determined rather than just a single value at a single wavelength. While at single wavelength different substances can lead to the same n or k, the refractive index dispersion is specific for each substance. Fitting these discrete n and k data with wavelength can then specify the medium substance. On the other hand, the mixture composition of a multi-element medium can also be determined by solving a system of linear equations corresponding to the different wavelengths. This method is similar to the one proposed in [18] however, we here determine the complex refractive index while they were only interested in the detection of the medium absorption coefficient $\alpha$. A ring resonator using the widespread silicon-on-insulator (SOI) technology, working in the near-infrared region around $\lambda$=1.55 μm is implemented. Our designed resonator is able to selectively detect methanol, ethanol, propanol and water in a single element medium with high accuracy. It is also able to determine the concentration composition of multi-element media. In general, there are many sources of error that can lead to inaccuracy in the detected concentrations for instance, temperature fluctuations.

Determining both n and k at multiple wavelengths increase the detection accuracy. This technique can be used with any chemical or biological substance. It doesn't require surface functionalization process. It is efficient even if the sample substance exhibit low absorption as both n and k are detected simultaneously. More importantly, multi-element sample is detected through single output.

Next, we integrated our proposed method with neural network techniques to eliminate electrical noise, enhancing accuracy and extending the detection range. This approach employs two types of neural networks: a convolutional neural network (CNN) and a deep neural network (DNN). The CNN isolates the transmission spectrum from noise, and this spectrum is then used to extract selected features, which are input into the DNN to determine the concentration of each chemical in the analyte. Both models were trained using simulated data from our designed SOI MRR. The CNN was trained on MRR transmission spectra combined with white Gaussian noise and Poisson noise to simulate various noise sources, while the DNN was trained on the extracted features.

This chapter is organized as follows; section 4.2 presents our method for on-chip complex refractive index sensor at multiple wavelengths for selective sensing along with the design and optimization of the SOI MRR. Section 4.3 demonstrates our method for selectively detecting a single-element medium and determining the concentration composition of a multi-element medium. In section 4.4 we verify our proposed method by experimental characterization of a fabricated MRR sensor and determine its cladding refractive index. Section 4.5 presents our method to enhance the performance of the sensor using deep learning. Finally, section 4.6 summarizes this chapter.

## 4.2 Complex Refractive Index Detection

### 4.2.1 Methods

To determine the medium complex refractive index n+jk over a range of wavelengths, a MRR with multiple resonances along this range is designed. Changing the real part of the medium refractive index results in a shift in the resonance wavelength $\lambda_{res}$. While changing the imaginary part of the medium index results in a change in the cavity losses and accordingly the field attenuation coefficient α. Hence, at each resonance the corresponding $n_i$ and $k_i$ of the medium are determined, where i=1-N is the resonance index and N the number of resonances in the measured wavelength range.

The transmission spectrum of the MRR with lossless coupling ($r^2+t^2=1$) is given by [18]:

$$T(\theta) = \frac{a^2+r^2-2ar\cos(\theta)}{1+a^2r^2-2ar\cos(\theta)} \qquad (4.1)$$

$$a = e^{-\alpha L} \qquad (4.2a) \qquad \text{and} \qquad \theta = \beta L = \frac{2\pi}{\lambda} n_{eff} L \qquad (4.2b)$$

$$\alpha = \alpha_{\text{int}} + \alpha_{abs} \quad (4.3a) \qquad \text{and} \qquad \alpha_{abs} = \frac{2\pi}{\lambda} k_{eff} \qquad (4.3b)$$

where α the field loss coefficient. The loss coefficient constitutes of two parts the intrinsic loss $\alpha_{int}$, such as waveguide surface roughness and bend loss, and medium absorption loss $\alpha_{abs}$. r the forward coupling coefficient, the fraction of field that transmit to the straight waveguide while t the cross-coupling coefficient, the fraction of field that is coupled to the ring cavity. θ the round trip phase where L the cavity length L=2πR, β the mode propagation constant, $n_{eff}$ the mode effective index and R the ring radius.

A reference medium with known refractive index $n_{ref}$ and $k_{ref}$ is used from which we calculate the change in the resonance wavelength and the change in the loss ($\Delta\lambda_{res}$ and $\Delta\alpha$). We first fit each resonance of the MRR spectrum both of the reference and the sensing medium with wavelength using equation (4.1) to obtain the resonant wavelengths $\lambda_{res,i}$ and the attenuation factors $\alpha_i$, from which the absorption loss coefficient $\alpha_i$ is calculated using equation (4.2a). When doing the fitting we expand the cosine term in equation (4.1) around the resonant wavelength $\lambda_{res}$ such that the fitting parameters are only $\lambda_{res,i}$ and $\alpha_i$ while the rest of the terms are known from the design, i.e. r and L=2πR, see equation (4.4). Note that, the group index $n_g$ in equation (4.a) can be easily calculated from the measured free spectral range (FSR), resonance spacing, using equation. (4.5).

$$cos(\theta) = cos(2\pi L n_{eff}/\lambda) \simeq 1 - \frac{2\pi^2 L^2 n_g^2}{\lambda_{res}^4}(\lambda - \lambda_{res})^2 + \frac{2\pi^4 L^4 n_g^4}{3\lambda_{res}^8}(\lambda - \lambda_{res})^4 \qquad (4.4)$$

$$FSR = \frac{\lambda^2}{n_g L} \qquad (4.5)$$

Then, the change in the mode effective index of the sensing medium from the reference, both the real $\Delta n_{eff,i}$ and the imaginary $\Delta k_{eff,i}$ part, are calculated from the resonance wavelength shift $\Delta\lambda_{res,i}$ (resonance condition) and the change in the loss coefficient $\Delta\alpha_i$, respectively using equation (4.6a) and equation (4.6b). This method is similar to the one used in [18]. However, they were fitting equation (4.1) with $\alpha_i$ as they were interested only in the detection of the medium absorption coefficient α.

$$\Delta n_{eff} = n_{eff,sens} - n_{eff,ref} = \frac{n_{eff}}{\lambda_{res}}\Delta\lambda_{res} \quad (4.6a)$$

$$\Delta k_{eff} = k_{eff,sens} - k_{eff,ref} = \frac{\lambda}{2\pi}\Delta\alpha \qquad (4.6b)$$

Next, we calculate the medium index change $\Delta n_{med,i}$ and $\Delta k_{med,I}$ from $\Delta n_{eff,i}$ and $\Delta k_{eff,i}$ as follows:

$$\Delta n_{eff} = S_{wg}\Delta n_{med} \qquad (4.7a)$$

$$\Delta k_{eff} = S_{wg}\Delta k_{med} \qquad (4.7b)$$

$$S_{wg} = \frac{dn_{eff}}{dn_{med}} = \frac{dk_{eff}}{dk_{med}} \qquad (4.7c)$$

where $S_{wg}$ the waveguide sensitivity. Finally, the absolute complex refractive index of the sensing medium is determined using the known reference medium index as:

$$n_{sens_med} = n_{ref_med} + \Delta n_{med} \qquad (4.8a)$$

$$k_{sens_med} = k_{ref_med} + \Delta k_{med} \qquad (4.8b)$$

### 4.2.2 MRR Design

In this section we design the MRR for the complex index detection, namely the waveguide width w and the ring radius R. The resonator is based on the standard SOI technology with thickness h=220 nm and is operating in the near-infrared region with 140 nm bandwidth from 1.46 μm to 1.6 μm.

The main performance parameters of the ring resonator sensor can be derived from the relations of equation (4.1-4.3). The wavelength sensitivity Sn, the attenuation factor

sensitivity $S_k$, the full width half maximum $\delta\lambda_{FWHM}$, the minimum/maximum detectable real refractive index change $dn_{min}/dn_{max}$ and the minimum/maximum detectable imaginary refractive index change $dk_{min}/dk_{max}$ are given by:

$$S_n = \frac{d\lambda}{dn} = \frac{\lambda}{n_g} S_{wg} \qquad (4.9a) \qquad S_k = \frac{da}{dk} = -aL\frac{2\pi}{\lambda} S_{wg} \qquad (4.9b)$$

$$\delta\lambda_{FWHM} = FSR\frac{1-a^2}{\pi a} \qquad (4.10)$$

$$\Delta n_{\min} = \frac{\Delta\lambda_{\min}}{S_n} = \frac{n_{eff}}{\lambda S_{wg}}\Delta\lambda_{\min} \qquad (4.11a) \qquad \Delta n_{\max} = \frac{FSR}{S_n} = \frac{\lambda}{S_{wg}L} \qquad (4.11b)$$

$$\Delta k_{\min} = \frac{1}{2S_n}\frac{a}{1+a^2}\Delta\lambda_{min} = \frac{1}{2}\frac{a}{1+a^2}\Delta n_{min} \qquad (4.12a) \qquad \Delta k_{\max} = \frac{1}{2\pi}\frac{da}{a}\bigg|_{\max}\Delta n_{\max} \qquad (4.12b)$$

The ring radius R determines the FSR of the cavity equation (4.5). Hence, increasing R will reduce the FSR, i.e. there will be more resonances in the operating wavelength range, and accordingly a greater number of points in the detected n and k. However, reducing the FSR will limit the maximum refractive index change that can be detected equation (4.11b). For a specific waveguide width there is a minimum radius $R_{min}$ to ensure low bend loss. So, to be able to detect wide range of $\Delta n$ we work on $R \approx R_{min}$. Note that, as the waveguide width decreases as this $R_{min}$ increase.

On the other hand, the width of the waveguide determines the waveguide sensitivity $S_{wg}$. We need to maximize $S_{wg}$ to maximize the sensor sensitivity both to changes in n and k as equation (4.9) depicts. This enhances the sensor performance as it will minimize the lowest detectable $dn_{min}$ and $dk_{min}$ according to equation (4.11a) and equation (4.12a). Figure 4.1(a)

shows the $S_{wg}$ of the SOI waveguide fundamental quasi-transverse electric (TE) mode versus the waveguide width w at $\lambda$=1.55 μm, where we can see that there is an optimum width $w_{opt}$=270 nm. For w=270 nm, $R_{min}$=32 μm for $\alpha_{bend}$>0.99, where $\alpha_{bend}$ field attenuation coefficient due to bend loss. In this case the maximum dn that can be detected is $dn_{max}$=0.01. Also note that, both $dk_{min}$ and $dk_{max}$ are proportional to $dn_{min}$ and $dn_{max}$, respectively. Thus, optimizing the real part parameters will also optimize the imaginary part parameters.

Here we design a MRR to detect four different elements namely, methanol, ethanol, propanol and water. The n and k data of these materials are obtained from [178], [179]. It can be seen that, in the wavelength range of interest $dn_{max}$=0.048. This detection range is not possible with $w_{opt}$=270 nm. So, we choose to increase the width slightly from the optimum value so that we can decrease R, increase FSR and accordingly $dn_{max}$. We finally choose to work with w=320 nm and R=7.6 μm resulting in FSR in the range of 9.85 nm to 12.13 nm with a total of 11 resonances (N=11) and $dn_{max}$=0.058. Figure 4.1(b) shows the mode profile of the used strip waveguide. We design the forward coupling r of both the reference and the sensing ring resonators such that they work around the critical coupling condition r≈α, to achieve the maximum possible extinction ratio (ER). The forward coupling is determined by the separation sep between the straight and the ring waveguide. In our case the attenuation coefficient α is in the range of 0.96-0.998 for the reference ring and 0.9395-0.9581 for the sensing ring. Hence, we choose sep=500 nm which corresponds to r between 0.95 and 0.996. We finally choose that the medium (cladding) index of the reference arm is $n_{ref}$=1.336 and $k_{ref}$=0.

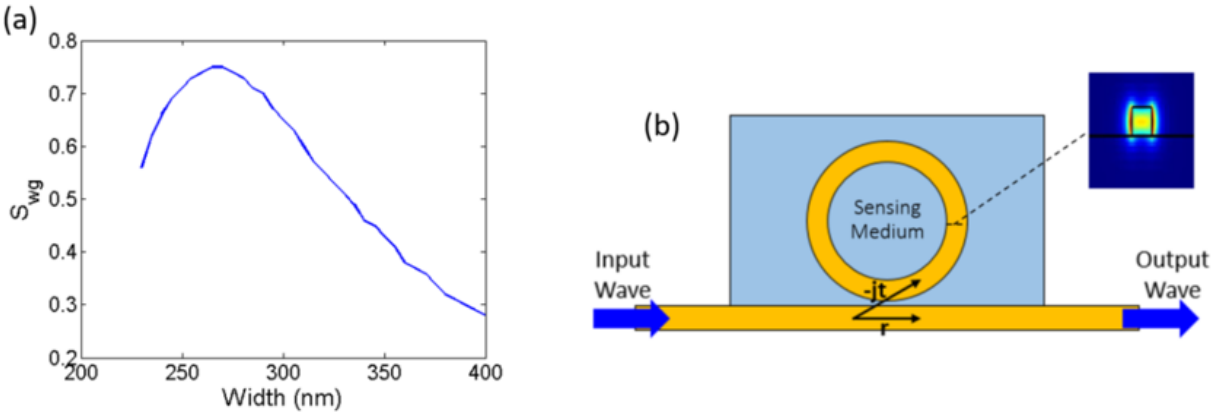



Figure 4.1: (a) $S_{wg}$ versus waveguide width w at λ=1.55 µm. (b) Schematic of the MRR sensor; inset: electric field intensity profile of the used SOI waveguide (w=320 nm) fundamental quasi-TE mode at λ=1.55 µm.

## 4.3 Results & Discussions

### 4.3.1 Single element medium

In this section we show how we selectively detect the substance of a single element medium. The MRR output spectrum of both the reference and sensing medium are determined using mode simulations. Ethanol, methanol, propanol and water are all used as sensing medium. Finite difference eigenmode (FDE) solver [29] is used to extract the waveguide complex effective index $n_{eff}(\lambda)+jk_{eff}(\lambda)$ corresponding to the reference and the sensing medium using the n and k data of the different elements over the wavelength range 1.46 µm to 1.6 µm. Then, the output transmission is determined using equation (4.1) and the obtained complex effective index. This data represents the measured spectrum of the sensor. Then we use the method described in section above to retrieve the $n_{med,i}$ and $k_{med,i}$ from these spectra. Figure 4.2 shows the ring resonator transmission spectra of the reference and the different sensing media namely, ethanol, methanol, propanol and water.

A resolution of 5 pm is used for the transmission spectra which correspond to the spectrometer resolution. All numerical calculations and fitting are done using MATLAB.

We first fit each resonance with equation (4.1) and get $\lambda_{res,i}$ and $\alpha_i$ from the spectra of Fig. 4.2 then calculate $\Delta n_{eff,i}$ and $\Delta k_{eff,i}$ using equation (4.6). The average fitting accuracy for the different resonances (N=19) is almost 100. Figure 4.3 shows $\Delta n_{eff}$ and $\Delta k_{eff}$ for the different sensing media as given from the mode simulations (actual or input) and as calculated from the transmission spectra after resonances fitting (our method). Table 4.1 shows the average error (for the different resonances) in the detected $\Delta n_{eff}$ and $\Delta k_{eff}$ for the different substances, showing maximum error of only 0.053 % and 1.23 %, respectively. Next, we determined the waveguide sensitivity $S_{wg}=dn_{eff}/dn_{med}=dk_{eff}/dk_{med}$ for our w×h=320 nm×220 nm SOI waveguide using FDE mode simulations. Figure 4.4 shows the $S_{wg}$ dependence with wavelength of the fundamental quasi-TE mode and its fitting with 4th degree polynomial from 1.46-1.6 μm.

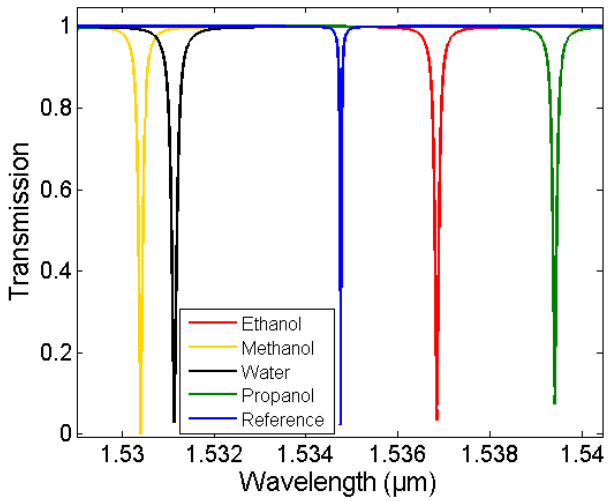



Figure 4.2: Transmission spectra of the micro-ring resonator at reference and different sensing media.

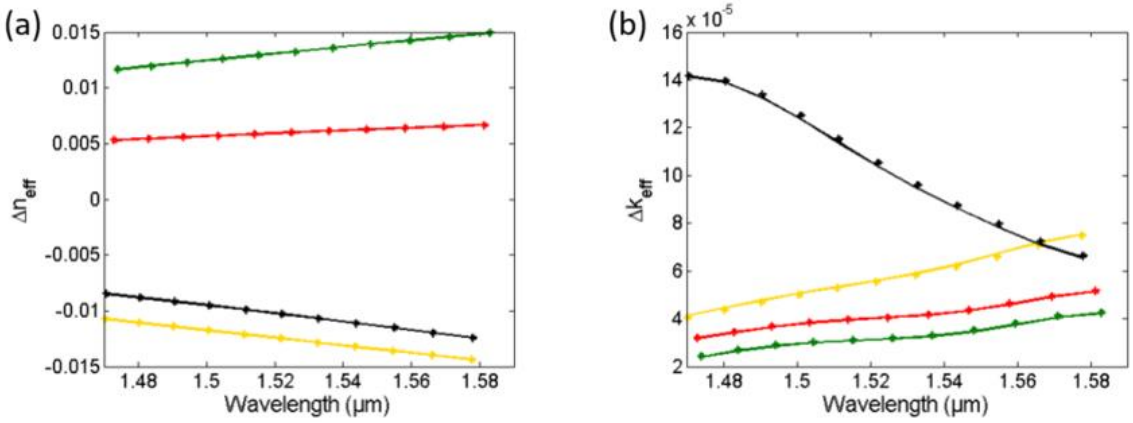



Figure 4.3: Actual (from FDE simulation) and detected (from our method) effective index change versus wavelength for different sensing media (a) Real part $\Delta n_{eff}$ and (b) Imaginary part $\Delta k_{eff}$.

Table 4.1: Average error in the detected effective index changes both real $\Delta n_{eff}$ and imaginary $\Delta k_{eff}$ for different sensing media.

| | Ethanol | Methanol | Propanol | Water |
|---|---|---|---|---|
| $\Delta n_{eff}$ average error (%) | 0.048 | 0.048 | 0.053 | 0.047 |
| $\Delta k_{eff}$ average error (%) | 0.4 | 1.23 | 1.13 | 1.17 |

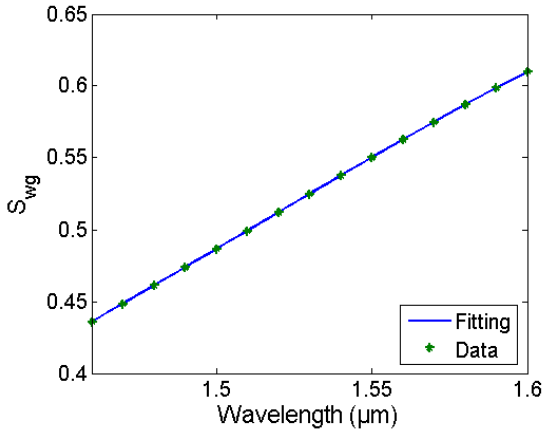



Figure 4.4: $S_{wg}$ of the fundamental quasi-TE mode versus wavelength of the used SOI waveguide with w=320 nm and its 4th degree polynomial fitting.

Finally, we calculate $n_{med,i}$ and $k_{med,i}$ from $\Delta n_{eff,i}$ and $\Delta k_{eff,i}$ using equation (4.7) and equation (4.8), knowing that the reference medium is $n_{med}$=1.336 and $k_{med}$=0. Figure 4.5 shows the input to the sensor n and k data of the different media, from [178], [179], and the detected values from the transmission spectra using our method. Table 4.2 shows the average error in the detected n and k for the different substances, showing maximum error of only 0.0047 % and 1.65 %, respectively.

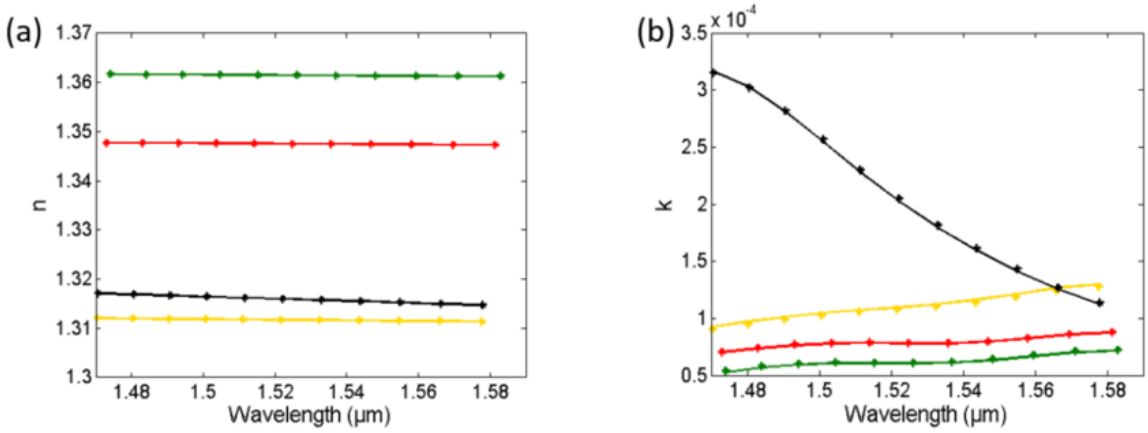



Figure 4.5: Actual (from data in [130-131]) and detected (from our method) medium refractive index versus wavelength for different sensing media (a) Real part n and (b) Imaginary part k.

Table 4.2: Average error in the detected refractive index both real n and imaginary k for different sensing media.

| | Ethanol | Methanol | Propanol | Water |
|---|---|---|---|---|
| n average error (%) | 0.0004 | 0.0047 | 0.0025 | 0.0033 |
| k average error (%) | 0.55 | 1.65 | 1.48 | 0.97 |

### 4.3.2 Multi-element medium

In this section we show how we can determine the concentration composition of a multi-element medium using the same MRR. The complex refractive index of a multi-element medium $n_{med}(\lambda_i)+jk_{med}(\lambda_i)$ at each wavelength $\lambda_i$ can be expressed as the linear combination of the different element refractive index $n_j(\lambda_i)+jk_j(\lambda_i)$ as follows:

$$n_{med}(\lambda_i) = \sum_{j=1-M} c_j * n_j(\lambda_i) \qquad (4.13a)$$

$$k_{med}(\lambda_i) = \sum_{j=1-M} c_j * k_j(\lambda_i) \qquad (4.13b)$$

where i=1-N the wavelength index with N the number of detected n and k points with wavelength (number of resonances), j=1-M the element index with M the number of elements present in the medium and $c_j$ the concentration of each element. Hence, to get the concentration composition $c_j$ of the medium we solve the 2N linear system of equations Ac=B where A is 2N×M matrix, B is 2N×1 matrix and c is M×1 matrix as shown:

$$A=\begin{bmatrix} n_1(\lambda_1) & n_2(\lambda_1) & .. & n_M(\lambda_1) \\ n_1(\lambda_2) & n_2(\lambda_2) & .. & n_M(\lambda_2) \\ .. & .. & .. & .. \\ n_1(\lambda_N) & n_2(\lambda_N) & .. & n_M(\lambda_N) \\ k_1(\lambda_1) & k_2(\lambda_1) & .. & k_M(\lambda_1) \\ k_1(\lambda_2) & k_2(\lambda_2) & .. & k_M(\lambda_2) \\ .. & .. & .. & .. \\ k_1(\lambda_N) & k_2(\lambda_N) & .. & k_M(\lambda_N) \end{bmatrix} \quad (4.14a), \quad B=\begin{bmatrix} n_{med}(\lambda_1) \\ n_{med}(\lambda_2) \\ .. \\ n_{med}(\lambda_N) \\ k_{med}(\lambda_1) \\ k_{med}(\lambda_2) \\ .. \\ k_{med}(\lambda_N) \end{bmatrix} \quad (4.14b) \text{ and } c=\begin{bmatrix} c_1 \\ c_2 \\ .. \\ c_M \end{bmatrix} (4.14c)$$

We tested different media with different concentration composition of the same four elements; ethanol, methanol, propanol and water. In this case N=11 and M=4. Again, FDE solver is used to determine the complex effective index of the multi-element medium using

the medium refractive index calculated from the n and k data of the different elements according to their concentration. Then, we get the corresponding ring-resonator transmission spectrum. To get the matrix c, we first determine the nmed,i and kmed,i from the ring resonator transmission spectra similar to method used in the previous section which gives the matrix B. Then, using the elements database [178], [179] we have the matrix A. Finally, we use the least square method to determine c with the constraints that $\Sigma c_j=1$ and $c_j$ takes values from 0 to 1. Table 4.3 shows the actual (input to the sensor) concentrations and the detected values using our proposed method as well as the percentage error for different media. It can be seen that, an error of less than 6.32% is achieved with an average error of 2.6%. It is important to note that, error increases as the concentration of any of these four elements increases. We can see that Medium 1, Medium 2 and Medium 5 that have concentrations greater than or equal to 0.4 show higher error in the detected concentration. This will be significantly improved using neural networks discussed in the next section.

Table 4.3. Actual (input) and detected (using our method) concentration composition for different multi-element media as well as the percentage error. $c_W/e_W$, $c_M/e_M$, $c_E/e_E$ and $c_P/e_P$ are the water, methanol, ethanol and propanol concentrations/error, respectively.

| | Medium 1 | Medium 2 | Medium 3 | Medium 4 | Medium 5 |
|---|---|---|---|---|---|
| Actual | $c_W$=0.1<br>$c_M$=0.15<br>$c_E$=0.25<br>$c_P$=0.5 | $c_W$=0.1<br>$c_M$=0.3<br>$c_E$=0.4<br>$c_P$=0.2 | $c_W$=0.2<br>$c_M$=0.2<br>$c_E$=0.3<br>$c_P$=0.3 | $c_W$=0.25<br>$c_M$=0.25<br>$c_E$=0.25<br>$c_P$=0.25 | $c_W$=0.5<br>$c_M$=0.1<br>$c_E$=0.25<br>$c_P$=0.15 |

| | | | | | |
|---|---|---|---|---|---|
| Detected | $c_W$=0.1<br>$c_M$=0.15<br>$c_E$=0.24<br>$c_P$=0.51 | $c_W$=0.1<br>$c_M$=0.3<br>$c_E$=0.42<br>$c_P$=0.19 | $c_W$=0.2<br>$c_M$=0.2<br>$c_E$=0.29<br>$c_P$=0.31 | $c_W$=0.25<br>$c_M$=0.25<br>$c_E$=0.26<br>$c_P$=0.25 | $c_W$=0.5<br>$c_M$=0.1<br>$c_E$=0.26<br>$c_P$=0.14 |
| Error (%) | $e_W$=0.56<br>$e_M$=2.45<br>$e_E$=4.59<br>$e_P$=1.67 | $e_W$=0.2<br>$e_M$=1.67<br>$e_E$=4.64<br>$e_P$=6.68 | $e_W$=0.4<br>$e_M$=1.58<br>$e_E$=2.86<br>$e_P$=2.08 | $e_W$=0.04<br>$e_M$=0.72<br>$e_E$=2.67<br>$e_P$=1.91 | $e_W$=0.33<br>$e_M$=5.28<br>$e_E$=5.25<br>$e_P$=6.32 |

## 4.4 Experimental Characterization

Our proposed selective MRR sensor design is verified experimentally. SOI MRRs with oxide cladding was fabricated in a MPW run on a 200 mm SOI wafer with a 220 nm silicon device layer. The device is characterized in the wavelength range between 1500 nm and 1580 nm using GCs provided by the fab PDK [180]. More details about the fabrication and characterization can be found in section 3.6.

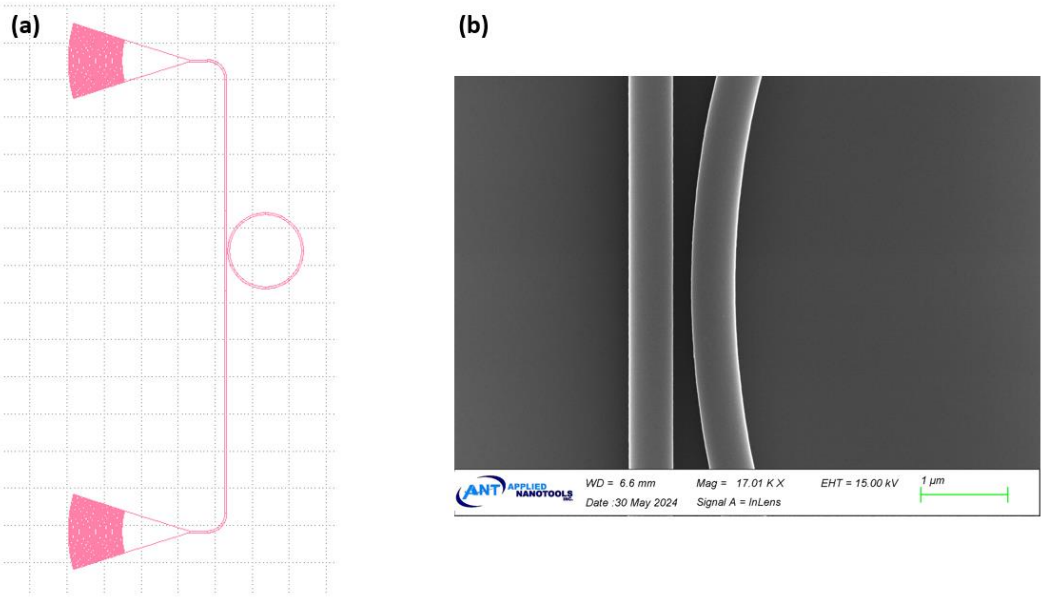



Figure 4.6: (a) Layout of the fabricated SOI MRR and (b) SEM image of the point coupler between bus waveguide and ring cavity.

In this MPW run we don't have access to the waveguide cladding as it must be covered with oxide. Hence, we tried to detect the refractive index of the silicon dioxide cladding using our proposed method. A MRR of a 500 nm width and 10 µm radius was designed, with bus waveguide and ring spacing of the directional coupler equal to 220 nm. Three copies of the same design were fabricated, named Design 1A, Design 1B and Design 1C. Another MRRs with the same waveguide width and radius but with directional coupler spacing of 200 nm and 150 nm were also fabricated, named Design 2 and Design 3, respectively. Due to high GC losses at low wavelengths, we focus on the range between 1510 nm and 1580 nm. With R=10 µm we have total of 8 resonances in this range hence we can get 6 different wavelength points for the detected $n_{SiO2}$, as the first and last

resonances are not used, see section 4.3.1. Figure 4.6 shows the layout and SEM image of the fabricated design. Figure 4.7 shows the measured transmission spectra of the five different MRRs after normalizing the output transmission by the GCs response and removing noise (digitally) to make spectrum clearer. From the spectra of the different MRR designs we can see that Design 2 and Design 3 have lower ER and larger FWHM due to different coupling coefficient indicating that Design 1 with 220 nm DC gap is closer to critically coupled condition.

Due to significantly low imaginary (k) part of $SiO_2$ refractive index ($<10^{-10}$) we here focus only on the real part (n). Figure 4.8 shows the detected silicon dioxide refractive index from the different fabricated MRRs together with index data from references [25], [181]. We can see that, different designs yield to slight differences between the detected silicon dioxide index which can be attributed to fabrication variations of the waveguide width. Design 1A shows lower refractive index and also higher slope (dispersion) with wavelength indicating smaller waveguide width in this case. While Design 1C and Design 2 show higher index and lower slope indicating larger waveguide width. However, in any case the detection error is still small less than 0.12 %.

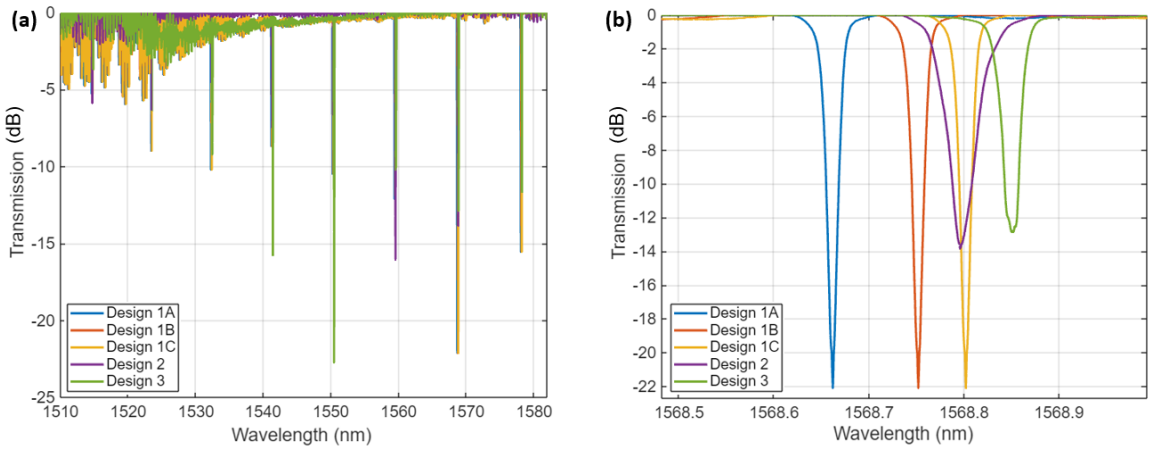



Figure 4.7: (a) Measured transmission spectra of the different fabricated MRR designs and (b) zoomed-in view around 1588nm wavelength.

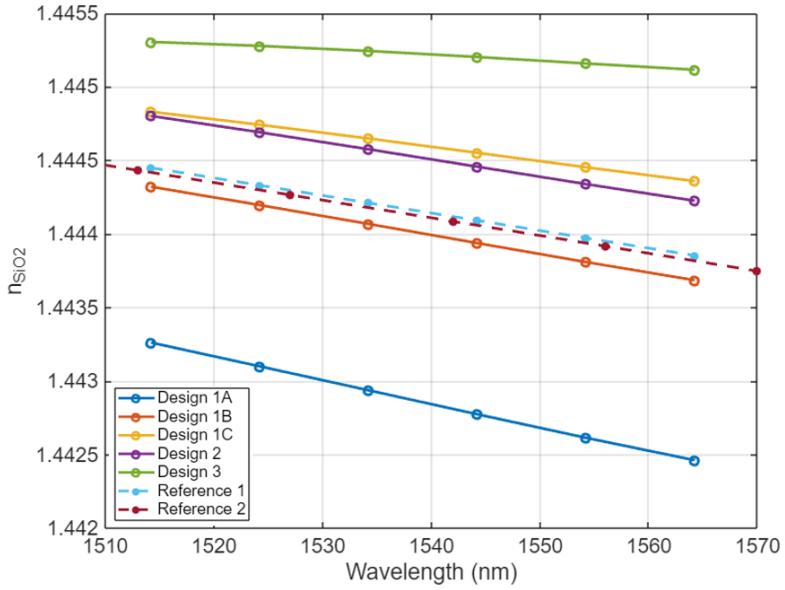



Figure 4.8: Refractive index of silicon dioxide from references [25], [181], and as detected from the five different MRR designs using our method.

## 4.5 Improving sensor's performance using deep learning

This section describes an enhancement to the proposed method using neural network techniques to eliminate electrical noise, improving accuracy and extending the detection range. The method employs two neural networks: a convolutional neural network (CNN) to remove noise and a deep neural network (DNN) to predict chemical concentrations. The process involves measuring the MRR output 10 times, removing noise with the CNN, then measuring resonance wavelengths and transmissions, obtaining medium n and k values, and using the DNN predict the concentration composition. This approach leverages the strengths of both neural network architectures for improved performance.

### 4.5.1 Data generation

To train these models, a substantial amount of simulation data is required. This involves simulating the transmission spectra of the MRR at various concentrations of the sensing medium. We utilized 6,221 unique complex refractive indices over the target wavelength range, corresponding to unique concentration combinations of the four chemicals. Using an FDE solver, we obtained the MRR transmission spectra for the different concentrations, thus creating a dataset suitable for training the machine learning models.

### 4.5.2 CNN for noise removal

Micro-ring resonators (MRRs) with high quality (Q) factors are inherently sensitive to noise, which can significantly impair their performance. Noise sources in these systems can be broadly categorized into additive and multiplicative noise. Among these, thermal noise, arising from fluctuations in thermal equilibrium, is the dominant source in MRRs

with high Q factors. This thermal noise exhibits a Brownian pattern consistent with random walk phenomena, which follows a Gaussian distribution. To account for this, additive Gaussian noise was introduced to model thermal noise and approximate the combined effects of other noise sources. For multiplicative noise, Poisson noise was utilized to mimic shot noise, which is known to follow a Poisson distribution.

For this study, the signal-to-noise ratio (SNR) was set at 11.8 dB—an exaggerated value chosen as a proof of concept to demonstrate the ability of the CNN to remove noise effectively. The Gaussian noise intensity was three times stronger than the Poisson noise, with a standard deviation set at 5% of the transmission for Gaussian noise and 1.6% for Poisson noise.

To remove noise from the transmission spectrum, a small autoencoder-based convolutional neural network (CNN) was designed. CNNs were chosen due to their capacity to identify and process both local and global patterns, making them particularly effective for noise removal tasks. The dataset, consisting of transmission data, was divided into manageable groups of 130 points to avoid the impracticality of processing 18,000 inputs corresponding to each point in the spectrum.

The architecture of the CNN was refined through trial and error, followed by a grid search to identify optimal hyperparameters. The encoder comprised five layers: the first two layers were one-dimensional convolutional layers (Conv1D) with 130 and 16 neurons, respectively. These were followed by two transpose Conv1D layers with 16 and 130 neurons. The final layer was an output Conv1D layer with a single neuron. Except for the

output layer, which used a sigmoid activation function, all layers employed the rectified linear unit (ReLU) activation function. The kernel size for all convolutional layers was set to nine. While smaller kernel sizes also yielded satisfactory results, larger sizes significantly reduced errors.

During training, a Gaussian noise layer with a standard deviation of 0.05 was added using the TensorFlow library to generate a new noisy spectrum at each epoch. A custom Poisson noise layer with an intensity of 0.016 was also implemented for this purpose. The model was trained using the Adam optimizer and a binary cross-entropy loss function over five epochs. Other loss functions such as root mean square error (RMSE) was used, but binary cross-entropy gives the best results. A train-test split was not used because the noise introduced during training was unique for every iteration.

The results demonstrated that the CNN effectively removed noise, achieving a binary cross-entropy loss of 0.0103. The comparison of denoised signals to the original and noisy signals confirmed the model's efficacy. For cases where extreme noise could not be entirely removed, averaging multiple denoised outputs—obtained by running the noise removal process multiple times—helped mitigate the effects of noise outliers. Accuracy improved with successive iterations, but returns diminished after 10 iterations, see Figure 4.9.

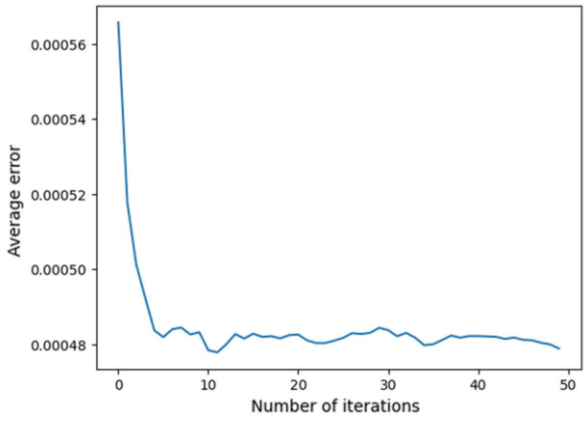



Figure 4.9: Average error as a function of number of iterations.

### 4.5.3 DNN for concentration measurement

Here a DNN is used to determine the chemical composition of the sensing medium. First, we obtain the clear MRR outputs spectrum after noise removal using the CNN. Then, the complex refractive index of the medium, $n_{med}(\lambda_i)$ and $k_{med}(\lambda_i)$, is determined from the spectrum using the same method of section 4.2.1. These $n_{med}(\lambda_i)$ and $k_{med}(\lambda_i)$ are the data used to train the DNN.

The dataset demonstrated significant potential during training. The hyperparameters of the DNN were optimized over several days using Weights and Biases sweeps, with the final parameters detailed in Table 4.4. A learning rate decay was applied after 100 epochs, and early stopping based on validation loss was implemented to prevent overfitting, typically occurring around the 1200th epoch. The output layer used a SoftMax activation function to ensure the concentration sum equaled one. For accuracy, 10% of the data was reserved for validation, and training included a Gaussian noise layer of 0.005% to align with refractive index error. Figure 4.10 illustrates the loss progression across training epochs.

This refined dataset significantly improved the neural network's performance in measuring concentration, achieving accuracy comparable to linear least squares methods, even with added noise. The average RMSE in the detected concentration composition of a multi-element sample from a noisy spectrum (measurements) using our proposed neural network-based method are summarized in Table 4.5. The range of the chemical concentrations used for each substance in these results is between 0.0357% and 75%.

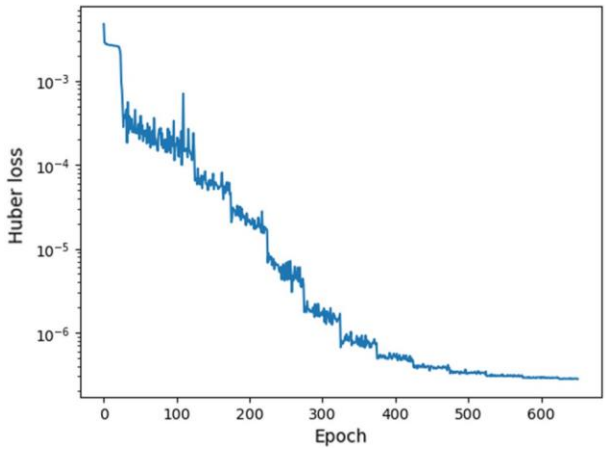



Figure 4.10: Loss versus training epoch.

Table 4.4: The hyperparameters of the DNN.

| Hyperparameters | Value |
|---|---|
| Learning rate | 4.7E−4 |
| Learning decay rate | 1/2 at plateau |
| Batch size | 10 |
| Depth | 9 Layers |

| | |
|---|---|
| Width | 28, 128, 256, 512, 256, 126, 64, 32, 4 |
| Loss function | Huber |
| Huber loss (δ) | 5 |
| Optimizer | Adam |

Table 4.5: The average RMSE of the different substances.

| | Methanol | Ethanol | Water | Propanol | Overall |
|---|---|---|---|---|---|
| RMSE (%) | 0.114 | 0.054 | 0.082 | 0.059 | 0.077 |

## 4.6 Conclusion

We propose a novel technique for selective on-chip optical detection. It is based on determining the complex refractive index n+jk of the medium at multiple wavelengths using an SOI MRR cavity. Using this technique the substance present in the sample can be specified by the refractive index dispersion of the medium. This technique doesn't require any surface functionalization process and can be used even with substances that have low absorption (k) coefficient. With this technique the concentration composition of a multi-element sample using single measurement. Our proposed technique was numerically demonstrated using methanol, ethanol, propanol and water as medium substances. Moreover, this technique was verified experimentally by detecting the refractive index of the silicon dioxide cladding of a MRRs fabricated using SOI MPW run. We then empowered our method capabilities through artificial intelligence (AI) algorithms. We

used neural network to achieve higher detection accuracy, wider dynamic range and more importantly to remove electrical noise. The method employs two machine learning models: a convolutional neural network (CNN) to eliminate noise and a deep neural network (DNN) to identify chemical concentrations. These models were trained using mixtures of the same four chemicals; water, ethanol, propanol, and methanol, but the same technique can be adapted for other chemical sets.

# 5 Chapter 5: On-Chip Sensing System Employing Wavelength Splitting for Noise Suppression

## 5.1 Introduction

For a reliable and high-performance RI sensor we need to have high sensitivity S, large detection range, low limit of detection (LoD) and high yield. LoD is the smallest index (concentration) change that can be detected. As mentioned in previous chapters, LoD doesn't depend solely on the optical sensor design. However, there are several other parameters that affect the LoD. For instance, the measurement system specifications, such as the source power level and detector's resolution and responsivity, as well as the optical and electrical noise. In order to, isolate these effects from the optical design of the sensor, intrinsic LoD (iLoD) is defined [3]. This iLoD only depends on the optical performance of the sensor; hence, it can be used to compare different optical sensor designs. iLoD doesn't account for any source of noise.

In sensing optical noise are phase changes in the mode field due to effects other than the medium index change. For instance, phase changes due to fabrication process variations and temperature fluctuations. This phase changes will result in a wavelength (or intensity) shift of the measured optical signal, leading to an error in the detected refractive index and limiting the sensor's accuracy and the actual LoD.

Silicon photonics, which is a widespread CMOS compatible technology, can offer high-sensitivity sensors with a small footprint due to its high index contrast. However, it is also

highly sensitive to phase errors (noise) [182]. Due to their submicron dimensions and high refractive index, silicon waveguides are sensitive to process variations. Also, due to the high thermo-optic coefficient of the silicon, they are sensitive to temperature variations. While there are plenty of designs that maximize the sensitivity of RI sensors, optical phase noise is a substantial obstacle to on-chip RI sensing, and very few studies have addressed this issue and tried to minimize its effect. In [183], the authors used a wide multimode waveguide width to reduce the phase sensitivity to waveguide width variations. However, this will limit the sensors' sensitivity to the medium index change as well. On the other hand, local heating with a feedback system was used to suppress phase errors due to temperature fluctuations, which requires additional space and power. Finally, MZI with different types of waveguide arms was also used to compensate for temperature changes [184].

In this chapter, we propose a system of four loop-terminated Mach-Zehnder interferometers (LT-MZIs) [143], [185] for on-chip RI sensing using wavelength splitting. Our system is capable to suppress optical phase noise completely and determine only the phase change due to the RI change of the medium under test. Hence, our proposed design is immune to temperature and fabrication variations, which can greatly degrade the performance and yield of any RI based sensor. This design maximizes the detection accuracy and minimizes the LoD significantly. The proposed system also doubles the sensitivity and increases the limited detection range of MZI based sensor's significantly. Our proposed system can be used to design various on-chip sensors with different waveguide platforms and operating wavelengths. A liquid sensor based on a CMOS-

compatible SOI platform, utilizing our four LT-MZI system and operating in the near-infrared region, is designed, and its noise suppression capability is confirmed. Simulation results for the sensor are obtained, and its performance parameters are compared with single LT-MZI and single MZI showing significant enhancement.

This chapter is organized as follows; section 5.2 discusses the limitations of conventional MZI sensor then, we present our four LT-MZI sensing system and analysis demonstrating how conventional MZI limitations are overcome. In section 5.3 a liquid sensor utilizing our proposed four LT-MZI structure is designed, optimized and a rigorous 3D-FDTD simulations results are presented. In addition, the proposed sensor was compared with conventional single MZI and single LT-MZI. In section 5.4 we verify our proposed method by experimental characterization of a fabricated four LT-MZI sensor system. Finally, section 5.5 provides a summary of the chapter.

## 5.2 Four LT-MZI Sensor

### 5.2.1 MZI RI Sensor Limitations

The wavelength sensitivity S, intrinsic limit of detection iLoD, and detection range DR of MZI are given by [3], [143]:

$$S_{MZI} = \frac{\lambda_{res}}{(n_{g,sens}L_{sens} - n_{g,ref}L_{ref})} L_{sens}S_{wg} \qquad (5.1)$$

$$iLoD_{MZI} = \frac{1}{FOM} = \frac{FWHM}{S_{MZI}} = \frac{\lambda}{\pi S_{wg}L_{sens}} \qquad (5.2)$$

$$DR_{MZI} = \frac{\Delta n_{max}}{\Delta n_{min} = \frac{\pi}{\sqrt{2}}} \quad (5.3)$$

where $n_{g,sens}$, $L_{sens}$ and $n_{g,ref}$, $L_{ref}$ are the mode group index and the length of the sensing and reference arm, respectively. $S_{wg}=dn_{eff,sens}/dn_{med}$ the sensing arm waveguide sensitivity. FOM is the figure of merit, and FWHM is the full width at half maximum of the interferometer spectrum. $\Delta n_{max}$ and $\Delta n_{min}$ are the maximum and minimum detectable index changes in the medium refractive index. As we can see, the maximum detectable index change in MZI is only 2.2 times its minimum detectable value, which is a very limited range. On the other hand, errors in the detected medium index change $\Delta n_{med}$ arise from changes in the mode effective index $\Delta n_{eff}$ due to effects other than medium index change, for instance, temperature fluctuations. Thus, when measuring only one parameter, the wavelength shift $\Delta\lambda_{res}$, we will not be able to differentiate the changes due to the medium and due to the other effects. Hence, to increase the detection accuracy, we need to have more parameters, independent of $\Delta\lambda_{res}$, that can be measured.

Here, we propose to use two different sensors having the same sensitivity magnitude but with opposite signs, i.e. $S_1 = -S_2$. In this case, initially, with no index change, both sensors will resonate at the same wavelength. When the sensing medium's refractive index changes, the sensor's spectrum will shift in opposite directions, resulting in a splitting of the resonance wavelength. Now, we have two independent variables, namely, the wavelength splitting and the average (or center) wavelength, that allow this system to be capable of suppressing any optical phase noise, as will be discussed in the next section. Wavelength splitting will also double the sensor's sensitivity.

### 5.2.2 General Analysis and Design

In this section, we are going to design a sensor that exhibits wavelength splitting using MZI. However, it is not possible to obtain this resonance splitting by simply adding or multiplying two conventional MZIs with opposite sensitivities. To achieve resonance splitting, we use loop-terminated MZI (LT-MZI) [143], [185]. LT-MZI, see Figure 5.1(a), is similar to MZI, but the second directional coupler (DC) outputs are connected through a loop that reflects back the EM wave, and the output is taken from the first DC. The transmission equation of LT-MZI is given by equation (5.2). The analysis of the LT-MZI and more details can be found in chapter 3. As we can see from Figure 5.1(b), by properly choosing the DC forward coupling coefficient α of the LT-MZI, we can suppress either the even resonances (peaks) or the odd resonances of the spectrum [143]. In this case, by simply adding two LT-MZIs, we can obtain resonance splitting. Also note that, in this case ($\alpha_{12}$=0.15), the FWHM is 1.46 times lower than the MZI of the same lengths, which further enhances the sensor's performance.

$$T_{LT-MZI} = |t_{LT-MZI}|^2 = A\left[\frac{H}{4} + cos(\Delta\phi)\right]^2 \quad (5.4a)$$

where $A = 16a_1^2a_2^2b_1^2b_2^2$ (5.4a)

and $H = -\frac{a_1a_2}{b_1b_2} - \frac{b_1b_2}{a_1a_2} + \frac{a_1b_2}{a_2b_1} + \frac{a_2b_1}{a_1b_2}$ (5.4c)

According to (5.1) for two LT-MZIs with the same sensing, reference waveguide, and sensing arm length to have the same but opposite sensitivity, the reference arm should satisfy:

$$L'_{ref} = 2\frac{n_{g,sens}}{n_{g,ref}}L_{sens} - L_{ref} \qquad (5.5)$$

where $L'_{ref}$ the second LT-MZI reference arm length.

On the other hand, in order to increase the limited DR of the MZI sensor, a cascaded MZI is proposed. Cascading a second MZI (MZI 2) of shorter length and equal sensitivity to the main MZI (MZI 1) will increase the sensor FSR and hence increase $\Delta n_{max}$. This will also decrease the FWHM, resulting in a lower $\Delta n_{min}$. Hence, increasing the DR. So, for MZI 2 with m times shorter arms' length than MZI 1, the sensor achieves more than m times enhancement in the DR. Note that, practically m≤4, as larger m will allow other peaks of MZI 1 to have high transmission within FSR2.

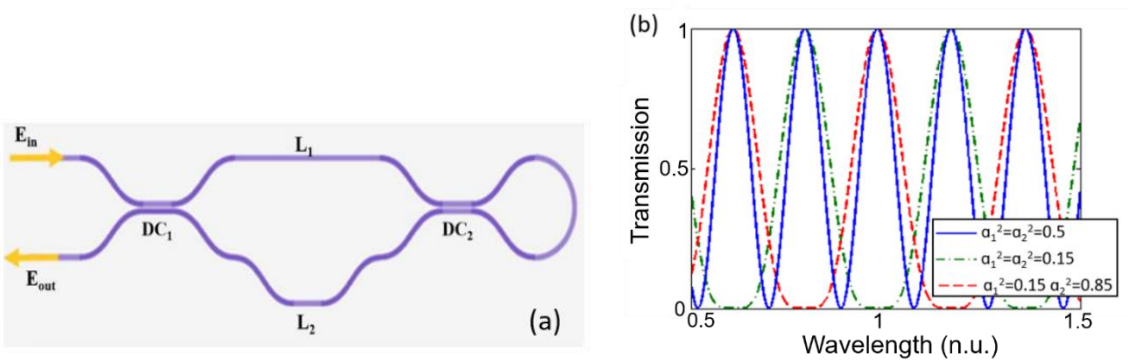



Figure 5.1: (a) Schematic of LT-MZI structure; (b) Typical LT-MZI transmission spectrum with $\alpha_{12}=\alpha_{22}=0.5$, $\alpha_{12}=\alpha_{22}=0.15$, $\alpha_{12}=0.15$ and $\alpha_{22}=0.85$.

Hence, our final design is shown in Figure 5.2, consisting of four LT-MZIs. The input wave is divided evenly using y-junction to two LT-MZIs having opposite wavelength sensitivities, and each one of these LT-MZIs is cascaded by another LT-MZI of quarter the

arms' length to expand the DR. All have the same reference and sensing waveguide structure. The lengths of the different LT-MZI arms are related as:

$L_{sens,1}=L_{sens,2}$ (5.6a)

$L_{sens,3}=L_{sens,4}=L_{sens,1}/4$ (5.6b)

$L_{ref,3}=L_{ref,1}/4$ (5.6c)

$L_{ref,4}=L_{ref,2}/4$ (5.6d)

Compared to single MZI, this design exhibits double the sensitivity, 1.46 times lower FWHM, resulting in almost 3 times enhancement in the iLoD and more than six times higher DR. Also note that, with this device, two independent parameters can be measured, one is the splitting wavelength ($\Delta\lambda_{split}$) and the other is the average wavelength ($\lambda_{aver}$). This makes our sensor capable of suppressing optical phase noise completely, as discussed below.

From the LT-MZI resonance condition [143] we can derive:

$$\Delta\lambda_{res} = \frac{\lambda_{res}}{(n_{eff,sens}L_{sens} - n_{eff,ref}L_{ref})}(aL_{sens} + bL_{sens} - cL_{ref}) \qquad (5.7)$$

where $\alpha=\Delta n_{eff,sens|med}$ the change in the sensing waveguide effective index due to the medium index change. $\Delta n_{eff,sens|med}$ is the parameter we need to determine to detect $\Delta n_{med}$. $b=\Delta n_{eff,sens|other}$ the change in the sensing waveguide effective index change due to any effect other than the medium index change (phase noise), $c=\Delta n_{eff,ref|other}$ the change in the

reference waveguide effective index due to any effect. Both b and c represent error terms (noise).

For the two sensors with the same sensitivities and opposite signs, having the same $n_{eff,sens}$, $n_{eff,ref}$, and $L_{sens}$ but different $L_{ref}$, the average wavelength $\lambda_{aver}$ and wavelength splitting $\Delta\lambda_{split}$ are given by:

$$\lambda_{aver} = \lambda_0 + \frac{c}{2}K(L_{ref,2} - L_{ref,1}) \quad (5.8)$$

$$\Delta\lambda_{split} = 2aL_{sens}K + 2bL_{sens}K - cK(L_{ref,2} + L_{ref,1}) \quad (5.9a)$$

$$\text{with } K = \frac{\lambda_{res}}{(n_{g,sens}L_{sens} - n_{g,ref}L_{ref})} \quad (5.9b)$$

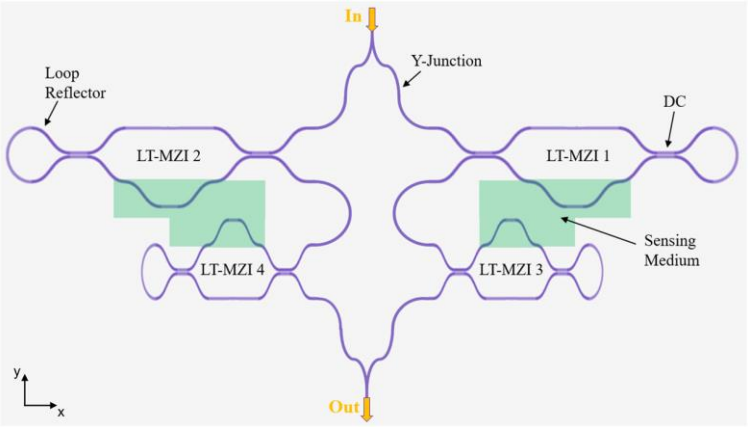



Figure 5.2: Schematic of the proposed four LT-MZI sensor.

When both interferometer arms (sensing and reference) have the same waveguide structure, i.e. $n_{eff,sens} = n_{eff,ref}$, we have b=c. In this case, using (7.8) and (7.9) we can determine both $\alpha = \Delta n_{eff|med}$ and $b = \Delta n_{eff|other}$ separately. This means that we can differentiate between the

change in the waveguide effective index due to medium index change (actual signal) and the change in the waveguide effective index due to any other effect (noise), such as temperature and fabrication variations. Thus, it allows us to completely compensate for any optical phase noise and exactly determine the medium refractive index ($n_{med}$). This technique is applied and discussed in the next section. This is not possible using a single MZI, as we only have a single equation in this case, the resonance shift (7.7), that includes both the signal and noise effect together, and we cannot separate them.

## 5.3 Liquid Sensor Design and Results

Now we utilize the widespread SOI platform, with a silicon thickness of 220 nm, to build a liquid sensor based on our proposed four LT-MZI sensor design (Figure 5.2) operating in the near-infrared range at $\lambda$ =1550 nm. As mentioned previously, the SOI platform can achieve high-sensitivity sensors, but is highly susceptible to optical phase noise. Using our four LT-MZI design, this problem is mitigated, and we benefit from silicon high sensitivity to medium index change. With FDE solver, we optimize the sensing waveguide width to maximize its sensitivity to the surrounding medium $S_{wg}$. While the optimum width for the fundamental quasi-transverse electric (TE) mode is 250 nm, resulting in $S_{wg}$ of 0.74, the mode is weakly confined, hence susceptible to high propagation losses. So, we choose $w_{sens}$ = 300 nm which results in $S_{wg}$ of 0.65, which is still high while maintaining appropriate confinement. This waveguide is used for both sensing and reference arms in the LT-MZI sensor with water cladding. For all other waveguides (interconnects) in our system, we use

a strip waveguide with the same thickness of h = 220nm and width $w_{ref}$ = 500 nm capped with $SiO_2$ cladding. FDE solver [29] is used to get $n_{eff}(\lambda)$ of both the reference waveguide and the sensing waveguide at different surrounding medium index $n_{med}$. A y-junction constructed from two S-bends with a length Ly-junction = 20 µm and a lateral shift Δx = 5 µm is designed to split/combine the waves from the different LT-MZIs, as shown in Figure 5.2. DCs with $\alpha_2$=0.15 and $\alpha_2$=0.85 at λ=1550 nm are designed with width w = 400 nm, gap g = 150 nm, and lengths of LDC = 7.15 µm and LDC = 2.42 µm, respectively. For the loop reflectors, we use a bend with a radius R = 5 µm. FDTD simulations were performed to optimize the different sensor components [161]. Then MATLAB modelling is used to determine the behavior of the whole sensor.

Table 5.1: Comparison of the sensor performance parameters between our proposed four LT-MZI, single LT-MZI, and single MZI with the same waveguide structure and lengths.

| | 4 LT-MZI | LT-MZI | MZI |
|---|---|---|---|
| S (nm/RIU) | 7890 | 4138 | 4138 |
| FWHM (nm) | 6.4 | 7 | 10.2 |
| FOM ($RIU^{-1}$) | 1233 | 591 | 406 |
| iLoD (RIU) | 8e-4 | 1.7e-3 | 2.5e-3 |
| DR | 10 | 2.4 | 1.6 |

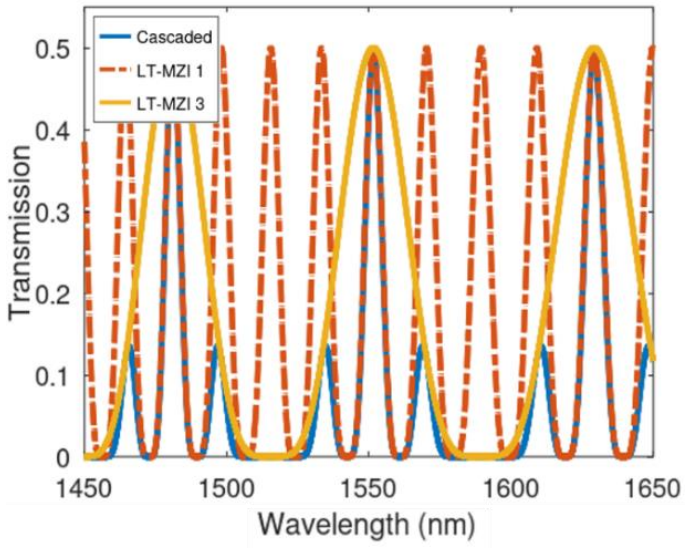



Figure 5.3: Transmission spectrum of LT-MZI 1 and LT-MZI 3 and their cascaded transmission with Lsens,1=500 µm, Lref,1=427.7 µm.

Figure 5.3 shows the transmission response of LT-MZI 1 and LT-MZI 3 using the Si strip waveguides and their cascaded transmission with $L_{sens,1}$=500 µm, $L_{ref,1}$= 427.7 µm, DCs coefficients of $\alpha_{12}$=0.15 and $\alpha_{22}$=1-$\alpha_{12}$. The lengths of the rest LT-MZIs are satisfying (7.5) and (7.6) and DCs coefficients are the same. Figure 5.4(a) shows the whole sensor transmission spectrum at different medium index changes $\Delta n_{med}$. From this, we get the wavelength splitting $\Delta\lambda_{split}$ at different medium index change $\Delta n_{med}$, Figure 5.4(b), the slope of which determines the sensitivity of our sensor at 7890 nm/RIU. Table 5.1 gives our four LT-MZI sensor performance parameters and compares it with the single LT-MZI design and the single MZI. We can see that, this sensor can achieve FOM as high as 1233 $RIU^{-1}$, corresponding to an iLoD of 8e-4, with only 500 µm of sensing arm length. Results show that this iLoD is 2 and 3 times lower than the single LT-MZI and single MZI, respectively. In addition, the sensitivity is doubled. Also, the DR is 4.2 and 6.3 times higher

than the single LT-MZI and MZI, respectively. As $\Delta n_{min}$ (iLoD) decreases, and the maximum detectable index change $\Delta n_{max}$ is doubled from 4e-3 to 8e-3 using the cascaded LT-MZIs. The footprint of the four LT-MZI is 2.5 times that of the single LT-MZI and MZI.

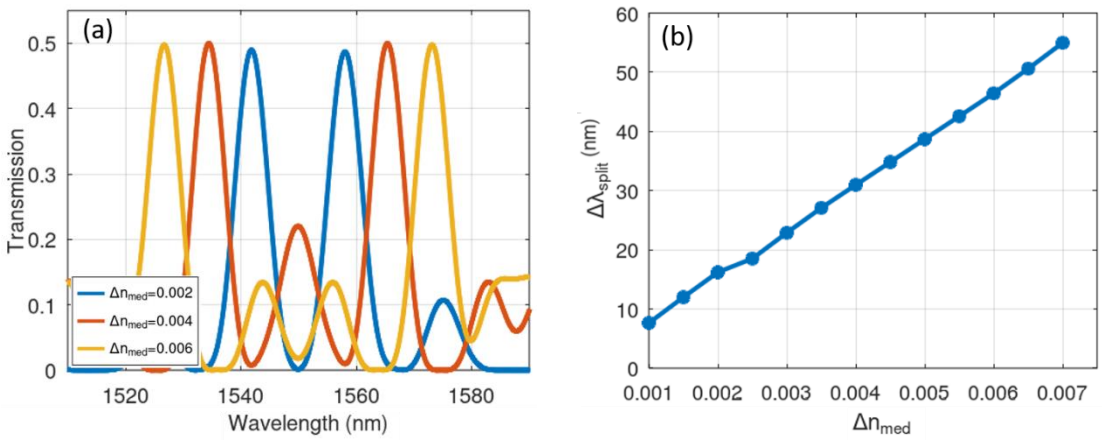



Figure 5.4: (a) Transmission spectrum of the whole sensor at different medium index change $\Delta n_{med}$ with $\Delta n_{eff,other}$=0. (b) Wavelength splitting $\Delta\lambda$split versus medium index change $\Delta n_{med}$.

More importantly, with this system, we measure two independent variables ($\Delta\lambda_{split}$ and $\lambda_{aver}$), which allow us to completely remove any optical phase noise and accurately determine the medium index, as discussed in the previous section. In Figure 5.4(a), the output spectrum is at zero phase noise $\Delta_{neff,other}$ = 0 and at different medium index change $\Delta n_{med}$. In this case, we see that the center wavelength $\lambda_{aver}$ doesn't change as the phase noise is fixed, and changing the medium index changes the wavelength splitting $\Delta\lambda_{split}$ according to (9). On the other hand, Figure 5.5(a) shows the spectrum of the LT-MZI system with $\Delta n_{med}$ = 0.002 (constant medium) and at different phase noise $\Delta n_{eff,other}$. As

expected from (7.8) different phase noise results in shifting the center wavelength $\lambda_{aver}$ while all have the same $\Delta\lambda_{split}$. Figure 5.5(b) shows the spectrum at different medium indices and different phase noises. Here we see that both the center wavelength and wavelength splitting change. These results show that we can clearly differentiate between changes in the medium (analyte) index and any other optical noise. This significantly enhances the detection accuracy and the actual LoD.

As mentioned, iLoD in Table 5.1 is only used to compare different optical sensors and doesn't account for many other factors, including optical phase noise. So, for the single LT-MZI and MZI, this iLoD assumes zero optical phase noises which is not the case. For instance, a silicon waveguide with a 220 nm height has around 4e-3 RIU/nm width sensitivity, i.e. $dn_{eff}/dw$ [182]. Hence, for just 0.5 nm width change $n_{eff}$ is increased by 2e-3. The same effective index change comes from a 3e-3 medium index change. Hence, in this case, the accuracy and LoD of single LT-MZI and MZI will be limited to 3e-3. Similarly, if we assume a 5-degree change in the temperature, the accuracy and LoD of single LT-MZI and single MZI will be limited to 1e-3. This LoD of single LT-MZI and MZI cannot be enhanced by simply increasing the length of the sensing arm.

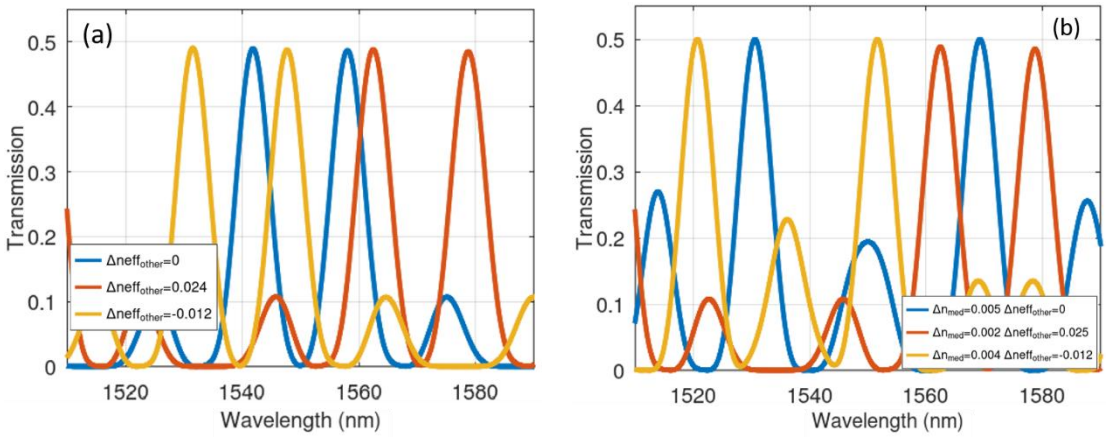



Figure 5.5: (a) Transmission spectrum of the whole sensor at different phase noise $\Delta n_{eff,other}$ with $\Delta n_{med}$=0.002. (b) Transmission spectrum of the whole sensor at different combinations of phase noise and medium index.

Table 5.2 compares our sensors' sensitivity and iLoD with other published works. Note that, the FOM of the sensor can be further increased (detecting lower $\Delta n_{med}$) by simply increasing the sensing arm length (7.2) up to several millimeters in a spiral configuration [186], [187]. There is a plethora of work done for RI sensing, and many review articles can be found in the literature [3], [52], [55], [56]. In this table, we have selected some of the works that have enough information for comparison. For instance, the actual LoD depends on the measurement setup; hence, this is not included here. Table 5.2 shows that our design achieves much better performance than all the other works, i.e. higher sensitivity and lower iLoD. Our phase sensitivity is only lower than the one in [59]. However, this sensitivity is calculated for a 2 mm MZI arm's length, while our sensitivity is only for 0.5 mm (4 times lower). For the same MZI arm's length, our phase sensitivity is 840(2π)/RIU which is almost 2 times higher than the one achieved in [59]. Also, the iLoD in [77] is lower than

our design, but again, this is for 1 cm length. For 1 cm, our sensor will achieve the same iLoD of 4e-5 RIU. Also, compared to the other works, we can see that our sensors' sensitivity is significantly higher, more than 10 times, than most of the studied works in [3], [52], [55], [56], [57], which contain various designs such as micro-ring resonator (MRR) [55], [56], plasmonic [52] and many other sensors [3], [57].

Table 5.2: Sensitivity and iLoD comparison of our proposed sensor design with other works.

| Ref. | Device | Sensitivity | iLoD |
|---|---|---|---|
| [77] | MZI | 740nm/RIU | 4e-5 $RIU^{-1}$ |
| [59] | MZI | 460(2π)/RIU | - |
| [149] | MRR | 70 nm/RIU | 1.1e-3 $RIU^{-1}$ |
| [74] | MRR | 298 nm/RIU | 1.5e-2 $RIU^{-1}$ |
| [188] | MRR | 345nm/RIU | 2.4e-3 $RIU^{-1}$ |
| [57] | MRR | 476 nm/RIU | 1.7e-3 $RIU^{-1}$ |
| This work | MZI | 7890/RIU<br>210(2π)/RIU | 8e-4 $RIU^{-1}$ |

Finally, we studied the effect of fabrication process variations on the directional couplers used in our design, as these are the only variations that cannot be compensated by our proposed system. We changed the width and thickness of the couplers waveguide by Δw and Δh, respectively. Then we get the changed splitting ratio and determine its effect on the spectrum and the performance of the sensor. We did simulation for all possible width

and thickness variations and got the corner cases, minimum and maximum splitting ratios possible. We tried 5 nm and 10 nm process variations for both Δw and Δh. Figure 5.6 below shows corner cases of the spectrum at $\Delta n_{med}$ =0.004. Also, Table 5.3 shows the change in DCs coefficients and the percentage change in the sensors' sensitivity, FWHM, and FOM at the different corners. We can see that, process variations didn't distort the output wavelength response. Variations will slightly increase the insertion loss. Results also show that, fabrication error of 10 nm can reduce the FOM up to only 9.4% from its original value. While for 5 nm, error changes are less than 4%. Note that, these are the corner results, hence the worst case of process variations. So, we conclude that our sensor can still operate effectively in the presence of fabrication errors, but with a small penalty in the optimum performance.

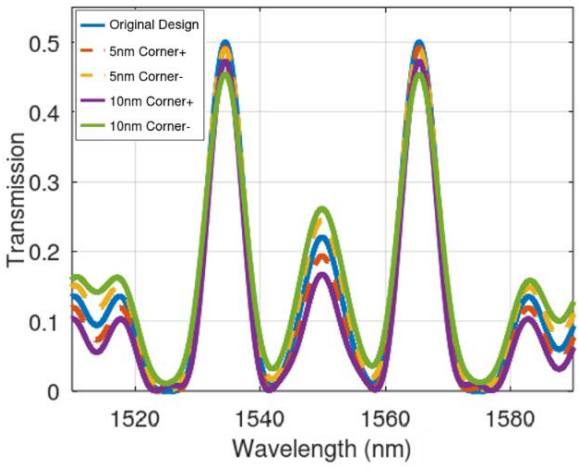



Figure 5.6: Transmission spectrum of the sensor at DC corners as well as the original design spectrum with $\Delta n_{med}$=0.004.

Table 5.3: Effect of fabrication process variations on DCs coefficients, the percentage change in the sensors' sensitivity, FWHM and FOM.

| | 5 nm Corner+ | 5 nm Corner- | 10 nm Corner+ | 10 nm Corner- |
|---|---|---|---|---|
| $\alpha_1^2/\alpha_2^2$ | 0.18/0.86 | 0.12/0.84 | 0.216/0.87 | 0.08/0.82 |
| ΔS (%) | -0.03 | 0 | 0 | 0.13 |
| ΔFWHM (%) | -3 | 4 | -5.2 | 10.5 |
| ΔFOM (%) | 3 | -3.9 | 5.4 | -9.4 |

## 5.4 Experimental Characterization

Here, our proposed four LT-MZI system design is verified experimentally. The design was fabricated on a SOI MPW run with a 220 nm silicon device layer. The device is characterized in the wavelength range between 1520 nm and 1580 nm using GCs provided by the fab PDK. More details about the fabrication and characterization can be found in section 3.5.

As we don't have access to the waveguide cladding in this MPW run and the cladding must be covered with oxide, actual sensing performance cannot be assessed. However, using this characterization we verify the validity of our idea. To mimic the phase change occurring due to medium index change, we changed one of the LT-MZI arm's length, representing the sensing arm. We used SOI waveguide of 450 nm width in all the LT-MZIs. We fabricated a cascaded LT-MZI design where no splitting occurs as a reference (named Reference Design). We also fabricated two different designs of our four LT-MZIs designs

with different increment ΔL in the sensing arm lengths of 5 μm (Design 1) and 10 μm (Design 2), respectively, which represent different sensing medium index change. The lengths of the different fabricated LT-MZIs are shown in Table 5.4. Note, that similar to equation (5.5) the reference arm lengths are designed to satisfy opposite sensitivities condition. Figure 5.7 shows the layout of the reference LT-MZI design together with its SEM image.

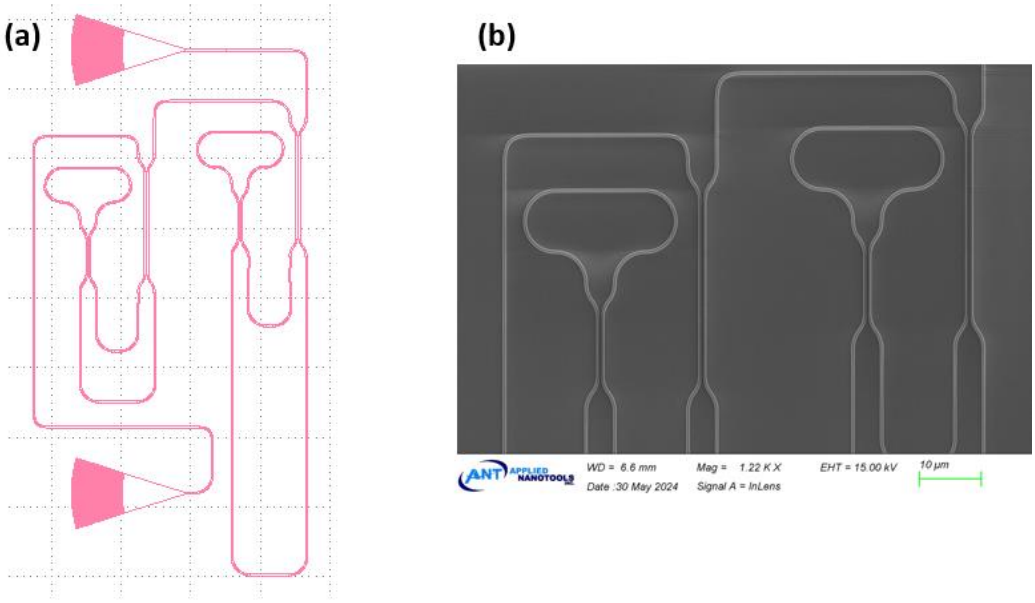



Figure 5.7: (a) Layout and (b) SEM images of the fabricated reference LT-MZI design.

Figure 5.8 shows the measured transmission spectra of the different LT-MZI designs after normalization by the GCs response (Reference Design transmission was scaled by 0.7 for clarity). We can clearly see the wavelength splitting in accordance to change in the sensing arm length ΔL (change on waveguide phase). A maximum transmission of nearly 0.5 (3-dB) is observed due to light splitting across the two sides of the four LT-MZI system. Higher losses up to 3.9 dB is observed at lower wavelengths which can be attributed to the

GC response. Also note that Design 2 peak at 1568 nm is higher than 0.5 because of the summation of different other minor peaks.

Table 5.4: Fabricated LT-MZIs arm lengths.

| | LT-MZI 1 | LT-MZI 2 | LT-MZI 3 | LT-MZI 4 |
|---|---|---|---|---|
| Reference Design | $L_{sens,1}$ = 160 μm<br>$L_{ref,1}$ = 8 μm | - | $L_{sens,3}$ = 40 μm<br>$L_{ref,3}$ = 2 μm | - |
| Design 1 | $L_{sens,1}$ = 170 μm<br>$L_{ref,1}$ = 8 μm | $L_{sens,2}$ = 170 μm<br>$L_{ref,2}$ = 312 μm | $L_{sens,1}$ = 42.5 μm<br>$L_{ref,1}$ = 2 μm | $L_{sens,1}$ = 42.5 μm<br>$L_{ref,1}$ = 78 μm |
| Design 2 | $L_{sens,1}$ = 175 μm<br>$L_{ref,1}$ = 8 μm | $L_{sens,2}$ = 175 μm<br>$L_{ref,2}$ = 312 μm | $L_{sens,1}$ = 43.75 μm<br>$L_{ref,1}$ = 2 μm | $L_{sens,1}$ = 43.75 μm<br>$L_{ref,1}$ = 78 μm |

Figure 5.9(a) shows the change in the splitting wavelength $\Delta\lambda$ across $\Delta L$ at different wavelengths as determined experimentally from the characterized designs and also as calculated from simulations. Figure 5.9(b) shows the corresponding sensitivity of the four LT-MZI system to length (or phase) changes across the wavelength. We can see that both simulation and experimental results have good matching.

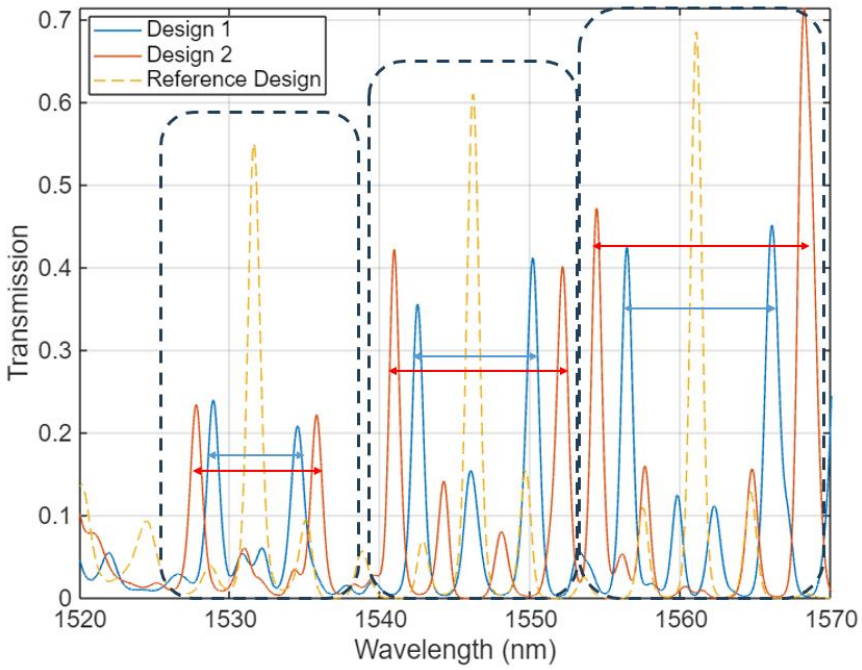



Figure 5.8: Measured transmission spectra of the different fabricated LT-MZI designs in Table 5.4, with arrows indicating the wavelength splitting and dotted boxes indicating different operating wavelengths.

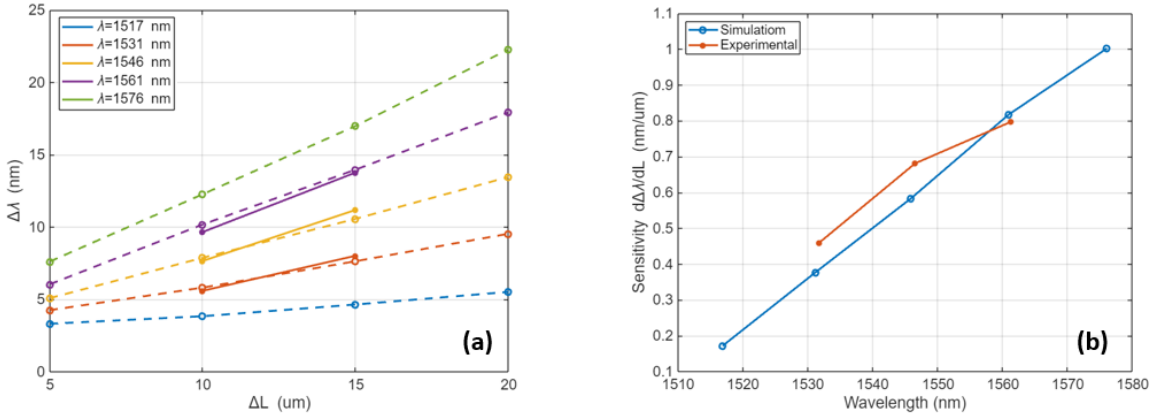



Figure 5.9: (a) Splitting wavelength versus sensing arm length at different wavelengths determined from both experimental (solid) and simulation (dotted) data. (b) Sensitivity (experimental and simulation) of the four LT-MZI to sensing arm length (or phase) changes versus operating wavelength.

## 5.5 Summary

In this chapter we have proposed a four LT-MZI sensing system based on the widespread SOI platform. The proposed structure increases the sensitivity, detection range, resolution and accuracy of conventional single stage MZI sensors. More importantly, it is capable of suppressing optical phase noise that significantly limits the performance of any optical RI sensor and especially the ones based on SOI platform. The system uses wavelength splitting to remove phase noise and cascaded LT-MZIs to expand the detection range. Wavelength splitting occurs with accordance to RI change using two LT-MZIs with opposite wavelength sensitivity. This wavelength splitting cannot be achieved using the conventional MZI. LT-MZI allows us to tune the output spectrum using its directional couplers coefficients hence, allow for wavelength splitting. By determining two independent parameters, namely the wavelength splitting and the average wavelength, the system is capable to differentiate between phase changes due to medium index change (analyte) and phase changes due to any other effects (noise). Using our four LT-MZI structure we have utilized a liquid sensor operating at near-IR. Our sensor achieves a high sensitivity of 7890 nm/RIU and an iLoD of 8e-4 with a sensing arm length of only 500 μm, which is 2 and 3 times higher, respectively, than a single MZI and much higher compared to many other proposed works. In addition, the detection range is 6.3 times higher than a single MZI. The sensor is immune to temperature and fabrication process variations, which significantly deteriorate silicon-based sensor designs. Finally, the proposed four LT-MZI system is fabricated using SOI MPW run and experimentally characterized, verifying the wavelength splitting idea and showing a good agreement with the theoretical simulations.

# 6 Chapter 6: Study of Silicon Nitride Waveguide Platform for On-Chip Virus Detection: Prospects for COVID-19

## 6.1 Introduction

CORONA virus disease of 2019 (COVID-19) pandemic was an exceptional threat to human lives all over the world. Its expeditious spread has led to millions of cases and hundreds of thousands of deaths in a few months. Almost all countries worldwide are forced to lock-down for several months to limit the spread of the virus, leading to devastating social and economic effects. In general, many people across the globe lose their life due to viral infection diseases [189]. Hence, simple, fast, cheap, and accurate detection of viruses are of great importance. Polymerase chain reaction (PCR) is one of the well-known methods used for virus detection and it is the primary method used currently for COVID-19 detection [190]. Although this technique is highly sensitive and accurate, it is expensive, time-consuming and involves complex procedures and sample preparation.

Optical refractive index (RI) sensors offer fast, compact and cheap detection with high sensitivities, as discussed before. However, RI sensors are not selective as they only detect the change in the medium (clad) refractive index which can occur due to different substances. A widely used technique to solve this problem in bio-sensing is surface

functionalization [171], [172]. In surface functionalization, the surface of the sensing waveguide is coated with specific molecules called binder or capture molecules and immobilized through a certain process. These immobilized molecules selectively capture the analyte molecules to be detected from the whole sample. After sensor exposure to the sample, a washing step is needed to make sure that only the analyte of interest is present in the sensors' medium (clad) hence, the detected refractive index change is due to this analyte alone.

As mentioned in chapter 2, RI sensors' performance is determined mainly by the figure of merit (FOM) which is the ratio between the sensor's sensitivity (S) and full width half maximum (FWHM) of the output spectrum, FOM=S/FWHM. The optical sensor sensitivity (S) is the product of the device sensitivity ($S_D$) and waveguide sensitivity ($S_{wg}$). Thus, to optimize the performance of any RI sensor, it is essential to maximize the waveguide sensitivity ($S_{wg}$).

Silicon nitride on insulator (SiNOI) waveguide platform, where insulator here is the silicon dioxide, offers numerous advantages for various applications [22], [23], [24], [99]. Similar to SOI platform SiNOI is CMOS compatible allowing for mass-scale and low cost fabrication [24], [99]. It also allows for monolithic integration with silicon devices and other electronic circuitry [22]. The lower refractive index contrast of the SiNOI waveguide compared to SOI reduce scattering loss due to surface roughness resulting in much lower propagation losses [22], [23], [24], while still maintaining device compactness. This lower index contrast also makes SiNOI devices more tolerant to fabrication errors [22], [23]. In addition, $Si_3N_4$ thermo-optical coefficient is one order of magnitude lower than Si [23],

hence $Si_3N_4$ based devices are less sensitive to temperature fluctuations. Moreover, SiNOI platform has wider transparency range, from visible to mid-infrared, compared to SOI platform [22], [23], [24], [99].

In this chapter, we present a detailed sensitivity analysis of silicon nitride (Si3N4) on silicon dioxide (SiO2) strip waveguide for virus detection. Leveraging the wide transparency range of the silicon nitride, in this study our main goal is to determine the most suitable wavelength range for on-chip RI virus sensing. Unlike gas or liquid (bulk) sensing, surface sensing applications like virus detection are highly dependent on the operating wavelength. The waveguide surface is assumed to be functionalized by the antibodies of the virus to be detected, using a process similar to that in [171], such that the medium index change is only due to this virus.

A Finite difference eigenmode solver (FDE) [29] is used to determine the waveguide dimensions that maximize the waveguide sensitivity ($S_{wg}$) to a virus layer attached to its surface. Both fundamental quasi-transverse electric (TE) and quasi-transverse magnetic (TM) mode are studied. Moreover, slot waveguide was also analyzed and compared to the strip waveguide. We examined wide range of operating wavelengths from the visible to the mid-infrared range. We found that $S_{wg}$ and FOM increase at lower wavelengths. This analysis is essential to construct an on-chip, low-cost, mass-scale fabrication, compact, and highly sensitive RI optical sensor for fast virus COVID-19 detection and generally any other virus. The optimized waveguides can be used in different integrated optical devices such as interferometers and resonators to construct the virus sensor. MZI sensors utilizing the optimized waveguides were designed reaching FOM=1231 RIU-1 at $\lambda$=450 nm with

500 μm arms' length. We also designed a MZI sensor with waveguide widths above 1 μm that can be easily fabricated in simple and cheap facilities. Finally, fiber edge coupling to the sensors chip was studied and optimized, showing higher coupling efficiencies than recently demonstrated fiber couplers. This analysis is generic for any biosensor that uses surface functionalization for selective sensing. One will only need to adjust the optimum waveguide dimensions according to the molecules' size and refractive index to be detected.

This chapter is organized as follows; section 6.2 presents the rigorous sensitivity analysis of the $Si_3N_4$ waveguide for virus sensing from visible to mid-infrared range. In section 6.3 we present the design of different virus sensors using our optimized waveguides from section 6.2. Section 6.4 includes the designs of edge couplers that we proposed to couple light from standard optical fibers to out $Si_3N_4$ sensors. At the end, section 6.5 summarizes this chapter.

## 6.2 Virus Sensing Waveguide Analysis and Discussion

Figure 6.1(a) shows the silicon nitride strip waveguide proposed for virus detection with width (w) and thickness (h) on $SiO_2$ substrate and water clad. The waveguide surface is functionalized for selective detection such that only the virus of interest will adhere to the surface and form a layer. We model this layer by a thickness $h_{vir}$ equal to the virus diameter and a refractive index $n_{layer}$, as shown in Figure 6.1(a). For COVID-19 the virus diameter is around 80 nm [191], [192] hence, we use $h_{vir}$=80 nm. The value of $n_{layer}$ changes between the refractive index of the water $n_{water}$ and the refractive index of the virus $n_{vir}$ according to

the virus coverage fraction r. The refractive index $n_{layer}$, of the water-virus mixture can be described using various models, including the refractive index average, dielectric permittivity average, Lorentz-Lorenz model, Bruggeman model [193], [194]. In our case, due to the small difference between $n_{water}$ and $n_{vir}$, these models yield nearly identical results, with less than 0.5% difference between them. Consequently, we model $n_{layer}$ by applying a weighted average of the refractive indices of the mixture, as follows:

$$n_{layer} = (1 - r) \times n_{water} + r \times n_{vir} \quad (6.1)$$

In this case, virus binding to the immobilized antibodies on the waveguide surface will change this layer refractive index which will accordingly change the waveguide mode effective index.

Different operating wavelengths are studied from visible range λ=450 nm (blue) and λ=650 nm (red), to near infrared λ=980 nm and λ=1550 nm, and MIR λ=3600 nm. Material dispersion is considered where silicon nitride, silicon dioxide and water refractive index data along the wavelength are obtained from [25], [195]. At each operating wavelength, we firstly define the single mode dimensions by determining the maximum width at different thicknesses using FDE solver [29], as shown in Figure 6.1(b) and Figure 6.1(c). Then we calculate the waveguide sensitivity ($S_{wg}$) at different waveguide dimensions (w and h), for both fundamental quasi-TE and fundamental quasi-TM modes, which we will denote as TE and TM for simplicity. Note that, unlike most RI sensors designs, here we calculate the surface waveguide sensitivity, $S_{wg}=dn_{eff}/dn_{layer}$, not the bulk sensitivity, $S_{wg}=dn_{eff}/dn_{clad}$, which is more accurate for viral detection. $S_{wg}$ is calculated with $n_{layer}$

around the water index, which means that the waveguides are optimized to have maximum sensitivity at minimum virus coverage (r close to 0). Accordingly, the exact virus refractive index $n_{vir}$ doesn't affect the obtained results.

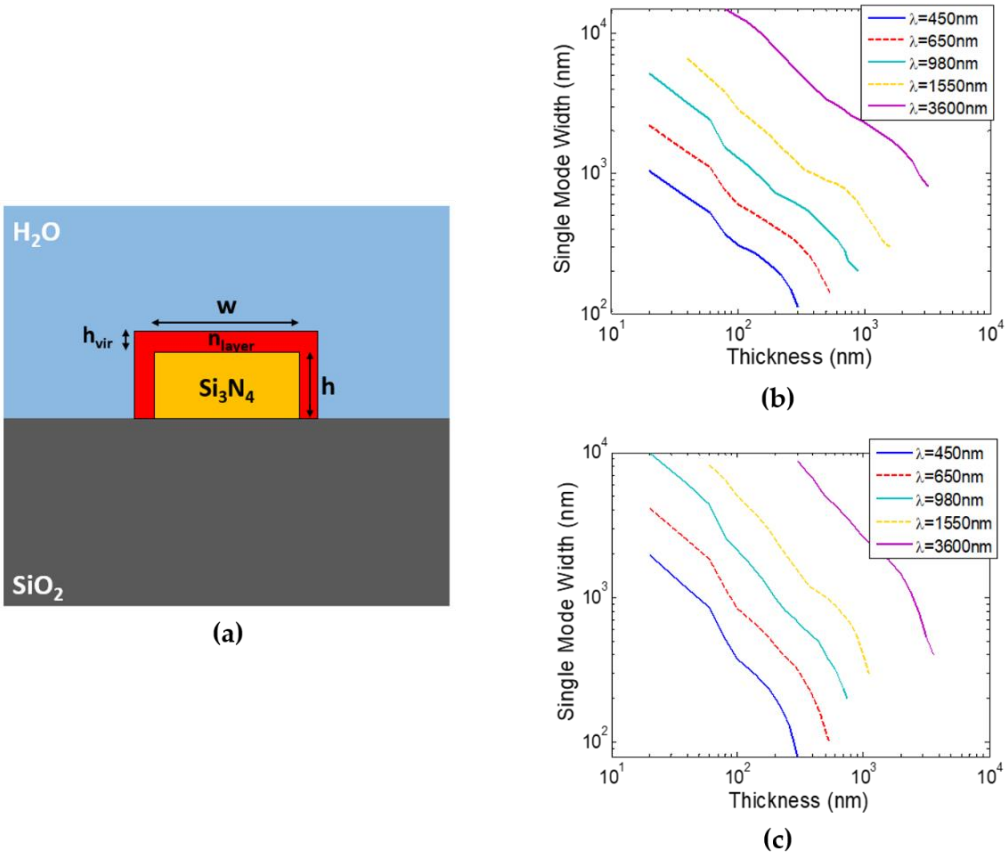



Figure 6.1: (a) Schematic of surface functionalized $Si_3N_4$ on $SiO_2$ strip waveguide for virus detection with a layer representing virus attachment. Strip waveguide single-mode width versus waveguide thickness at different wavelengths of: (b) TE mode and (c) TM mode.

Figure 6.2(a) and Figure 6.2(b) show the waveguide sensitivity of the TE mode and TM mode respectively, for different widths and thicknesses at λ=450 nm. Results show that for each waveguide thickness, there is an optimum width that maximizes the waveguide

sensitivity. Such behavior is expected [71]. For large waveguide widths most of the mode field is confined inside the silicon nitride core, resulting in low sensitivity. As the width decreases, the mode becomes less confined and the evanescent field in the cladding increase, increasing the sensitivity. However, for small widths, near cut-off, more field moves to the higher (than clad) refractive index substrate which again decreases the sensitivity [71]. Results also show that as the waveguide thickness increases, the optimum width ($w_{opt}$) decreases and optimum sensitivity increases, which is also expected [71]. Hence, waveguides with a higher aspect ratio (AR), AR=h/$w_{opt}$, can achieve higher $S_{wg}$ reaching 0.513 for the TE mode with w=104 nm and h=300 nm (AR=2.88). This behavior is similar for all wavelengths. However, high AR waveguide sensors are more challenging to fabricate and expensive as they need a fine mask and complex lithography system to obtain the small waveguide widths needed.

Moreover, at high AR the waveguide sensitivity is very sensitive to width variations. For example, changing the width by only 20 nm for the TE mode with AR=2.88 will significantly reduce $S_{wg}$ to lower than 0.15, i.e. 3.4 times reduction. Also, this optimum width ($w_{opt}$=104 nm) is close to the mode cut-off width ($w_{cut\text{-}off}$=78 nm) and multimode width ($w_{MM}$=110 nm). Figure 6.2(c) shows that for high thicknesses with low optimum widths (high AR) TE mode can achieve higher $S_{wg}$ than TM mode while, for lower thicknesses (higher $w_{opt}$ and low AR) TM mode exhibit higher $S_{wg}$ than TE mode. This is because the TE/TM modes have field discontinuity at the core edges in the x/y direction; hence, decreasing the width/thickness will increase the evanescent field's amount and,

hence, sensitivity. Hence, TM mode is optimum for cheap and easy to fabricate large feature size sensors.

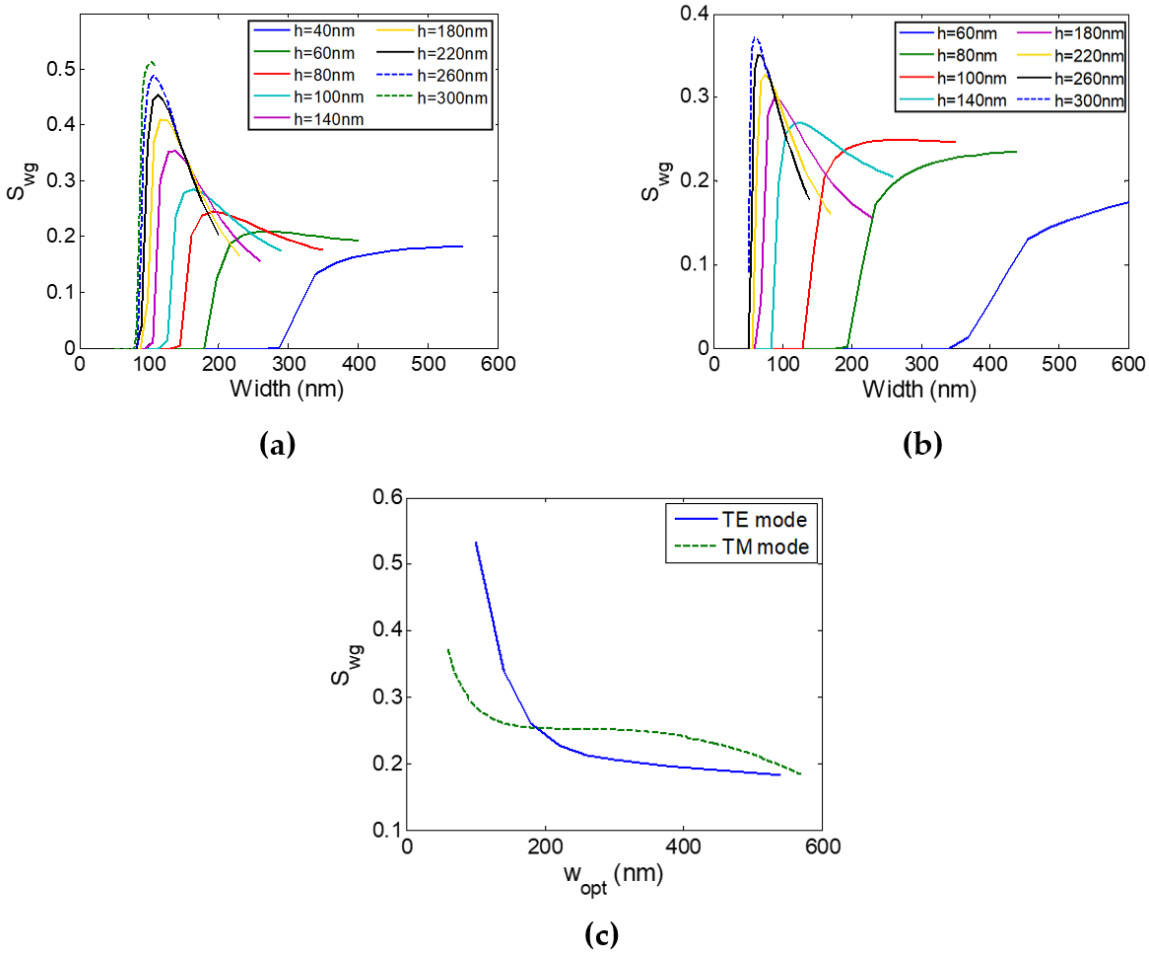



Figure 6.2: Strip waveguide sensitivity ($S_{wg}$) versus waveguide width at different thicknesses at λ=450 nm of: (a) TE mode and (b) TM mode. (c) Strip waveguide sensitivity ($S_{wg}$) of both TE and TM modes versus the optimum waveguide width ($w_{opt}$) at λ=450 nm.

Figure 6.3 shows the optimum $S_{wg}$ and the optimum width ($w_{opt}$) dependence with the AR (or h) at different wavelengths for the TE and TM mode, respectively. We can see that both

the operating wavelength and AR have a significant effect on the obtained $S_{wg}$. In addition, the highest $S_{wg}$ is obtained at the lowest operating wavelength ($\lambda$=450 nm) and it decreases monotonically as the wavelength increases. It is important to note that, scaling the waveguide dimensions with the wavelength does not result in the same waveguide sensitivity. This is mainly due to the unchanged layer's thickness that changes its refractive index (representing virus attachment). Results also show that TE mode achieves the highest possible $S_{wg}$ at every wavelength.

We have also examined $Si_3N_4$ slot waveguides (TE mode only). Slot waveguides sensitivity increase as slot width decrease. Here we used a slot width ($w_{slot}$) of 200nm as this is the smallest width that can still allow waveguide functionalization [31], [196]. Figure 6.4(a) shows the maximum slot waveguide sensitivities at $\lambda$=1.55 µm and $\lambda$=3.6 µm for the TE mode. At $\lambda$=1.55 µm $S_{wg}$ = 0.22 while at $\lambda$=3.6 µm $S_{wg}$=0.192. In order to obtain slot mode in the visible range, the slot width should be less than 200nm hence not suitable for virus detection (functionalization). Consequently, strip waveguides are more suitable for virus detection as they can achieve higher $S_{wg}$ at lower wavelengths leading to much higher FOM. In addition, the functionalization process in tiny a 200nm slot is challenging. Moreover, strip waveguide offers a simple sensor design as for example there is no need for a strip to slot mode converter.

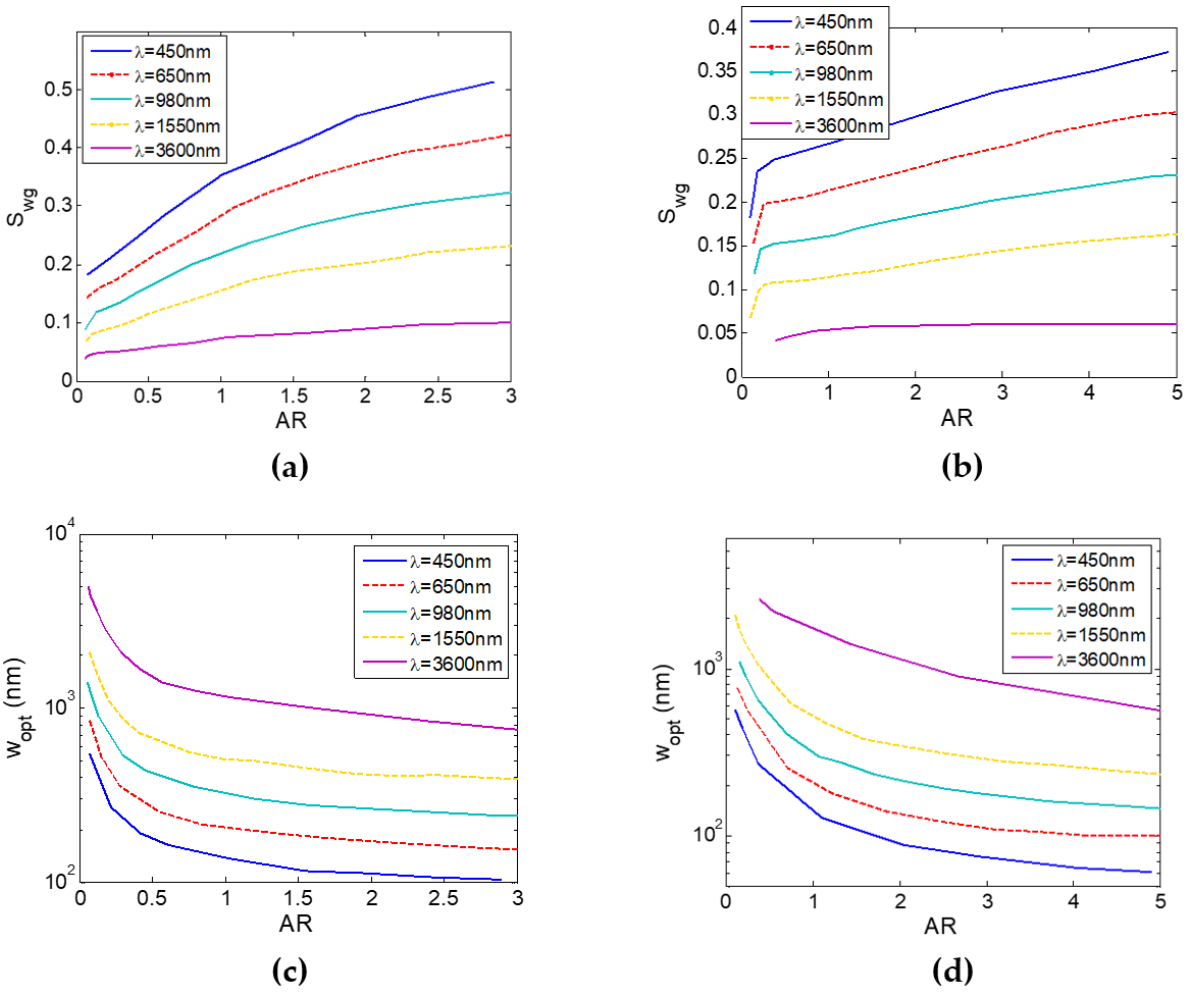



Figure 6.3: Strip waveguide sensitivity ($S_{wg}$) versus waveguide aspect ratio (AR) for: (a) TE mode and (b) TM mode at different operating wavelengths. Optimum width (wopt) versus waveguide aspect ratio (AR) for: (c) TE mode and (d) TM mode at different operating wavelengths.

We also tested the optimized waveguides for different virus diameters (i.e. different $h_{vir}$) from 60 nm to 200 nm. Figure 6.4(b) shows that the waveguide sensitivity increases with the virus size. It is important to note that, the optimum waveguide dimensions don't change

significantly from the one obtained for $h_{vir}$ = 80 nm, by changing the virus size. Hence, the same waveguide can be used for a range of viruses with different diameters.

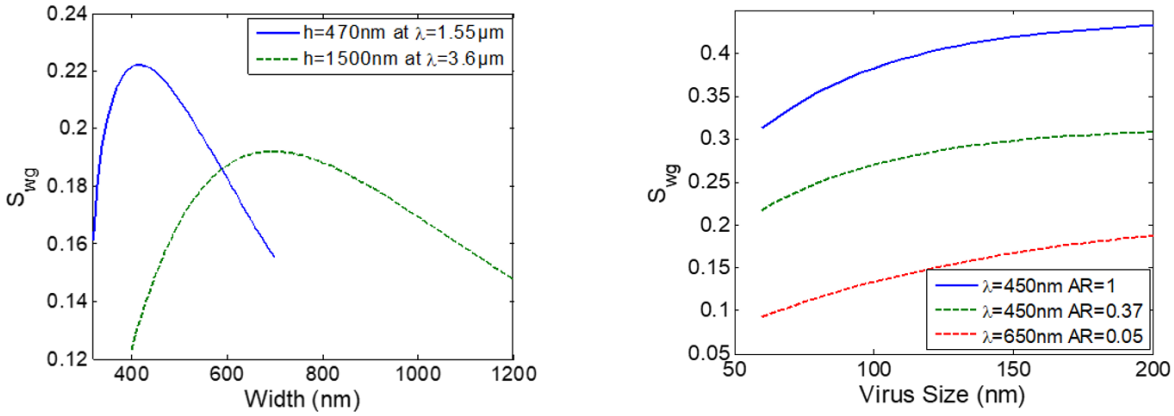



Figure 6.4: (a) Slot waveguide sensitivity $S_{wg}$ versus waveguide width at the optimum wave-guide thickness at λ=1.55μm and λ=3.6μm with $w_{slot}$=200 nm. (b) Strip waveguide sensitivity $S_{wg}$ of TE mode (solid) and TM mode (dashed) versus virus size for different optimized designs.

Previous results, assume infinitesimal functionalization layer. Figure 6.5 shows the effect of functionalization layer thickness on the $S_{wg}$ for three different designs, with $n_{funct}$=1.6. As expected, increasing this layer will reduce the $S_{wg}$, due to lower field overlap. There are lot of research trying to minimize the thickness of functionalization, reaching values less than 5 nm [91].

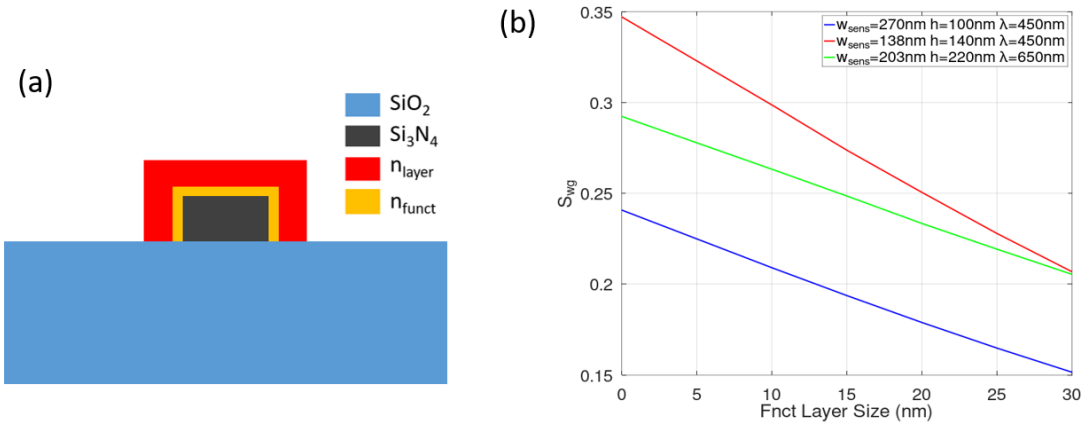



Figure 6.5: (a) Schematic of $Si_3N_4$ on $SiO_2$ strip waveguide with functionalization layer virus attachment layer and (b) $S_{wg}$ versus functionalization layer thickness for three different designs.

While waveguide sensitivity is an important parameter, the RI optical sensors' overall performance is determined by the FOM. In both interferometers and resonators, the sensitivity (S) is proportional to $S_{wg}\times\lambda$. However, the FOM is proportional to $S_{wg}/\lambda$ because the FWHM is proportional to $\lambda^2$. Hence, operating the sensor at lower wavelengths will achieve the highest performance as $S_{wg}$ increase and $\lambda$ decrease maximizing the FOM. Figure 6.6 shows fitted curves of both $S_{wg}\times\lambda$ and $S_{wg}/\lambda$ terms versus wavelength for different AR. It can be seen that, FOM increased around 8 times from NIR ($\lambda$=1.55 µm) to visible ($\lambda$=450 nm) wavelength for both TE and TM modes while the sensitivity decreased only 1.4 times. Moreover, working in the visible wavelength range have another advantage for biosensing as in this range, the losses due to water absorption are minimized. From more than 200 dB/cm mode loss at $\lambda$=3.6 µm to less than $3\times10^{-3}$ dB/cm in the visible range.

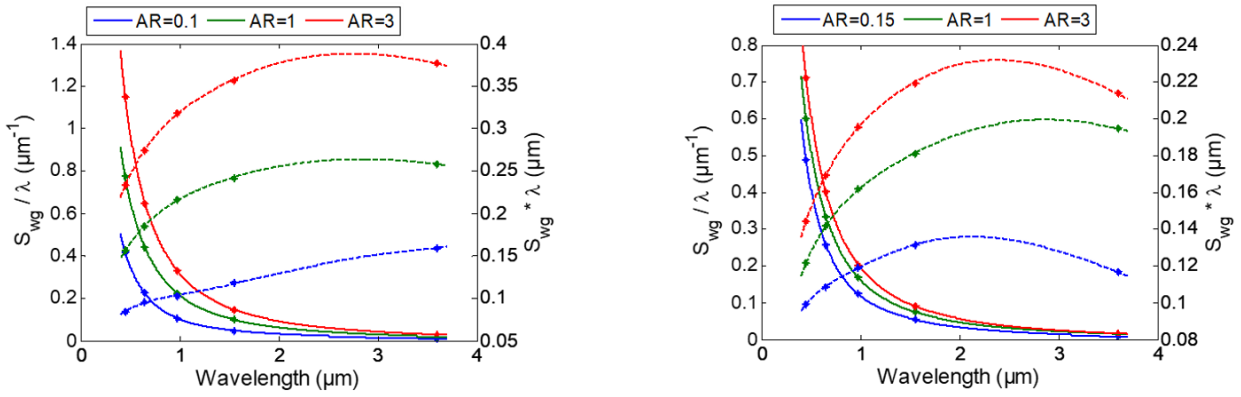



Figure 6.6: Fitted $S_{wg}/\lambda$ (solid line) and $S_{wg}\times\lambda$ (dashed line) of strip waveguide versus wavelength at different AR with (*) representing the simulated results for: (a) TE mode and (b) TM mode.

## 6.3 Virus Sensors Design

Different MZI sensors have been designed to convert the change in the waveguide's effective index to a sensible quantity for COVID-19 detection. The $Si_3N_4$ waveguide surface will be functionalized with the virus antibodies. In this case, the COVID virus in the sample will be selectively captured by the waveguide. COVID virus binding will change the refractive index of the 80 nm layer covering the waveguide core. Accordingly, a wavelength shift ($\Delta\lambda$) in the transmission spectrum of the MZI will occur, from which the virus concentration can then be determined.

From previous chapters, the sensitivity and FOM of the MZI sensor are:

$$S_{\lambda,MZI} = \frac{\lambda_{res}}{(n_{g,sens}L_{sens} - n_{g,ref}L_{ref})} L_{sens}S_{wg} \qquad (6.2a)$$

$$FOM = \frac{S}{FWHM} = \frac{\pi S_{wg} L_{sens}}{\sqrt{2}\lambda} = \frac{1}{\Delta n_{min}} \quad (6.2b)$$

FOM is the main performance parameter of any RI sensor as it determines the minimum detectable refractive index change, $\Delta n_{min}$. Table 6.1 shows the dimensions, FOM, and $\Delta n_{min}$ of different symmetric MZI (s-MZI) sensors designs, $L_{sens}=L_{ref}=L_{MZI}$, at different wavelengths with AR≈1, $L_{MZI}$=500μm and ideal y-junction for comparison. The optimized strip waveguides from the previous analysis with the virus layer around the core are used as the sensing arm with width $w_{sens}$. Oxide capped waveguides are used as the reference arm with width $w_{ref}$. These results are obtained using a FDE solver to determine $n_{eff}(\lambda)$ of the sensing and reference waveguides. Next, integrated photonics circuit simulator [29] is used to determine $T(\lambda)$ at different $n_{layer}$ (virus concentration) from which S, FWHM and FOM are then calculated. Results exhibit the same response as Figure 6.6 with around 8 times greater FOM at the lower (blue) wavelength. Table 6.1 also shows FOM at λ=450 nm for different AR, reaching a maximum of 2462 $RIU^{-1}$ at AR=2.88 which corresponds to a limit of detection as low as $4e^{-4}$. As mentioned before, at higher waveguide widths, TM mode was used as it can reach higher $S_{wg}$ than TE mode, see Figure 6.2(c). Note that, MIR range was discarded due to its low $S_{wg}$ and high (water absorption) losses.

Table 6.1: FOM and $\Delta n_{min}$ of S-MZI sensors with $L_{MZI}$ = 1000 μm.

| λ (nm) | AR | Dimensions (nm) | FOM ($RIU^{-1}$) | $\Delta n_{min}$ |
|---|---|---|---|---|
| 450 | 0.13* | $w_{sens}$=550, $w_{ref}$=520 and h=70 | 1002 | 1e-3 |
| | 0.37* | $w_{sens}$=270, $w_{ref}$=300 and h=100 | 1106 | 9e-4 |

| | | | | |
|---|---|---|---|---|
| | 1 | $w_{sens}$=138, $w_{ref}$=145 and h=140 | 1624 | 6.2e-4 |
| | 2.88 | $w_{sens}$=104, $w_{ref}$=104 and h=300 | 2462 | 4e-4 |
| 650 | 1.1 | $w_{sens}$=203, $w_{ref}$=230 and h=220 | 908 | 1.1e-3 |
| 980 | 1.2 | $w_{sens}$=300, $w_{ref}$=360 and h=360 | 488 | 2e-3 |
| 1550 | 1 | $w_{sens}$=512, $w_{ref}$=850 and h=500 | 200 | 5e-3 |

* Representing TM mode.

Although small waveguide dimensions in the visible range exhibit high sensing performance, the fabrication of such waveguides require complex and expensive lithography systems like electron beam or deep UV lithography. Hence, we want to determine a sensor's performance with a feature size above 1µm, which will allow for easy and cheap fabrication. While lower wavelengths exhibit higher performance, the blue wavelength has almost zero sensitivity for small AR waveguides with $w_{opt}$>1µm. Hence, we choose to compare two designs both with TM mode. The first design is operating at low (red) wavelength λ=650 nm exhibiting $S_{wg}$ of 0.115, and the second is operating at a higher wavelength at λ=980 nm but demonstrating slightly higher $S_{wg}$ of 0.13. Table 6.2 shows the dimensions, FOM, and $\Delta n_{min}$ of both MZI sensor designs. We can see that the first design operating at lower (red) wavelength with AR=0.05 has a higher FOM of 316 $RIU^{-1}$ even if it exhibits slightly lower $S_{wg}$.

Table 6.2: FOM and $\Delta n_{min}$ of S- MZI sensors with large feature size using TM mode at $L_{MZI}$ = 1000 µm.

| λ (nm) | Dimensions (nm) | FOM ($RIU^{-1}$) | $\Delta n_{min}$ |
|---|---|---|---|
| 650 | $w_{sens}$=1500, $w_{ref}$=1000 and h=80 | 316 | 3.2e-3 |
| 980 | $w_{sens}$=1500, $w_{ref}$=1100 and h=160 | 254 | 3.9e-3 |

The minimum detectable index change of the virus layer can be converted to minimum detectable virus coverage $r_{min}$ using equation (6.1). Table 6.3 shows $\Delta n_{min}$ and $r_{min}$ for the MZI sensors designs at the blue wavelength with different AR and the design at red wavelength optimized for large dimensions (AR=0.05) with $L_{MZI}$=1000μm. Results shows that the proposed sensors can detect a surface coverage as low as 0.26%. Note that, lower virus concentrations (coverage r) can be detected by increasing the FOM by increasing the MZI sensor length as given in equation (6.2b).

It is important to note that, silicon nitride waveguides with film thickness great than 300 nm suffer large stress, and different techniques are used to overcome this problem [197], [198], [199]. However, our analysis shows that thin silicon nitride waveguides, with h<300 nm, in the visible range are of better sensing performance. In this case, such stress is reduced and a homogeneous index and thickness can be obtained using low-pressure chemical vapor deposition (LP-CVD) [199].

Table 6.3: $\Delta n_{min}$ and $r_{min}$ of S-MZI sensors with $L_{MZI}$ = 1000 μm.

| λ (nm) | AR | $\Delta n_{min}$ | $r_{min}$ (%) |
|---|---|---|---|
| 450 | 0.13* | $1.0\times10^{-3}$ | 0.65 |
| | 0.37* | $9\times10^{-4}$ | 0.58 |

| | | | |
|---|---|---|---|
| | 1 | $6.2\times10^{-4}$ | 0.40 |
| | 2.88 | $4\times10^{-4}$ | 0.26 |
| 650 | 0.05* | $3.2\times10^{-3}$ | 1.87 |

*Representing TM mode.

Finally, different MZI configurations were studied and compared for sensing, namely symmetric MZI (s-MZI), asymmetric MZI (a-MZI) and loop terminated MZI (LT-MZI) shown in Figure 6.7. The simulated results of the different configurations are summarized in Table 6.4 for the design of TM mode with $w_{sens}$=270 nm, h=100 nm and $w_{ref}$=300 nm at $\lambda$=450 nm and $L_{sens}$=1000 μm. While s-MZI ($L_{sens}=L_{ref}$) sensitivity is determined only by its waveguide structures, i.e. $\Delta n_{eff}=n_{eff,sens}-n_{eff,ref}$, a-MZI ($L_{sens}=L_{ref}+\Delta L$) sensitivity can be engineered using $\Delta L=L_{sens}-L_{ref}$, according to equation (6.2a). However, both structures will exhibit almost the same FOM for the same $L_{sens}$. On the other hand, LT-MZI is a recently proposed design [200] that consists of a conventional MZI with a loop connecting the output directional coupler arms, reflecting back the wave to the interferometer. For the same waveguide structure and length $L_{sens}$, LT-MZI will exhibit the same sensitivity with the conventional MZI while the FWHM will reduce to half resulting in twice the FOM. The LT-MZI directional couplers are also assumed to be ideal 3-dB couplers. The asymmetric LT-MZI can also be used to control the sensitivity using $\Delta L$ as in the a-MZI case.

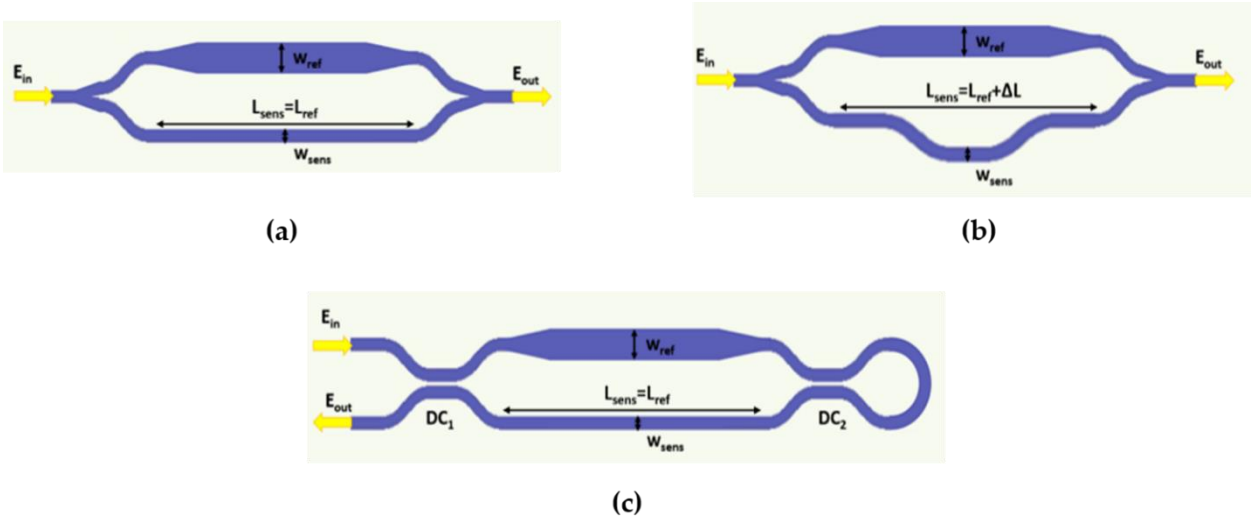



Figure 6.7: Schematic of MZI sensor configurations: (a) s-MZI, (b) a-MZI and (c) LT-MZI.

Table 6.4: S, $\Delta n_{min}$, and FOM of the TM mode for different MZI sensors configurations with wsens=270 nm, h=100 nm and $w_{ref}$ =300 nm at λ=450 nm and $L_{sens}$=1000µm.

| | | S (nm/RIU) | FOM ($RIU^{-1}$) | $\Delta n_{min}$ |
|---|---|---|---|---|
| s-MZI | | 3098 | 1106 | 9e-4 |
| a-MZI | ΔL=30µm | 1316 | 1080 | 9.3e-4 |
| | ΔL=5µm | 5579 | 1060 | 9.4e-4 |
| LT-MZI | | 3098 | 2212 | 4.5e-4 |

## 6.4 Edge Fiber Coupling of Designed Sensors

Recently, many efforts have been done to couple light from fiber to Si3N4 platform in the NIR range (around 1.55μm) [121], [201], [202], reaching a measured coupling efficiency as low as -1.75 dB [201], using bottom multilayer reflector and apodized grating coupler. However, few work have been published for coupling in the visible wavelength range [203], [204], [205].

In this section, we study the coupling from single-mode fibers [206] in the visible region (blue and red) to the silicon nitride chip through edge coupling again using an FDE solver. We focus on the coupling to the TM mode sensors designs mentioned in the previous section, which can achieve high coupling efficiencies and high FOM with waveguide widths larger than 250 nm, see Table 6.1 and Table 6.2. Design 1: $w_{sens}$=550 nm, h=70 nm (AR=0.13) and Design 2: $w_{sens}$=270 nm, h=100 nm (AR=0.37) both at λ=450 nm. While Design 3: $w_{sens}$=1500 nm, h=80 nm (AR=0.05) at λ=650 nm for large feature size sensor (w>1 μm). Figure 6.8(a) shows the coupling efficiency at blue and red wavelengths to a waveguide with thickness h=20nm and h=40nm, respectively. A maximum coupling efficiency of 93% and 92.7% can be achieved from the fiber to the waveguide TM mode at λ=450nm with w=600 nm and at λ=650nm with w=565 nm, respectively. Note that, waveguides with higher thicknesses exhibit significantly lower coupling for w>250nm. Hence, there is a mismatch between the waveguide dimensions with optimum fiber coupling ($w_{cpl}$, $h_{cpl}$) and optimum sensing ($w_{sens}$, $h_{sens}$), as shown in Figure 6.8(b). Accordingly, the coupling between these two waveguides was studied and the insertion loss (IL) was determined for the different sensing waveguides. For each design we optimize

the waveguide-waveguide coupling, $w_{cpl} \times h_{cpl} \rightarrow w_{out} \times h_{sens}$, by changing the output waveguide width ($w_{out}$), which can then be converted to $w_{sens}$ with significantly low losses using a taper. Hence, for the blue wavelength the IL from the fiber to the optimum coupling waveguide, $w_{cpl}$=600 nm and $h_{cpl}$=20 nm, is 0.3 dB. The waveguide-waveguide coupling for the sensing waveguide with $h_{sens}$=70 nm (Design 1) shows IL=3.8 dB with optimum $w_{out}$ of 2600 nm. While for the sensing waveguide with $h_{sens}$=100 nm (Design 2) IL=4.8 dB at $w_{out}$=2700 nm. Hence, the overall fiber coupling loss to the sensing waveguide is 6.1 dB and 5.1 dB, respectively. For the red wave-length (Design 3), with w>1 µm, the fiber-coupling waveguide IL is 1.37 dB at $w_{cpl}$=1030 nm and $h_{cpl}$=40 nm. While waveguide-waveguide coupling exhibits IL=1.1 dB for $w_{out}$=1940 nm to the sensing waveguide with $h_{sens}$=80 nm, resulting in an overall coupling loss of 2.47 dB.

These designs exhibit higher coupling efficiencies than most fiber couplers proposed for the Si3N4 platform at the same wavelength range [203], [204], [205]. This is mainly due to the different waveguide dimensions, as the optimum waveguides for sensing have small core thickness dimensions. Thus, exhibiting large mode size which leads to better matching with the fiber mode. The recently proposed fiber couplers and our proposed ones are summarized in Table 6.5.

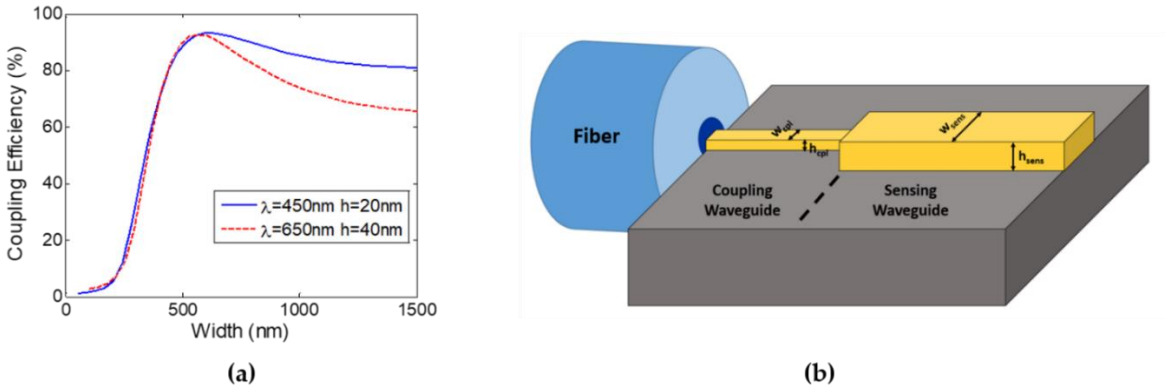



Figure 6.8: (a) Fiber to strip waveguide coupling efficiency versus waveguide width at λ=450 nm and λ=650 nm with h=20 nm and h=40 nm, respectively. (b) Schematic of two-step fiber edge coupling.

Table 6.5: Comparison of our proposed fiber couplers and recently demonstrated ones, showing the output waveguide (w×h) in each case with (S) and (M) denoting simulated and measured results, respectively.

| | λ (nm) | Output waveguide (nm) | Technique | Coupling Loss (dB) |
|---|---|---|---|---|
| [203] | 660 | 700×100 | Grating | 4.2 (M) |
| [204] | 532 | 350×180 | Grating | 6 (S) |
| | 640 | 340×220 | Grating | 6.6 (S)<br>7.5 (M) |
| [205] | 430 - 648 | 340×135 | Edge | 8-8.9 (M) |
| | 450 | 550×70 | Edge with h step | 4.1 (S) |
| | | 270×100 | Edge with h step | 5.1 (S) |

| | | | | |
|---|---|---|---|---|
| Our Designs | 650 | 1500×80 | Edge with h step | 2.47 (S) |

## 6.5 Summary

In this chapter we determined the best operating wavelength range for on-chip RI virus sensing. In our study, we utilized the wide wavelength transparency range and the CMOS compatibility of $Si_3N_4$ on $SiO_2$ waveguide platform. A rigorous sensitivity analysis for the silicon nitride waveguide is presented, including different waveguide dimensions, $Si_3N_4$ waveguide widths and heights, and wavelength range from visible to mid-infrared. Integrated RI sensors offer fast, cheap and simple detection when compared to the existing methods like the PCR, which is expensive and involves complex procedures. Our study shows that the Si3N4 strip waveguide sensors can achieve high sensitivity, and with surface functionalization, it can detect only a specific virus for high selectivity. As an example, in this study, we focus on COVID-19 detection. Our analysis determines the waveguide dimensions that maximize the sensitivity to a virus layer attached to its surface. The optimum dimensions were determined for different wavelengths from the visible to the MIR and for both fundamental quasi-TE and quasi-TM modes. In addition, we compared the silicon nitride strip waveguide to the slot waveguide. Results show that $Si_3N_4$ strip waveguide operating at low wavelengths is the best choice for virus detection as it maximizes the waveguide sensitivity and the FOM as well as minimizes water absorption significantly compared to infrared which results in better sensor resolution and accuracy.

The results show that $Si_3N_4$ strip waveguides operating at low wavelengths are the optimal choice for virus detection. They maximize waveguide sensitivity and figure of merit (FOM), while significantly reducing water absorption compared to infrared, resulting in improved sensor resolution and accuracy. MZI sensors were designed offering a FOM as high as 2462 $RIU^{-1}$ for $L_{MZI}$=1000 μm at λ=450 nm which yields a refractive index detection as low as 4e-4, which can be furtherly enhanced with longer sensing arm length. Finally, different edge couplers used to couple light from the fiber to our proposed sensors was designed, showing coupling loss less than 5.1 dB and only 2.47 dB at λ=650 nm.

# 7 Chapter 7: Conclusions, Future Work and Publications

## 7.1 Conclusions

Chemical and biological detection is crucial in fields such as healthcare, diagnostics, food industry, environmental monitoring, safety, and industrial process control. Conventional methods often suffer from low sensitivity, limited dynamic range, complex preparation, and bulky equipment, necessitating an affordable, compact, simple, real-time, and accurate detection method.

Refractive index (RI) sensors are well-suited for integrated on-chip optical sensing due to their high sensitivity to small volume samples. Minor changes in the refractive index over short distances can result in significant phase shifts in the propagating wave, leading to exceptionally high sensitivities. Utilizing CMOS-compatible technologies, RI sensors enable low-cost and large-scale fabrication. Moreover, they facilitate the monolithic integration of electrical circuitry with the optical sensor, resulting in compact, distributed, real-time, and remote sensing capabilities.

The performance of a refractive index (RI) sensor is primarily defined by its sensitivity and limit of detection (LoD). Enhancing the accuracy and resolution of RI sensors requires increasing their sensitivity and reducing the LoD. Additionally, optical phase errors caused by fabrication processes and temperature variations significantly impact sensor performance. To optimize RI sensor design, three key aspects must be considered: the

waveguide platform and its dimensions, the optical device design and dimensions, and the operating wavelength.

Two critical areas of study and two primary challenges in RI sensing need to be addressed:

1) Identify the optimal platform and device for RI sensing.
2) Determine the best operating wavelength for surface sensing.
3) Enhancement of the sensor's robustness against process and temperature variations.
4) Addressing sensor's selectivity to accurately identify medium substances.

Chapter 2 provides a comprehensive comparison of various on-chip refractive index (RI) platforms and optical devices, including a detailed analysis of their performance parameters. This chapter addresses the challenge of identifying the most suitable platform and device for RI sensing. As a continuation, in chapter 3 we tackle the issue of high LoD and large size of interferometer-based sensors. We introduce a compact SOI gas sensor utilizing a slot-based LT-MZI design and achieving a low LoD. This design was also engineered to be able to compensate for temperature fluctuations.

Chapter 4 proposes a method for selective on-chip RI sensing by detecting both the real and imaginary parts of the medium's refractive index at multiple wavelengths, allowing for the determination of the dispersion of the detected substance. This method also enables the determination of concentration composition in multi-element mediums. Artificial intelligence algorithms are employed to significantly enhance detection accuracy, range, and noise removal from the measured spectrum. Thus, addressing the issue of RI sensors selectivity.

Chapter 5 presents a novel approach for optical phase noise suppression using wavelength splitting with a system of four LT-MZIs. This proposed sensing system doubles sensitivity, increases detection range, and significantly improves the limit of detection (LoD) compared to conventional MZI-based designs, addressing the challenge of robustness against process and temperature variations.

Finally, chapter 6 focuses on the silicon nitride platform for on-chip RI virus sensing. It includes an extensive sensitivity analysis of the silicon nitride waveguide across a wide wavelength range, from visible to mid-infrared. Based on this study, a RI-based virus sensor and edge coupler designs for light coupling from fibers are developed. Addressing the need to determine the optimal wavelength for surface sensing.

## 7.2 Future Work

### 7.2.1 Enhancing Performance Through the Application of AI

The idea proposed in chapter 4 where we use a MRR to selectively detect the sensing medium through detecting the complex refractive index at different wavelengths is very powerful and can be extended to cover different substances for various applications. Specially by exploiting artificial intelligence algorithm. While we used these ANNs to improve the detection accuracy, detection range and remove the electrical noise, one important addition is to train these ANN so that they can compensate for optical phase errors. As we discussed temperature fluctuation is a major issue in RI sensors that limits the LoD specially, in SOI based sensors due to the high TO coefficient of silicon. Using

our proposed idea in chapter 4, ANN can be trained to separate between the signal, which is medium index change, and the noise due to temperature making the design immune to any temperature changes and achieving low LoD.

AI can also be using with our four LT-MZI sensor system proposed in chapter 5. One of the assumptions used in this design is that the arms of the cascaded LT-MZIs (LT-MZIs on the same side of y-junction) are affected by the same error (noise), either due to fabrication change or temperature change. However, even if these LT-MZIs don't see the same error in principle we can still compensate for such error, by the relative intensities of the major and minor peaks in the four LT-MZI system output spectrum which can be challenging. Using ANN, it can easily compensate for such error by training the network using the different spectrum peaks.

### 7.2.2 Consider other designs immune to fabrication and temperature variations

We have proposed in chapter 5 the four LT-MZI sensing system to compensate for fabrication and temperature variations. This is done by detecting two independent parameters from the spectrum, the center and splitting wavelength. Other designs were proposed in the literature to either compensate for fabrication variations or temperature variations. For instance, fabrication or temperature variations can be compensated using MZI by designing different waveguide structures in both arms. In addition, MRR measured using different polarizations, TE and TM, was proposed to compensate for temperature

variations. However, these techniques can compensate for only one change (error) either fabrication change or temperature change. So, a good design merging the two techniques can compensate for both variations. The design can be constructed of a two identical MZI sensors where one of them excited by a TE mode and the other one use polarization rotator to rotate the input TE mode to TM and then pass it to the other MZI sensors. The MZI arms are designed to compensate for fabrication variations. In this case, hopefully we could have a high-performance sensor compensating for both fabrication and temperature variations. We can also implement our four LT-MZI design using fabrication tolerant LT-MZIs so that it can be tolerant even to different variations in the different waveguide arms.

### 7.2.3 Experimental verification

While we have fabricated and experimentally characterized most of our proposed devices, verifying the basic concept and the reliability of the designs. The sensing performance of our proposed sensors is not characterized yet as we didn't have access to the waveguide cladding in the previous MPW runs. The characterization of the sensing performance of the sensor designs proposed in this thesis are currently in progress. We have fabricated all the proposed sensor designs in applied nanotools (ANT) using silicon nitride (SiN) MPW run, where the oxide cladding over the sensing waveguide was selectively removed to allow for sensitivity characterization. Figure 7.1 shows a sample of the fabricated die layout. Optical setup for on-chip characterization of this SiN die was built, where an infrared tunable laser and a germanium photodetector are used for wavelength sweeping. The in and out coupling of the optical signal is done thorugh on-chip GCs. Microfluidic channel that can be integrated with the SiN die are also fabricated. More details about the

fabricated designs, optical setup and preliminary measurements can be found in Appendix A.

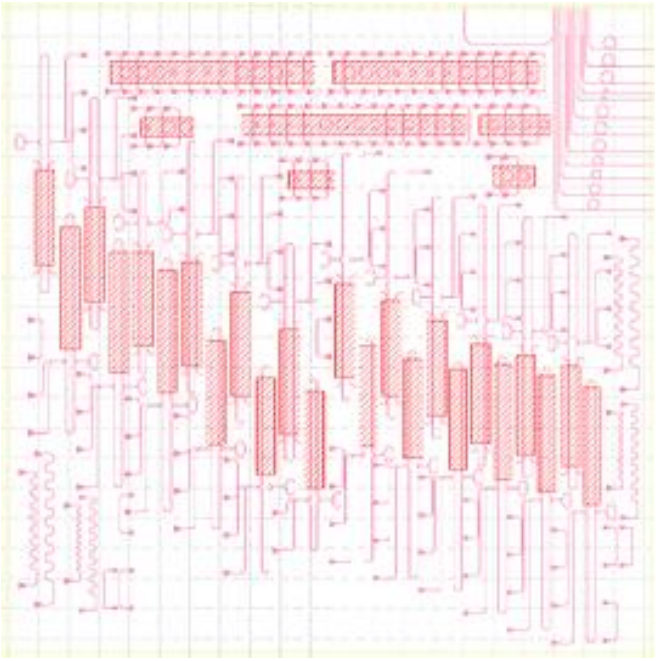


Figure 7.1: Layout of the fabricated SiN die.

## 7.3 Publications

### 7.3.1 Journal papers

1) R. S. E. Shamy, M. A. Swillam, and X. Li, “Comparative Study of Photonic Platforms and Devices for On-Chip Sensing,” *Photonics*, vol. 10, no. 11, p. 1233, Nov. 2023, doi: 10.3390/photonics10111233.
2) R. S. E. Shamy, M. A. Swillam, M. M. ElRayany, A. Sultan, and X. Li, “Compact gas sensor using Silicon-on-Insulator Loop-Terminated Mach–Zehnder

interferometer," *Photonics*, vol. 9, no. 1, p. 8, Dec. 2021, doi: 10.3390/photonics9010008.

3) R. S. E. E. Shamy, M. A. Swillam, and X. Li, "Optimization of Silicon Nitride Waveguide platform for On-Chip virus Detection," *Sensors*, vol. 22, no. 3, p. 1152, Feb. 2022, doi: 10.3390/s22031152.
4) R. S. E. Shamy, M. A. Swillam, and X. Li, "On-chip complex refractive index detection at multiple wavelengths for selective sensing," *Scientific Reports*, vol. 12, no. 1, Jun. 2022, doi: 10.1038/s41598-022-13033-3.
5) T. J. Mikhail, R. E. Shamy, M. A. Swillam, and X. Li, "Enhanced performance of on-chip integrated biosensor using deep learning," *Optical and Quantum Electronics*, vol. 55, no. 11, Sep. 2023, doi: 10.1007/s11082-023-05258-x.

### 7.3.2 Conference papers

1) El Shamy, Raghi S., Mohamed A. Swillam, and Xun Li. "Micro-ring Resonator for Complex Refractive Index Detection at Multiple Wavelengths." *Frontiers in Optics*. Optica Publishing Group, 2022.
2) El Shamy, Raghi, Xun Li, and Mohamed Swillam. "On-Chip Virus Sensing Using Silicon Nitride Waveguide." *2022 Photonics North (PN)*. IEEE, 2022.
3) El Shamy, Raghi S., et al. "On-Chip Loop-Terminated Mach-Zehnder Interferometer Gas Sensor." *2022 Photonics North (PN)*. IEEE, 2022.

4) El Shamy, Raghi S., et al. "Visible and near-infrared photonic components library based on silicon nitride platform." *Integrated Optics: Devices, Materials, and Technologies XXVII*. SPIE, 2023.
5) Mikhail, Thomas J., et al. "Prediction of medium chemical concentration with micro-ring resonators and deep learning." *Smart Photonic and Optoelectronic Integrated Circuits 2023*. Vol. 12425. SPIE, 2023.

# APPENDIX A

We fabricated our proposed sensor designs in applied nanotools (ANT) silicon nitride (SiN) MPW run. ANT MPW platform consists of 400 nm SiN device layer. In this MPW run electron beam lithography (EBeam) is used for waveguide patterning. The minimum feature size is 120 nm and the minimum feature spacing is 120 nm. Fully-etched devices are created using anisotropic inductively coupled plasma - reactive ion etching (ICP-RIE) process. We have designed all necessary components for our sensors:

- Splitters (DCs and MMIs)
- Mode converters (strip to slot wgs)
- Bend waveguides

Sensors designs including strip and slot-based LT-MZIs proposed in chapter 3, MRRs for complex refractive index sensing proposed in chapter 4, and four LT-MZI sensing system proposed in chapter 5 together with conventional MZIs for comparison, are all included in this run. Figure A.1 shows the layout of different fabricated sensors and Figure A.2 shows SEM images of fabricated devices.

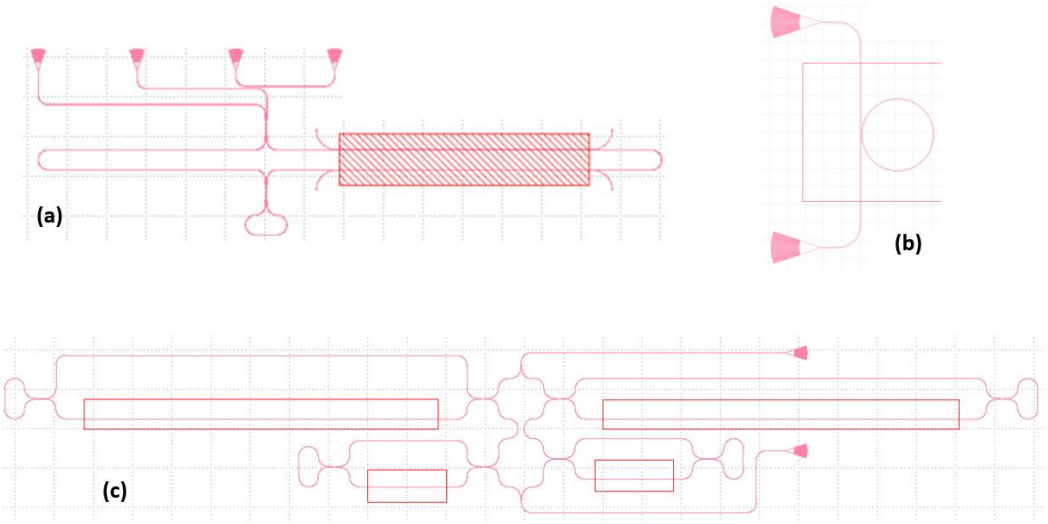



Figure A.1: Layout of the fabricated (a) slot-based LT-MZI sensor, (b) MRR sensor and (c) four LT-MZI sensor.

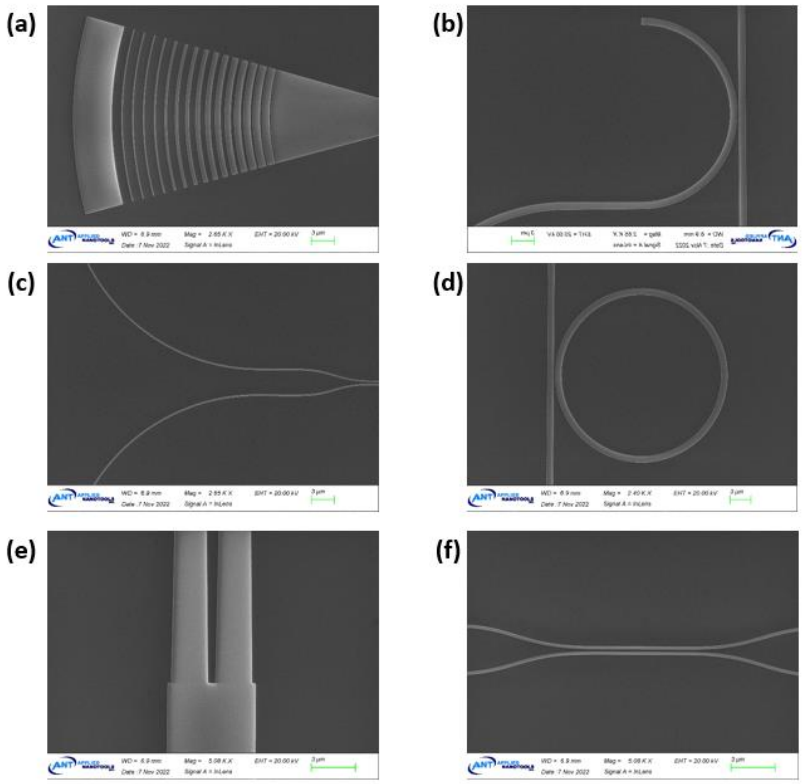
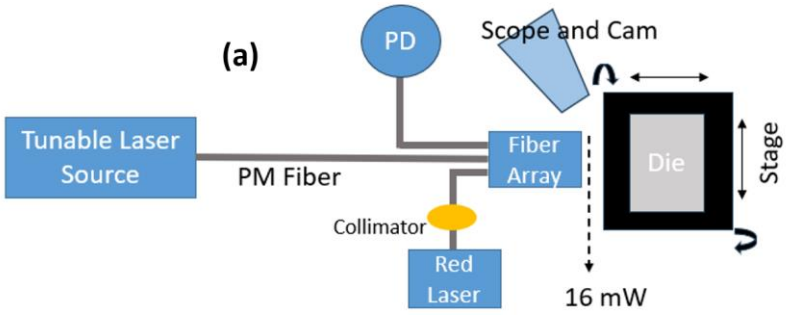



Figure A.2: SEM images of fabricated devices (a) GC, (b) half ring DC, (c) MZI DC splitter, (d) MRR, (e) MMI, (f) straight DC.

We have built optical setup for the device characterization of this SiN die. The optical setup shown in Figure A.3 includes an infrared tunable laser (Thorlabs TLX1 - Benchtop Tunable Laser Source, C-Band) ranging from 1528 nm to 1565 nm, with a narrow 10 kHz linewidth and output power up to 16 mW, and a germanium photodetector (Thorlabs S122C - Standard Photodiode Power Sensor) for wavelength sweeping. A python code has been developed for wavelength sweeping automation. A polarization maintaining (PM) fiber array containing 6 fibers with a 250 µm fiber spacing is laying above the die vertically and used to couple IR light into and out of the die through grating couplers with the same spacings. The fibers are single mode at 1550nm with mode field diameter (MFD) of around 10 µm. GCs working around 1550 nm is provided by the fab with around 12 dB IL. The setup also includes an optical microscope connected with IR or Visible camera to align the input IR beam with the input GC of the targeted device. Figure A.4 shows images of the devices on the SiN die captured from the camera. Different positioners are used for active alignment. Using these positioners, we tune the die stage in the x, y, z direction as well as the different angles to maximize detected power in the PD.



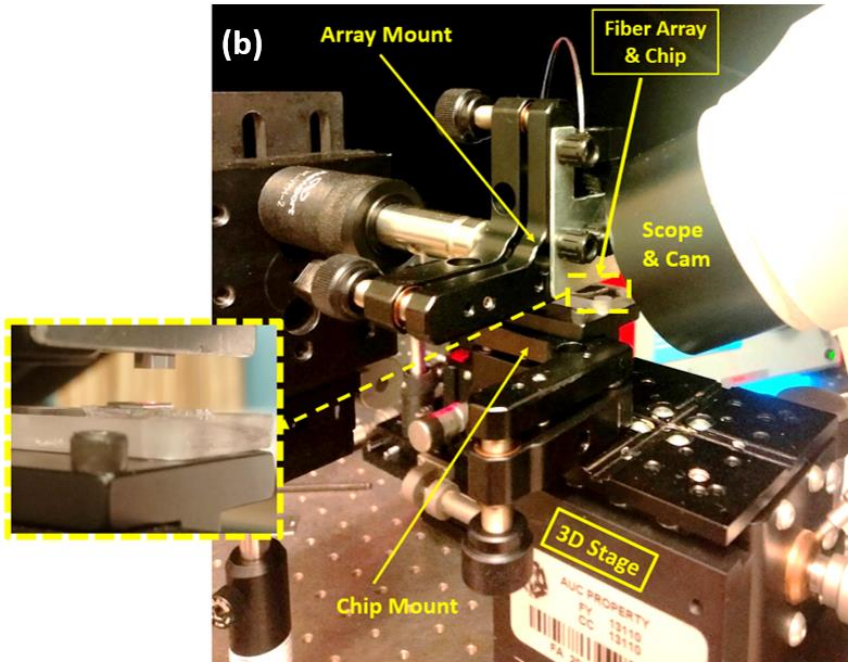



Figure A.3: (a) Schematic of the optical setup for experimental characterization. (b) Image of the real optical setup.

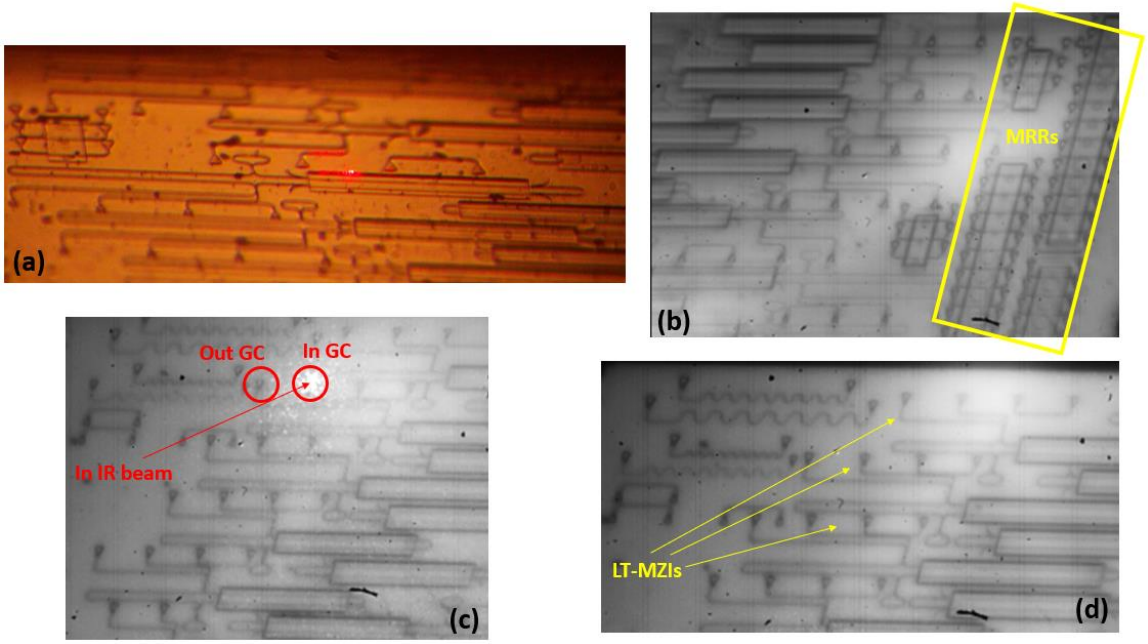



Figure A.4: (a) Images of the devices on the SiN die from the captured from the camera.

Figure A.5 shows the measured transmission of a GC and a LT-MZI with wavelength resolution of 1.2 nm and 0.4 nm, respectively. We can see the bell-shaped response of the GC and the LT-MZI transmission with FSR of 4-5 nm close to the designed FSR of 4.5 nm. Results show that the detected power is significantly low with IL of 54 dB. From the fab GC we expect IL due to in and out coupling of around 24 dB, see Figure A.5(c). This large IL is due to the large distance between the fiber array and the GCs. The used fiber array is polished at zero angle while the optimum angle for coupling to the GCs is 8$^{o}$, so we tilt the fiber array using the fiber array mount. However, due to the fiber array chip thickness the distance between the fiber and the die is 1-2 mm, which increase the IL to more than 30 dB (verified by 3D- FDTD simulations).

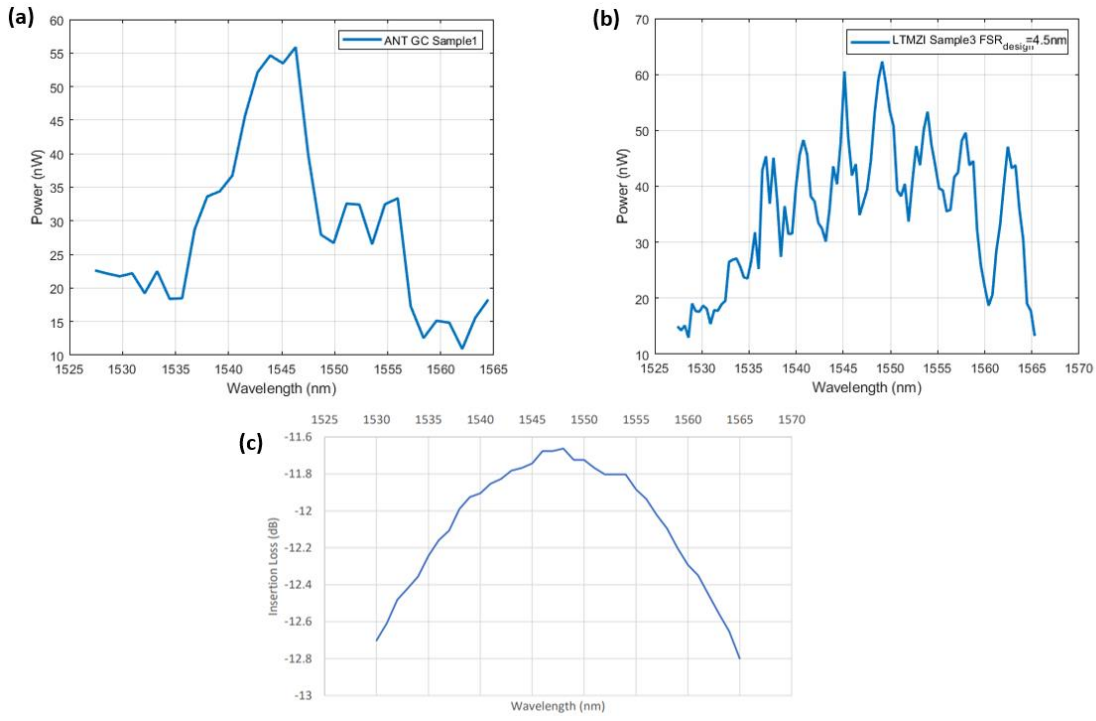



Figure A.5: Measured transmission spectrum of (a) GC and (b) LT-MZI.

We used an alternative setup to tackle this problem. The modified setup is shown in Figure A.6 where we use a lens (with 78 mm focal length) to focus the input beam exactly on the die. The fiber array and chip are two focal lengths from the lens, L=78×2 mm, so that we obtain exact image at the die, maintaining the same MFD at the fiber array and the fibers spacings (250 μm). Figure A.8 shows the measured transmission of a GC and a LT-MZI using the modified setup. Results show that the detected output power is higher than the previous setup by around 4 to 6 dB. However, the response shows a decaying power with wavelength indicating that the incident angle to the GCs is not the optimum one ($8^{o}$). Also, ripples due to Fabry-Perot effect from the fiber array tip can be seen. We can see an envelope in the LT-MZI transmission spectrum with FSR around 5 nm which is close to the actual design FSR.

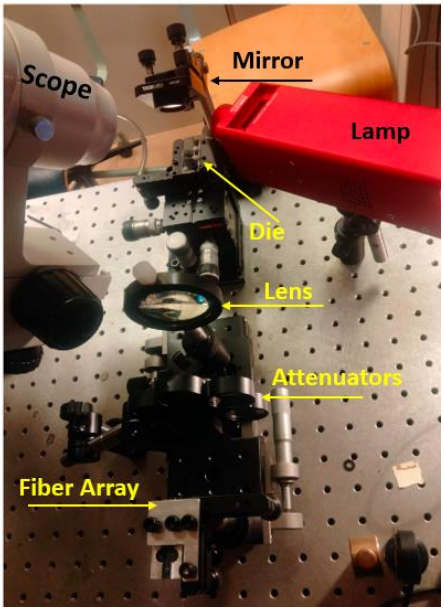



Figure A.6: Image of modified setup.

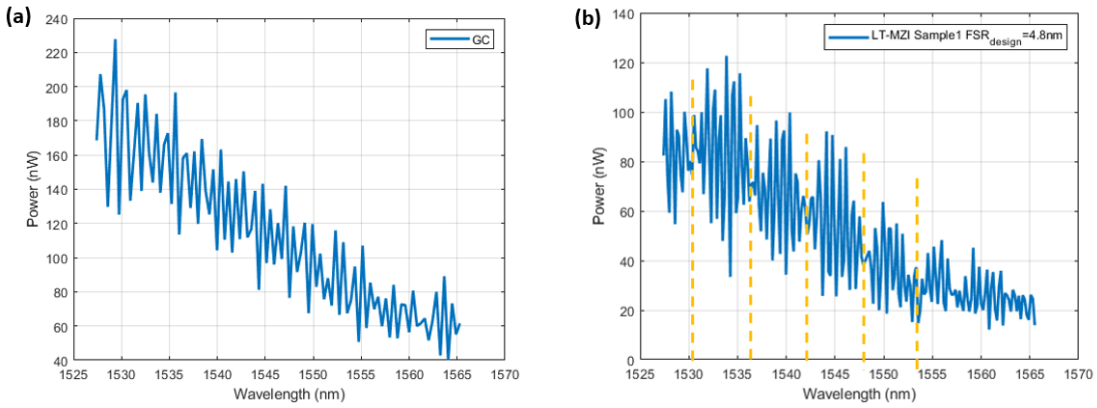



Figure A.7: Measured transmission spectrum of (a) GC and (b) LT-MZI.

Finally, microfluidic channels that can be integrated with our SiN chip to bring the sensing medium (analyte) to our on-chip sensors are also fabricated, see Figure A.8. The preparation of micro-fluidic channel is as follows; Silicon master mold is prepared in cleanroom using standard photolithography. PDMS (Dow Sylgard 184), is mixed with its curing/cross-linking agent in a ratio of 10:1 and poured on the mold then cured at 115 °C for 15 minutes. The channel part is then cut and transferred to substrate. To improve adhesion of PDMS substrate is heated to 75 °C for 10 minutes. Needles are inserted to the ends of the channels (input/output ports)



Figure A.8: Image of fabricated microfluidic channel on top of silicon wafer.

# BIBLIOGRAPHY

[1] Y. Shi *et al.*, "On-Chip Optical Detection of Viruses: A Review," *Adv. Photonics Res.*, vol. 2, no. 4, p. 2000150, 2021.

[2] H. Zhu, H. Zhang, S. Ni, M. Korabečná, L. Yobas, and P. Neuzil, "The vision of point-of-care PCR tests for the COVID-19 pandemic and beyond," *TrAC Trends Anal. Chem.*, vol. 130, p. 115984, 2020.

[3] E. Luan, H. Shoman, D. M. Ratner, K. C. Cheung, and L. Chrostowski, "Silicon photonic biosensors using label-free detection," *Sensors*, vol. 18, no. 10, p. 3519, 2018.

[4] A. Correa-Mena, L. González, L. Quintero-Rodríguez, and I. Zaldivar-Huerta, "Review on integrated optical sensors and its applications," presented at the 2017 IEEE Mexican Humanitarian Technology Conference (MHTC), IEEE, 2017, pp. 170–173.

[5] M. C. Estevez, M. Alvarez, and L. M. Lechuga, "Integrated optical devices for lab-on-a-chip biosensing applications," *Laser Photonics Rev.*, vol. 6, no. 4, pp. 463–487, 2012.

[6] C. Peng *et al.*, "Optical waveguide refractive index sensor for biochemical sensing," *Appl. Sci.*, vol. 13, no. 6, p. 3829, 2023.

[7] Y. Chen, J. Liu, Z. Yang, J. S. Wilkinson, and X. Zhou, "Optical biosensors based on refractometric sensing schemes: A review," *Biosens. Bioelectron.*, vol. 144, p. 111693, 2019.

[8] V. Stojanović *et al.*, “Monolithic silicon-photonic platforms in state-of-the-art CMOS SOI processes,” *Opt. Express*, vol. 26, no. 10, pp. 13106–13121, 2018.

[9] X. Fan, I. M. White, S. I. Shopova, H. Zhu, J. D. Suter, and Y. Sun, “Sensitive optical biosensors for unlabeled targets: A review,” *Anal. Chim. Acta*, vol. 620, no. 1–2, pp. 8–26, 2008.

[10] Y. Xie, M. Zhang, and D. Dai, “Design rule of Mach-Zehnder interferometer sensors for ultra-high sensitivity,” *Sensors*, vol. 20, no. 9, p. 2640, 2020.

[11] P. Kumaar and A. Sivasubramanian, “Optimization of the transverse electric photonic strip waveguide biosensor for detecting diabetes mellitus from bulk sensitivity,” *J. Healthc. Eng.*, vol. 2021, 2021.

[12] C. Zhao, L. Xu, and L. Liu, “Ultrahigh sensitivity Mach− Zehnder interferometer sensor based on a weak one-dimensional field confinement silica waveguide,” *Sensors*, vol. 21, no. 19, p. 6600, 2021.

[13] J. T. Robinson, L. Chen, and M. Lipson, “On-chip gas detection in silicon optical microcavities,” *Opt. Express*, vol. 16, no. 6, pp. 4296–4301, 2008.

[14] S. T. Fard *et al.*, “Performance of ultra-thin SOI-based resonators for sensing applications,” *Opt. Express*, vol. 22, no. 12, pp. 14166–14179, 2014.

[15] L. Jin, M. Li, and J.-J. He, “Analysis of wavelength and intensity interrogation methods in cascaded double-ring sensors,” *J. Light. Technol.*, vol. 30, no. 12, pp. 1994–2002, 2012.

[16] J. Hodgkinson and R. P. Tatam, “Optical gas sensing: a review,” *Meas. Sci. Technol.*, vol. 24, no. 1, p. 012004, 2012.

[17] G. Farca, S. Shopova, and A. Rosenberger, “Cavity-enhanced laser absorption spectroscopy using microresonator whispering-gallery modes,” *Opt. Express*, vol. 15, no. 25, pp. 17443–17448, 2007.

[18] A. Nitkowski, L. Chen, and M. Lipson, “Cavity-enhanced on-chip absorption spectroscopy using microring resonators,” *Opt. Express*, vol. 16, no. 16, pp. 11930–11936, 2008.

[19] A. R. Bastos *et al.*, “Integrated optical Mach-Zehnder interferometer based on organic-inorganic hybrids for photonics-on-a-chip biosensing applications,” *Sensors*, vol. 18, no. 3, p. 840, 2018.

[20] X. Wang, X. Guan, Q. Huang, J. Zheng, Y. Shi, and D. Dai, “Suspended ultra-small disk resonator on silicon for optical sensing,” *Opt. Lett.*, vol. 38, no. 24, pp. 5405–5408, 2013.

[21] L. Huang *et al.*, “Improving the detection limit for on-chip photonic sensors based on subwavelength grating racetrack resonators,” *Opt. Express*, vol. 25, no. 9, pp. 10527–10535, 2017.

[22] M. A. Porcel *et al.*, “Silicon nitride photonic integration for visible light applications,” *Opt. Laser Technol.*, vol. 112, pp. 299–306, 2019.

[23] Q. Wilmart *et al.*, “A versatile silicon-silicon nitride photonics platform for enhanced functionalities and applications,” *Appl. Sci.*, vol. 9, no. 2, p. 255, 2019.

[24] R. Baets *et al.*, “Silicon Photonics: silicon nitride versus silicon-on-insulator,” presented at the Optical fiber communication conference, Optica Publishing Group, 2016, pp. Th3J-1.

[25] E. D. Palik, *Handbook of optical constants of solids*, vol. 3. Academic press, 1998.

[26] A. Kumar, K. Thyagarajan, and A. K. Ghatak, “Analysis of rectangular-core dielectric waveguides: an accurate perturbation approach,” *Opt. Lett.*, vol. 8, no. 1, pp. 63–65, 1983.

[27] E. A. Marcatili, “Dielectric rectangular waveguide and directional coupler for integrated optics,” *Bell Syst. Tech. J.*, vol. 48, no. 7, pp. 2071–2102, 1969.

[28] R. Knox and P. Toulios, “Integrated circuits for the millimeter through optical frequency range,” presented at the Proc. Symp. Submillimeter Waves, Brooklyn, NY, 1970, pp. 497–515.

[29] *Lumerical MODE Solutions*. ANSYS. [Online]. Available: https://www.ansys.com/products/optics/mode

[30] M. S. Murib, D. Martens, and P. Bienstman, “Label-free real-time optical monitoring of DNA hybridization using SiN Mach–Zehnder interferometer-based integrated biosensing platform,” *J. Biomed. Opt.*, vol. 23, no. 12, pp. 127002–127002, 2018.

[31] Q. Liu *et al.*, “Highly sensitive Mach–Zehnder interferometer biosensor based on silicon nitride slot waveguide,” *Sens. Actuators B Chem.*, vol. 188, pp. 681–688, 2013.

[32] M. R. Bryan, D. J. Steiner, J. S. Cognetti, and B. L. Miller, “Design, manufacture, and testing of a silicon nitride ring resonator-based biosensing platform,” presented at the Chemical, Biological, Radiological, Nuclear, and Explosives (CBRNE) Sensing XIX, SPIE, 2018, pp. 104–108.

[33] G. A. Besselink, D. H. Geuzebroek, F. Schreuder, F. Falke, A. Leinse, and R. G. Heideman, “Microring resonator biosensing platform for sensitive detection of thrombin,” presented at the Microfluidics, BioMEMS, and Medical Microsystems XVII, SPIE, 2019, pp. 177–180.

[34] M. J. Al Mahmod, R. Hyder, and M. Z. Islam, “A highly sensitive metal–insulator–metal ring resonator-based nanophotonic structure for biosensing applications,” *IEEE Sens. J.*, vol. 18, no. 16, pp. 6563–6568, 2018.

[35] T. Yoshie, L. Tang, and S.-Y. Su, “Optical microcavity: Sensing down to single molecules and atoms,” *Sensors*, vol. 11, no. 2, pp. 1972–1991, 2011.

[36] R. Samson, G. R. Navale, and M. S. Dharne, “Biosensors: frontiers in rapid detection of COVID-19,” *3 Biotech*, vol. 10, no. 9, p. 385, 2020.

[37] O. Bratash, A. Buhot, L. Leroy, and E. Engel, “Optical fiber biosensors toward in vivo detection.,” *Biosens. Bioelectron.*, p. 116088, 2024.

[38] L. Li *et al.*, “Optical fiber optofluidic bio-chemical sensors: a review,” *Laser Photonics Rev.*, vol. 15, no. 7, p. 2000526, 2021.

[39] B. Song, H. Zhang, B. Liu, W. Lin, and J. Wu, “Label-free in-situ real-time DNA hybridization kinetics detection employing microfiber-assisted Mach-Zehnder interferometer,” *Biosens. Bioelectron.*, vol. 81, pp. 151–158, 2016.

[40] S. Gao *et al.*, “High-sensitivity DNA biosensor based on microfiber Sagnac interferometer,” *Opt. Express*, vol. 25, no. 12, pp. 13305–13313, 2017.

[41] R. M. André *et al.*, “Simultaneous measurement of temperature and refractive index using focused ion beam milled Fabry-Perot cavities in optical fiber micro-tips,” *Opt. Express*, vol. 24, no. 13, pp. 14053–14065, 2016.

[42] X. Liu, M. Jiang, Q. Sui, S. Luo, and X. Geng, “Optical fiber Fabry–Perot interferometer for microorganism growth detection,” *Opt. Fiber Technol.*, vol. 30, pp. 32–37, 2016.

[43] X. Li, Y. Shao, Y. Yu, Y. Zhang, and S. Wei, “A highly sensitive fiber-optic Fabry–Perot interferometer based on internal reflection mirrors for refractive index measurement,” *Sensors*, vol. 16, no. 6, p. 794, 2016.

[44] S. Tang *et al.*, “Fabry-Perot interferometer based on a fiber-tip fixed-supported bridge for fast glucose concentration measurement,” *Biosensors*, vol. 12, no. 6, p. 391, 2022.

[45] B. Kavitha, N. Radhika, S. S. Gorthi, and S. Asokan, “Etched fiber Bragg grating sensor for quantification of DNA,” *IEEE Sens. J.*, vol. 21, no. 2, pp. 1588–1595, 2020.

[46] D. Sun, T. Guo, Y. Ran, Y. Huang, and B.-O. Guan, “In-situ DNA hybridization detection with a reflective microfiber grating biosensor,” *Biosens. Bioelectron.*, vol. 61, pp. 541–546, 2014.

[47] E. Kretschmann and H. Raether, “Radiative decay of non radiative surface plasmons excited by light,” *Z. Für Naturforschung A*, vol. 23, no. 12, pp. 2135–2136, 1968.

[48] J. Homola, S. S. Yee, and G. Gauglitz, “Surface plasmon resonance sensors: review,” *Sens Actuators B Chem*, vol. 54, no. 1–2, pp. 3–15, 1999.

[49] M. A. Otte, B. Sepulveda, W. Ni, J. P. Juste, L. M. Liz-Marzán, and L. M. Lechuga, “Identification of the optimal spectral region for plasmonic and nanoplasmonic sensing,” *ACS Nano*, vol. 4, no. 1, pp. 349–357, 2010.

[50] M. H. Lee, H. Gao, and T. W. Odom, “Refractive index sensing using quasi one-dimensional nanoslit arrays,” *Nano Lett.*, vol. 9, no. 7, pp. 2584–2588, 2009.

[51] J. Henzie, M. H. Lee, and T. W. Odom, “Multiscale patterning of plasmonic metamaterials,” *Nat. Nanotechnol.*, vol. 2, no. 9, pp. 549–554, 2007.

[52] N. Kazanskiy, S. Khonina, and M. A. Butt, “Plasmonic sensors based on Metal-insulator-metal waveguides for refractive index sensing applications: A brief review,” *Phys. E Low-Dimens. Syst. Nanostructures*, vol. 117, p. 113798, 2020.

[53] R. S. El Shamy, D. Khalil, and M. A. Swillam, “Mid infrared optical gas sensor using plasmonic Mach-Zehnder interferometer,” *Sci. Rep.*, vol. 10, no. 1, p. 1293, 2020.

[54] Y. Gao, Q. Gan, Z. Xin, X. Cheng, and F. J. Bartoli, “Plasmonic Mach–Zehnder interferometer for ultrasensitive on-chip biosensing,” *ACS Nano*, vol. 5, no. 12, pp. 9836–9844, 2011.

[55] L. Liu, Z. Hu, M. Ye, Z. Yu, C. Ma, and J. Li, “On-chip refractive index sensor with ultra-high sensitivity based on sub-wavelength grating racetrack microring resonators and Vernier effect,” *IEEE Photonics J.*, vol. 14, no. 5, pp. 1–7, 2022.

[56] N. Kazanskiy, M. A. Butt, and S. Khonina, “Silicon photonic devices realized on refractive index engineered subwavelength grating waveguides-A review,” *Opt. Laser Technol.*, vol. 138, p. 106863, 2021.

[57] V. Mere, H. Muthuganesan, Y. Kar, C. Van Kruijsdijk, and S. K. Selvaraja, “On-chip chemical sensing using slot-waveguide-based ring resonator,” *IEEE Sens. J.*, vol. 20, no. 11, pp. 5970–5975, 2020.

[58] D. Yuan, Y. Dong, Y. Liu, and T. Li, “Mach-Zehnder interferometer biochemical sensor based on silicon-on-insulator rib waveguide with large cross section,” *Sensors*, vol. 15, no. 9, pp. 21500–21517, 2015.

[59] A. Densmore *et al.*, “Spiral-path high-sensitivity silicon photonic wire molecular sensor with temperature-independent response,” *Opt. Lett.*, vol. 33, no. 6, pp. 596–598, 2008.

[60] K. E. Zinoviev, A. B. González-Guerrero, C. Domínguez, and L. M. Lechuga, "Integrated bimodal waveguide interferometric biosensor for label-free analysis," *J. Light. Technol.*, vol. 29, no. 13, pp. 1926–1930, 2011.

[61] R. Dwivedi and A. Kumar, "Refractive index sensing using silicon-on-insulator waveguide based directional coupler," *IEEE Sens. Lett.*, vol. 2, no. 3, pp. 1–4, 2018.

[62] A. Yalcin *et al.*, "Optical sensing of biomolecules using microring resonators," *IEEE J. Sel. Top. Quantum Electron.*, vol. 12, no. 1, pp. 148–155, 2006.

[63] M. Iqbal *et al.*, "Label-free biosensor arrays based on silicon ring resonators and high-speed optical scanning instrumentation," *IEEE J. Sel. Top. Quantum Electron.*, vol. 16, no. 3, pp. 654–661, 2010.

[64] S. M. Grist *et al.*, "Silicon photonic micro-disk resonators for label-free biosensing," *Opt. Express*, vol. 21, no. 7, pp. 7994–8006, 2013.

[65] E. Krioukov, D. Klunder, A. Driessen, J. Greve, and C. Otto, "Sensor based on an integrated optical microcavity," *Opt. Lett.*, vol. 27, no. 7, pp. 512–514, 2002.

[66] V. M. Passaro, R. Loiacono, G. D'Amico, and F. De Leonardis, "Design of Bragg grating sensors based on submicrometer optical rib waveguides in SOI," *IEEE Sens. J.*, vol. 8, no. 9, pp. 1603–1611, 2008.

[67] A. S. Jugessur, J. Dou, J. S. Aitchison, R. M. De La Rue, and M. Gnan, "A photonic nano-Bragg grating device integrated with microfluidic channels for bio-sensing applications," *Microelectron. Eng.*, vol. 86, no. 4–6, pp. 1488–1490, 2009.

[68] P. Prabhathan, V. Murukeshan, Z. Jing, and P. V. Ramana, "Compact SOI nanowire refractive index sensor using phase shifted Bragg grating," *Opt. Express*, vol. 17, no. 17, pp. 15330–15341, 2009.

[69] E. Chow, A. Grot, L. Mirkarimi, M. Sigalas, and G. Girolami, "Ultracompact biochemical sensor built with two-dimensional photonic crystal microcavity," *Opt. Lett.*, vol. 29, no. 10, pp. 1093–1095, 2004.

[70] S. Mandal and D. Erickson, "Nanoscale optofluidic sensor arrays," *Opt. Express*, vol. 16, no. 3, pp. 1623–1631, 2008.

[71] Y. Shi, K. Ma, and D. Dai, "Sensitivity enhancement in Si nanophotonic waveguides used for refractive index sensing," *Sensors*, vol. 16, no. 3, p. 324, 2016.

[72] V. R. Almeida, Q. Xu, C. A. Barrios, and M. Lipson, "Guiding and confining light in void nanostructure," *Opt. Lett.*, vol. 29, no. 11, pp. 1209–1211, 2004.

[73] X. Wang *et al.*, "A silicon photonic biosensor using phase-shifted Bragg gratings in slot waveguide," *J. Biophotonics*, vol. 6, no. 10, pp. 821–828, 2013.

[74] T. Claes, J. G. Molera, K. De Vos, E. Schacht, R. Baets, and P. Bienstman, "Label-free biosensing with a slot-waveguide-based ring resonator in silicon on insulator," *IEEE Photonics J.*, vol. 1, no. 3, pp. 197–204, 2009.

[75] T. Li, D. Gao, D. Zhang, and E. Cassan, "High-$ Q $ and high-sensitivity one-dimensional photonic crystal slot nanobeam cavity sensors," *IEEE Photonics Technol. Lett.*, vol. 28, no. 6, pp. 689–692, 2015.

[76] C. F. Carlborg *et al.*, “A packaged optical slot-waveguide ring resonator sensor array for multiplex label-free assays in labs-on-chips,” *Lab. Chip*, vol. 10, no. 3, pp. 281–290, 2010.

[77] A. M. Taha, B. Paredes, A. Khilo, and M. S. Dahlem, “SOI-based centimeter-scale Mach-Zehnder interferometers for fluid sensing,” presented at the Integrated Optics: Devices, Materials, and Technologies XXI, SPIE, 2017, pp. 133–137.

[78] J. Flueckiger *et al.*, “Sub-wavelength grating for enhanced ring resonator biosensor,” *Opt. Express*, vol. 24, no. 14, pp. 15672–15686, 2016.

[79] S. TalebiFard *et al.*, “Optimized sensitivity of silicon-on-insulator (SOI) strip waveguide resonator sensor,” *Biomed. Opt. Express*, vol. 8, no. 2, pp. 500–511, 2017.

[80] X. Jiang, J. Ye, J. Zou, M. Li, and J.-J. He, “Cascaded silicon-on-insulator double-ring sensors operating in high-sensitivity transverse-magnetic mode,” *Opt. Lett.*, vol. 38, no. 8, pp. 1349–1351, 2013.

[81] F. Prieto *et al.*, “An integrated optical interferometric nanodevice based on silicon technology for biosensor applications,” *Nanotechnology*, vol. 14, no. 8, p. 907, 2003.

[82] D. M. Kita, J. Michon, S. G. Johnson, and J. Hu, “Are slot and sub-wavelength grating waveguides better than strip waveguides for sensing?,” *Optica*, vol. 5, no. 9, pp. 1046–1054, 2018.

[83] M. A. Butt, N. L. Kazanskiy, S. N. Khonina, G. S. Voronkov, E. P. Grakhova, and R. V. Kutluyarov, "A review on photonic sensing technologies: status and outlook," *Biosensors*, vol. 13, no. 5, p. 568, 2023.

[84] T. Claes, W. Bogaerts, and P. Bienstman, "Experimental characterization of a silicon photonic biosensor consisting of two cascaded ring resonators based on the Vernier-effect and introduction of a curve fitting method for an improved detection limit," *Opt. Express*, vol. 18, no. 22, pp. 22747–22761, 2010.

[85] V. M. Passaro, B. Troia, and F. De Leonardis, "A generalized approach for design of photonic gas sensors based on Vernier-effect in mid-IR," *Sens. Actuators B Chem.*, vol. 168, pp. 402–420, 2012.

[86] J. Hu and D. Dai, "Cascaded-ring optical sensor with enhanced sensitivity by using suspended Si-nanowires," *IEEE Photonics Technol. Lett.*, vol. 23, no. 13, pp. 842–844, 2011.

[87] X. Jiang, Y. Chen, F. Yu, L. Tang, M. Li, and J.-J. He, "High-sensitivity optical biosensor based on cascaded Mach–Zehnder interferometer and ring resonator using Vernier effect," *Opt. Lett.*, vol. 39, no. 22, pp. 6363–6366, 2014.

[88] O. A. Marsh, Y. Xiong, and N. Y. Winnie, "Slot waveguide ring-assisted Mach–Zehnder interferometer for sensing applications," *IEEE J. Sel. Top. Quantum Electron.*, vol. 23, no. 2, pp. 440–443, 2016.

[89] “Optiqua,” Optiqua. Accessed: Dec. 23, 2024. [Online]. Available: https://optiqua.com/technology/mach-zehnder-interferometer/

[90] “Genalyte,” Genalyte. Accessed: Dec. 23, 2024. [Online]. Available: www.genalyte.com

[91] L. S. Puumala *et al.*, “Biofunctionalization of multiplexed silicon photonic biosensors,” *Biosensors*, vol. 13, no. 1, p. 53, 2022.

[92] G. Roelkens, “III-V-on-silicon photonic integrated circuits for optical communication and sensing,” presented at the Frontiers in Optics, Optica Publishing Group, 2016, pp. FF5F-4.

[93] Z. Zhou, B. Yin, and J. Michel, “On-chip light sources for silicon photonics,” *Light Sci. Appl.*, vol. 4, no. 11, pp. e358–e358, 2015.

[94] C. Lerma Arce, D. Witters, R. Puers, J. Lammertyn, and P. Bienstman, “Silicon photonic sensors incorporated in a digital microfluidic system,” *Anal. Bioanal. Chem.*, vol. 404, pp. 2887–2894, 2012.

[95] L. Luan, M. W. Royal, R. Evans, R. B. Fair, and N. M. Jokerst, “Chip scale optical microresonator sensors integrated with embedded thin film photodetectors on electrowetting digital microfluidics platforms,” *IEEE Sens. J.*, vol. 12, no. 6, pp. 1794–1800, 2011.

[96] A. Himeno, K. Kato, and T. Miya, “Silica-based planar lightwave circuits,” *IEEE J. Sel. Top. Quantum Electron.*, vol. 4, no. 6, pp. 913–924, 1998.

[97] A. Rahim *et al.*, “Expanding the silicon photonics portfolio with silicon nitride photonic integrated circuits,” *J. Light. Technol.*, vol. 35, no. 4, pp. 639–649, 2017.

[98] C. R. Doerr, “Integrated photonic platforms for telecommunications: InP and Si,” *IEICE Trans. Electron.*, vol. 96, no. 7, pp. 950–957, 2013.

[99] T. Sharma *et al.*, “Review of recent progress on silicon nitride-based photonic integrated circuits,” *Ieee Access*, vol. 8, pp. 195436–195446, 2020.

[100] K. Suzuki *et al.*, “Wavelength (DE) MUX-and-switch based on 5.5%-Δ-silica PLC/silicon photonics hybrid platform,” *J. Light. Technol.*, vol. 40, no. 6, pp. 1810–1814, 2022.

[101] L. Wan *et al.*, “Demonstration of versatile whispering-gallery micro-lasers for remote refractive index sensing,” *Opt. Express*, vol. 26, no. 5, pp. 5800–5809, 2018.

[102] M. A. Butt, M. Shahbaz, and R. Piramidowicz, “Racetrack ring resonator integrated with multimode interferometer structure based on low-cost silica–titania platform for refractive index sensing application,” presented at the Photonics, MDPI, 2023, p. 978.

[103] Y. Zuta, I. Goykhman, B. Desiatov, and U. Levy, “On-chip switching of a silicon nitride micro-ring resonator based on digital microfluidics platform,” *Opt. Express*, vol. 18, no. 24, pp. 24762–24769, 2010.

[104] M. Liu, L. Ma, C. Qiu, and Z. He, “Refractive Index Sensor Based on Few-mode Silicon-Nitride Micro-Ring Resonator,” presented at the 2019 24th OptoElectronics and

Communications Conference (OECC) and 2019 International Conference on Photonics in Switching and Computing (PSC), IEEE, 2019, pp. 1–3.

[105] K. B. Gylfason *et al.*, “On-chip temperature compensation in an integrated slot-waveguide ring resonator refractive index sensor array,” *Opt. Express*, vol. 18, no. 4, pp. 3226–3237, 2010.

[106] P. Liu and Y. Shi, “Simultaneous measurement of refractive index and temperature using a dual polarization ring,” *Appl. Opt.*, vol. 55, no. 13, pp. 3537–3541, 2016.

[107] A. Arbabi and L. L. Goddard, “Measurements of the refractive indices and thermo-optic coefficients of Si 3 N 4 and SiO x using microring resonances,” *Opt. Lett.*, vol. 38, no. 19, pp. 3878–3881, 2013.

[108] H. Zengzhi *et al.*, “High confinement factor ridge slot waveguide for optical sensing,” *IEEE Photonics Technol. Lett.*, vol. 27, no. 22, pp. 2395–2398, 2015.

[109] S. Y. Siew *et al.*, “Review of silicon photonics technology and platform development,” *J. Light. Technol.*, vol. 39, no. 13, pp. 4374–4389, 2021.

[110] Y. Su, Y. Zhang, C. Qiu, X. Guo, and L. Sun, “Silicon photonic platform for passive waveguide devices: materials, fabrication, and applications,” *Adv. Mater. Technol.*, vol. 5, no. 8, p. 1901153, 2020.

[111] L. G. Carpenter *et al.*, “Towards low propagation losses in active photonic multi-project wafer runs,” presented at the Integrated Photonics Research, Silicon and Nanophotonics, Optica Publishing Group, 2021, pp. ITu3A-5.

[112] L. Carroll *et al.*, “Photonic packaging: transforming silicon photonic integrated circuits into photonic devices,” *Appl. Sci.*, vol. 6, no. 12, p. 426, 2016.

[113] X. Mu, S. Wu, L. Cheng, and H. Fu, “Edge couplers in silicon photonic integrated circuits: A review,” *Appl. Sci.*, vol. 10, no. 4, p. 1538, 2020.

[114] Cheng, “Grating couplers on silicon photonics: Design principles, emerging trends and practical issues,” *Micromachines*, vol. 11, no. 7, 2020.

[115] M. Bahadori, M. Nikdast, Q. Cheng, and K. Bergman, “Universal design of waveguide bends in silicon-on-insulator photonics platform,” *J. Light. Technol.*, vol. 37, no. 13, pp. 3044–3054, 2019.

[116] Y. A. Vlasov and S. J. McNab, “Losses in single-mode silicon-on-insulator strip waveguides and bends,” *Opt. Express*, vol. 12, no. 8, pp. 1622–1631, 2004.

[117] P. Muñoz *et al.*, “State of the art of silicon nitride photonics integration platforms,” presented at the 2017 19th International Conference on Transparent Optical Networks (ICTON), IEEE, 2017, pp. 1–4.

[118] M. J. Shaw, J. Guo, G. A. Vawter, S. Habermehl, and C. T. Sullivan, “Fabrication techniques for low-loss silicon nitride waveguides,” presented at the Micromachining Technology for Micro-Optics and Nano-Optics III, SPIE, 2005, pp. 109–118.

[119] T. Lin *et al.*, “Ultra-broadband and highly efficient silicon nitride bi-layer grating couplers,” *Opt. Commun.*, vol. 530, p. 129209, 2023.

[120] X. Zhu *et al.*, “Low-loss fiber-to-chip edge coupler for silicon nitride integrated circuits,” *Opt. Express*, vol. 31, no. 6, pp. 10525–10532, 2023.

[121] Y. Chen, R. Halir, Í. Molina-Fernández, P. Cheben, and J.-J. He, “High-efficiency apodized-imaging chip-fiber grating coupler for silicon nitride waveguides,” *Opt. Lett.*, vol. 41, no. 21, pp. 5059–5062, 2016.

[122] Y. Hibino, “Silica-based planar lightwave circuits and their applications,” *MRS Bull.*, vol. 28, no. 5, pp. 365–371, 2003.

[123] M. Popovic, K. Wada, S. Akiyama, H. A. Haus, and J. Michel, “Air trenches for sharp silica waveguide bends,” *J. Light. Technol.*, vol. 20, no. 9, pp. 1762–1772, 2002.

[124] M. Takahashi, Y. Uchida, S. Yamasaki, J. Hasegawa, and T. Yagi, “Compact and low-loss coherent mixer based on high Δ ZrO 2-SiO 2 PLC,” *J. Light. Technol.*, vol. 32, no. 17, pp. 3081–3088, 2014.

[125] H. Takahashi, “High performance planar lightwave circuit devices for large capacity transmission,” *Opt. Express*, vol. 19, no. 26, pp. B173–B180, 2011.

[126] M. Kawachi, “Silica waveguides on silicon and their application to integrated-optic components,” *Opt. Quantum Electron.*, vol. 22, pp. 391–416, 1990.

[127] M. Kawachi, “Recent progress in silica-based planar lightwave circuits on silicon,” *IEE Proc.-Optoelectron.*, vol. 143, no. 5, pp. 257–262, 1996.

[128] R. Palmer *et al.*, “Low-loss silicon strip-to-slot mode converters,” *IEEE Photonics J.*, vol. 5, no. 1, pp. 2200409–2200409, 2013.

[129] K. Debnath *et al.*, “Low-loss slot waveguides with silicon (111) surfaces realized using anisotropic wet etching,” *Front. Mater.*, vol. 3, p. 51, 2016.

[130] V. H. Chen, J. R. Ong, and C. E. Png, “Polarisation independent silicon-on-insulator slot waveguides,” *Sci. Rep.*, vol. 6, no. 1, p. 37760, 2016.

[131] V. M. Passaro and M. La Notte, “Optimizing SOI slot waveguide fabrication tolerances and strip-slot coupling for very efficient optical sensing,” *Sensors*, vol. 12, no. 3, pp. 2436–2455, 2012.

[132] N. Feilchenfeld *et al.*, “Silicon photonics and challenges for fabrication,” presented at the Advanced Etch Technology for Nanopatterning VI, SPIE, 2017, p. 101490D.

[133] S. Maier, “Plasmonics: Fundamentals and Applications,” 2007.

[134] P. Berini, “Bulk and surface sensitivities of surface plasmon waveguides,” *New J. Phys.*, vol. 10, no. 10, p. 105010, 2008.

[135] Y. Chowdhury, “Plasmonic waveguides: design and comparative study,” 2011.

[136] Y. Xu *et al.*, “Optical refractive index sensors with plasmonic and photonic structures: promising and inconvenient truth,” *Adv. Opt. Mater.*, vol. 7, no. 9, p. 1801433, 2019.

[137] A. Sánchez-Ramírez, J. M. Luque-González, J. G. Wangüemert-Pérez, A. Ortega-Moñux, and Í. Molina-Fernández, “Comparing dielectric and plasmonic waveguides in a Mach–Zehnder architecture with coherent readout for sensing applications,” *Opt. Laser Technol.*, vol. 168, p. 109813, 2024.

[138] L. Gao *et al.*, “On-chip plasmonic waveguide optical waveplate,” *Sci. Rep.*, vol. 5, no. 1, p. 15794, 2015.

[139] K. J. Ooi, P. Bai, H. S. Chu, and L. K. Ang, “Ultracompact vanadium dioxide dual-mode plasmonic waveguide electroabsorption modulator,” *Nanophotonics*, vol. 2, no. 1, pp. 13–19, 2013.

[140] M. G. Saber *et al.*, “A CMOS compatible ultracompact silicon photonic optical add-drop multiplexer with misaligned sidewall Bragg gratings,” *IEEE Photonics J.*, vol. 9, no. 3, pp. 1–10, 2017.

[141] X. Wang *et al.*, “Precise control of the coupling coefficient through destructive interference in silicon waveguide Bragg gratings,” *Opt. Lett.*, vol. 39, no. 19, pp. 5519–5522, 2014.

[142] X. Wang, “Silicon photonic waveguide Bragg gratings,” 2013.

[143] R. S. El Shamy, A. E. Afifi, M. M. Badr, and M. A. Swillam, “Modelling, characterization, and applications of silicon on insulator loop terminated asymmetric Mach Zehnder interferometer,” *Sci. Rep.*, vol. 12, no. 1, p. 3598, 2022.

[144] R. S. El Shamy, M. A. Swillam, and X. Li, “On-chip complex refractive index detection at multiple wavelengths for selective sensing,” *Sci. Rep.*, vol. 12, no. 1, p. 9343, 2022.

[145] X. Tu *et al.*, “Thermal independent silicon-nitride slot waveguide biosensor with high sensitivity,” *Opt. Express*, vol. 20, no. 3, pp. 2640–2648, 2012.

[146] K. Zinoviev, L. G. Carrascosa, J. Sánchez del Río, B. Sepúlveda, C. Domínguez, and L. M. Lechuga, “Silicon photonic biosensors for lab-on-a-chip applications,” 2008.

[147] J. J. Lillie, M. A. Thomas, N.-M. Jokerst, S. E. Ralph, K. A. Dennis, and C. L. Henderson, “Multimode interferometric sensors on silicon optimized for fully integrated complementary-metal-oxide-semiconductor chemical-biological sensor systems,” *JOSA B*, vol. 23, no. 4, pp. 642–651, 2006.

[148] D. Sun, Y. Ran, and G. Wang, “Label-free detection of cancer biomarkers using an in-line taper fiber-optic interferometer and a fiber Bragg grating,” *Sensors*, vol. 17, no. 11, p. 2559, 2017.

[149] K. D. Vos, I. Bartolozzi, E. Schacht, P. Bienstman, and R. Baets, “Silicon-on-Insulator microring resonator for sensitive and label-free biosensing,” *Opt. Express*, vol. 15, no. 12, pp. 7610–7615, 2007.

[150] S. TalebiFard *et al.*, “Optimized sensitivity of silicon-on-insulator (SOI) strip waveguide resonator sensor,” *Biomed. Opt. Express*, vol. 8, no. 2, pp. 500–511, 2017.

[151] Z. Xie *et al.*, “Highly-sensitive optical biosensor based on equal FSR cascaded microring resonator with intensity interrogation for detection of progesterone molecules,” *Opt. Express*, vol. 25, no. 26, pp. 33193–33201, 2017.

[152] O. Parriaux and G. Veldhuis, “Normalized analysis for the sensitivity optimization of integrated optical evanescent-wave sensors,” *J. Light. Technol.*, vol. 16, no. 4, p. 573, 1998.

[153] G. Veldhuis, O. Parriaux, H. Hoekstra, and P. Lambeck, “Sensitivity enhancement in evanescent optical waveguide sensors,” *J. Light. Technol.*, vol. 18, no. 5, pp. 677–682, 2000.

[154] A. Densmore *et al.*, “A silicon-on-insulator photonic wire based evanescent field sensor,” *IEEE Photonics Technol. Lett.*, vol. 18, no. 23, pp. 2520–2522, 2006.

[155] F. Dell’Olio and V. M. Passaro, “Optical sensing by optimized silicon slot waveguides,” *Opt. Express*, vol. 15, no. 8, pp. 4977–4993, 2007.

[156] D. B. Mortimore, “Fiber loop reflectors,” *J. Light. Technol.*, vol. 6, no. 7, pp. 1217–1224, 1988.

[157] C. Han, H. Ding, and F. Lv, “Demonstration of a refractometric sensor based on an optical micro-fiber three-beam interferometer,” *Sci. Rep.*, vol. 4, no. 1, p. 7504, 2014.

[158] J.-J. Guo, Y. Yang, and G.-D. Peng, “Analysis of polarization-independent tunable optical comb filter by cascading MZI and phase modulating Sagnac loop,” *Opt. Commun.*, vol. 284, no. 21, pp. 5144–5147, 2011.

[159] X. Jiang *et al.*, “Wavelength and bandwidth-tunable silicon comb filter based on Sagnac loop mirrors with Mach-Zehnder interferometer couplers,” *Opt. Express*, vol. 24, no. 3, pp. 2183–2188, 2016.

[160] D. Patel, V. Veerasubramanian, S. Ghosh, A. Samani, Q. Zhong, and D. V. Plant, “High-speed compact silicon photonic Michelson interferometric modulator,” *Opt. Express*, vol. 22, no. 22, pp. 26788–26802, 2014.

[161] *Lumerical FDTD*. ANSYS. [Online]. Available: https://www.ansys.com/products/optics/fdtd

[162] Z. Wang, N. Zhu, Y. Tang, L. Wosinski, D. Dai, and S. He, "Ultracompact low-loss coupler between strip and slot waveguides," *Opt. Lett.*, vol. 34, no. 10, pp. 1498–1500, 2009.

[163] H. Zhang *et al.*, "CMOS-compatible fabrication of silicon-based sub-100-nm slot waveguide with efficient channel-slot coupler," *IEEE Photonics Technol. Lett.*, vol. 24, no. 1, pp. 10–12, 2011.

[164] Q. Deng, Q. Yan, L. Liu, X. Li, J. Michel, and Z. Zhou, "Robust polarization-insensitive strip-slot waveguide mode converter based on symmetric multimode interference," *Opt. Express*, vol. 24, no. 7, pp. 7347–7355, 2016.

[165] *Lumerical INTERCONNECT*. ANSYS. [Online]. Available: https://www.ansys.com/products/optics/interconnect

[166] B. H. Sang and T.-I. Jeon, "Pressure-dependent refractive indices of gases by THz time-domain spectroscopy," *Opt. Express*, vol. 24, no. 25, pp. 29040–29047, 2016.

[167] "Applied Nanotools." [Online]. Available: https://www.appliednt.com/

[168] University of British Columbia, "Microsystems and Nanotechnology (MiNa) Group."

[169] "Maple Leaf Photonics," Maple Leaf Photonics. [Online]. Available: http://mapleleafphotonics.com/

[170] D. Duval, J. Osmond, S. Dante, C. Dominguez, and L. M. Lechuga, “Grating couplers integrated on Mach-Zehnder interferometric biosensors operating in the visible range,” *IEEE Photonics J.*, vol. 5, no. 2, pp. 3700108–3700108, 2013.

[171] H. Sakamoto, Y. Minpou, T. Sawai, Y. Enami, and S. Suye, “A novel optical biosensing system using Mach–Zehnder-type optical waveguide for influenza virus detection,” *Appl. Biochem. Biotechnol.*, vol. 178, pp. 687–694, 2016.

[172] A. Ymeti *et al.*, “Fast, ultrasensitive virus detection using a young interferometer sensor,” *Nano Lett.*, vol. 7, no. 2, pp. 394–397, 2007.

[173] C. Yu, A. Ganjoo, H. Jain, C. Pantano, and J. Irudayaraj, “Mid-IR biosensor: detection and fingerprinting of pathogens on gold island functionalized chalcogenide films,” *Anal. Chem.*, vol. 78, no. 8, pp. 2500–2506, 2006.

[174] M. Angelopoulou, S. Kakabakos, and P. Petrou, “Label-free biosensors based onto monolithically integrated onto silicon optical transducers,” *Chemosensors*, vol. 6, no. 4, p. 52, 2018.

[175] S. Mukherji and N. Punjabi, “Label—Free integrated optical biosensors for multiplexed analysis,” *J. Indian Inst. Sci.*, vol. 92, no. 2, pp. 253–294, 2012.

[176] A. M. Armani and K. J. Vahala, “Heavy water detection using ultra-high-Q microcavities,” *Opt. Lett.*, vol. 31, no. 12, pp. 1896–1898, 2006.

[177] X. Zhang *et al.*, “On-chip integrated optofluidic complex refractive index sensing using silicon photonic crystal nanobeam cavities,” *Opt. Lett.*, vol. 41, no. 6, pp. 1197–1200, 2016.

[178] T. L. Myers *et al.*, “Accurate measurement of the optical constants n and k for a series of 57 inorganic and organic liquids for optical modeling and detection,” *Appl. Spectrosc.*, vol. 72, no. 4, pp. 535–550, 2018.

[179] S. Kedenburg, M. Vieweg, T. Gissibl, and H. Giessen, “Linear refractive index and absorption measurements of nonlinear optical liquids in the visible and near-infrared spectral region,” *Opt. Mater. Express*, vol. 2, no. 11, pp. 1588–1611, 2012.

[180] “SiEPIC EBeam PDK & Library.” [Online]. Available: https://github.com/SiEPIC/SiEPIC_EBeam_PDK

[181] C. Tan, “Determination of refractive index of silica glass for infrared wavelengths by IR spectroscopy,” *J. Non-Cryst. Solids*, vol. 223, no. 1–2, pp. 158–163, 1998.

[182] R. S. El Shamy, M. A. Swillam, and X. Li, “Comparative Study of Photonic Platforms and Devices for On-Chip Sensing,” presented at the Photonics, MDPI, 2023, p. 1233.

[183] F. Horst, W. M. Green, S. Assefa, S. M. Shank, Y. A. Vlasov, and B. J. Offrein, “Cascaded Mach-Zehnder wavelength filters in silicon photonics for low loss and flat pass-band WDM (de-) multiplexing,” *Opt. Express*, vol. 21, no. 10, pp. 11652–11658, 2013.

[184] Q. Deng, L. Liu, R. Zhang, X. Li, J. Michel, and Z. Zhou, “Athermal and flat-topped silicon Mach-Zehnder filters,” *Opt. Express*, vol. 24, no. 26, pp. 29577–29582, 2016.

[185] R. S. El Shamy, A. A. Osama, A. E. Afifi, and M. A. Swillam, “A compact 100 GHz femtojoule silicon-organic hybrid modulator based on a novel Mach–Zehnder interferometer design,” *J. Opt.*, vol. 23, no. 9, p. 095801, 2021.

[186] D. Kohler *et al.*, “Biophotonic sensors with integrated Si3N4-organic hybrid (SiNOH) lasers for point-of-care diagnostics,” *Light Sci. Appl.*, vol. 10, no. 1, p. 64, 2021.

[187] B. Ouyang *et al.*, “Highly sensitive silicon Mach-Zehnder interferometer-based ultrasound sensor,” presented at the Integrated Optics: Devices, Materials, and Technologies XXIV, SPIE, 2020, pp. 79–87.

[188] L. Chrostowski *et al.*, “Silicon photonic resonator sensors and devices,” presented at the Laser Resonators, Microresonators, and Beam Control XIV, SPIE, 2012, pp. 387–402.

[189] “WHO 2017.” [Online]. Available: http://www.who.int/mediacentre/factsheets/fs310/en/

[190] M. Piatak Jr *et al.*, “High levels of HIV-1 in plasma during all stages of infection determined by competitive PCR,” *Science*, vol. 259, no. 5102, pp. 1749–1754, 1993.

[191] G. Tavazzi *et al.*, “Myocardial localization of coronavirus in COVID-19 cardiogenic shock,” *Eur. J. Heart Fail.*, vol. 22, no. 5, pp. 911–915, 2020.

[192] A.-R. Sahin *et al.*, “2019 novel coronavirus (COVID-19) outbreak: a review of the current literature,” *Ejmo*, vol. 4, no. 1, pp. 1–7, 2020.

[193] P. Brocos, Á. Piñeiro, R. Bravo, and A. Amigo, “Refractive indices, molar volumes and molar refractions of binary liquid mixtures: concepts and correlations,” *Phys. Chem. Chem. Phys.*, vol. 5, no. 3, pp. 550–557, 2003.

[194] V. D. Bruggeman, “Berechnung verschiedener physikalischer Konstanten von heterogenen Substanzen. I. Dielektrizitätskonstanten und Leitfähigkeiten der Mischkörper aus isotropen Substanzen,” *Ann. Phys.*, vol. 416, no. 7, pp. 636–664, 1935.

[195] H. R. Philipp, “Optical properties of silicon nitride,” *J. Electrochem. Soc.*, vol. 120, no. 2, p. 295, 1973.

[196] C. A. Barrios *et al.*, “Label-free optical biosensing with slot-waveguides,” *Opt. Lett.*, vol. 33, no. 7, pp. 708–710, 2008.

[197] K. Luke, A. Dutt, C. B. Poitras, and M. Lipson, “Overcoming Si 3 N 4 film stress limitations for high quality factor ring resonators,” *Opt. Express*, vol. 21, no. 19, pp. 22829–22833, 2013.

[198] C. J. Krückel, A. Fülöp, T. Klintberg, J. Bengtsson, P. A. Andrekson, and V. Torres-Company, “Linear and nonlinear characterization of low-stress high-confinement silicon-rich nitride waveguides,” *Opt. Express*, vol. 23, no. 20, pp. 25827–25837, 2015.

[199] A. Subramanian *et al.*, “Low-loss singlemode PECVD silicon nitride photonic wire waveguides for 532–900 nm wavelength window fabricated within a CMOS pilot line,” *IEEE Photonics J.*, vol. 5, no. 6, pp. 2202809–2202809, 2013.

[200] M. M. El-Rayany, R. S. El Shamy, and M. A. Swillam, “A compact silicon-on-insulator gas sensor,” presented at the Silicon Photonics XIV, SPIE, 2019, pp. 189–196.

[201] J. Hong, A. M. Spring, F. Qiu, and S. Yokoyama, “A high efficiency silicon nitride waveguide grating coupler with a multilayer bottom reflector,” *Sci. Rep.*, vol. 9, no. 1, p. 12988, 2019.

[202] J. C. Mak, W. D. Sacher, H. Ying, X. Luo, P. G.-Q. Lo, and J. K. Poon, “Multi-layer silicon nitride-on-silicon polarization-independent grating couplers,” *Opt. Express*, vol. 26, no. 23, pp. 30623–30633, 2018.

[203] S. Romero-García, F. Merget, F. Zhong, H. Finkelstein, and J. Witzens, “Silicon nitride CMOS-compatible platform for integrated photonics applications at visible wavelengths,” *Opt. Express*, vol. 21, no. 12, pp. 14036–14046, 2013.

[204] J. H. Song *et al.*, “Grating devices on a silicon nitride technology platform for visible light applications,” *OSA Contin.*, vol. 2, no. 4, pp. 1155–1165, 2019.

[205] W. D. Sacher *et al.*, “Visible-light silicon nitride waveguide devices and implantable neurophotonic probes on thinned 200 mm silicon wafers,” *Opt. Express*, vol. 27, no. 26, pp. 37400–37418, 2019.

[206] “THORLABS,” THORLABS. [Online]. Available: https://www.thorlabs.com/